

PLEASANT VALLEY FIRE DEPARTMENT

EMS PROTOCOLS



2026

INTRODUCTION

The following protocols have been developed by The Pleasant Valley Fire Department and Dr Steven Orr, MD, Medical Director for Pleasant Valley Fire Department. These protocols define the standard of care for EMS providers who work for the Pleasant Valley Fire Department.

It is not possible to write a protocol for every scenario EMS providers may encounter and providers are expected to operate in the patient's best interest. Providers are also expected to document their clinical reasoning and judgment for all actions taken.

These protocols are written in a bullet point arrangement and are intended to highlight decision points in patient care and interventions to perform. While they are written in a BLS interventions before ALS interventions format. Headers of "BLS or ALS" or "ALS ONLY" will be noted above listed assessments or interventions. "BLS or ALS" means the following assessments or interventions may be accomplished by either an EMT or Paramedic. "ALS ONLY" means the following assessments or interventions are to be accomplished by only a Paramedic. It is understood that clinical needs may require the Paramedic to move immediately to ALS interventions.

ORGANIZATION

Protocols are organized into Medical or Trauma categories. Medical protocols are subdivided into Combined, Adult, Pediatric, or Obstetrical. Trauma protocols are presented in a combined format as well. Combined Medical and Combined Trauma protocols are protocols that encompass adult and pediatric treatments, as those treatments are similar. Medication doses in the Combined protocols are identified as **ADULT** or **PEDS**. A Medication Formulary is included as well as an Appendix.

COLOR KEY

In order to keep clutter to a minimum, these protocols are written in color type:

Any assessment or intervention in black type (normal, CAPITALIZED, or **BOLD**) can be accomplished by EMT or Paramedic

ANY INTERVENTION IN CAPITALIZED BOLD GREEN TYPE MUST BE ACCOMPLISHED BY A PARAMEDIC

ANY INTERVENTION IN CAPITALIZED BOLD RED TYPE THAT IS UNDERLINED REQUIRES MEDICAL CONTROL CONTACT

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PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

PROTOCOL AUTHORIZATION



PROTOCOL AUTHORIZATION

The following set of medical protocols has been developed for use by the Pleasant Valley Fire Department's Emergency Medical Services. These Protocols have been reviewed and approved for use by Dr Steven Orr, M.D. who serves as the Agency's Medical Director

Steven Orr, M.D.

Date



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

BIOHAZARD EXPOSURE PROTECTION



BIOHAZARD EXPOSURE PROTECTION

While the best care and treatment of patients shall be provided by the paramedics of the Pleasant Valley Fire Department, protection from risk of exposure to infectious diseases must be addressed.

The paramedic will use the following precautions when establishing an IV line, administering I.M. or Sub-q medications, or using Lancet needles to obtain blood sample

1. Wear protective gloves.
2. Prepare site with antiseptic solution.
3. Minimize blood loss through catheter.
4. Cover site post injection with sterile cover.
5. Dispose of used needles in appropriate containers.

The paramedic will use the following precautions when performing invasive procedures, such as intubations:

1. Wear protective gloves.
2. Wear protective eyewear and face mask if able.
3. Dispose of contaminated waste in appropriate containers.

Crews shall clean and disinfect all surfaces and the floor after each patient transport.

EMS crew safety is of primary concern.

In the event of an exposure, refer to department S.O.G. on exposure.



DO NOT RESUSCITATE



DO NOT RESUSCITATE

Outside the hospital Do Not Resuscitate (DNR) requests are orders by a patient's physician to refrain from initiating cardiopulmonary resuscitative measures in the event of a cardiac or respiratory arrest. DNR requests are compatible with maximal therapeutic care and the patient may receive vigorous support (i.e. IV's & drugs) until the point of cardiac or respiratory arrest.

The EMS provider will honor the DNR request when properly executed and presented. Guidelines for acceptance of DNR requests include:

GUIDELINES

1. If any doubt exists about the validity of the DNR form or the intent of the patient or the appropriate surrogate, resuscitation must be initiated and **MEDICAL CONTROL CONTACTED**.
2. The patient must not be pregnant
3. Alternately worded directives to limit care, (i.e. Supportive Care Only, Limited Terminal Care, Justified Use of Conservative Care) are not acceptable substitutes for DNR directives. If presented with an alternate form, **MEDICAL CONTROL MUST BE CONTACTED**.
4. The DNR form must be signed and dated by the patient or appropriate surrogate, the witness, and the patient's physician.
5. The revocation provision must remain unsigned in order for this DNR form to remain valid.
6. The DNR form must be with the patient at the time of the EMS providers' arrival.
7. The form must be an original and seen by the EMS provider.
8. No cardiac monitoring is necessary. No Medical Control contact is necessary.
9. **If a DNR Request Form is presented after BLS or ALS resuscitative procedures (including the AED) have begun, all resuscitative efforts shall cease.**

DOCUMENTATION

The EMS report shall include the existence of a DNR and the DNR request form number. If a copy can be obtained it shall be attached to the EMS report. The original DNR form shall remain at the facility or home. If the patient is transported, the DNR form will accompany the patient.



DOA POLICY



DOA POLICY

For patients with a **MEDICAL** etiology:

1. CPR and ALS treatment are to be withheld only if the patient is obviously dead per the following criteria:
 - a. If ANY of the following clinical signs of irreversible death:
 - i. Rigor mortis
 - ii. Dependent lividity
 - iii. Tissue decomposition
 - iv. Fetal death < 20 weeks
 - v. Injury incompatible with life:
 1. Decapitation
 2. Incineration
 - b. AND if ALL of the following are present:
 - i. Pulseless
 - ii. No heart tone on auscultation
 - iii. Apnea
 - iv. No pupillary response
2. The patient may be pronounced DOA

IF PRESENTED WITH DO NOT RESUSCITATE ORDERS, REFER TO THE DO NOT RESUSCITATE POLICY

For patients with a **TRAUMATIC** etiology:

1. CPR and ALS treatment are to be withheld only if the patient is obviously dead per the following criteria:
 - a. If ANY of the following clinical signs of irreversible death:
 - i. Rigor mortis
 - ii. Dependent lividity
 - iii. Tissue decomposition
 - iv. Fetal death < 20 weeks
 - v. Injury incompatible with life:
 1. Decapitation
 2. Incineration
 - b. AND if ALL of the following are present:
 - i. Pulseless
 - ii. No heart tone on auscultation
 - iii. Apnea
 - iv. No pupillary response
 - v. Severe mechanism of injury
2. The patient may be pronounced DOA

CONTRAINDICATIONS to pronouncing a patient DOA:

1. Hypothermia
2. Drowning with hypothermia and submersion < 60 minutes
3. Pregnant patient with estimated gestational age > 20 weeks
4. Lightning strike

NOTES

1. It is intended for this policy to be utilized by an EMT or a Paramedic
2. An ePCR detailing the situation and the specific DOA criteria met is to be completed by the pronouncing provider
3. If resuscitation efforts are in progress and the patient meets the above criteria, CONSIDER discontinuing the resuscitation efforts.
4. It is not the policy of PVFD to move or transport deceased patients. These patients should not be moved to an ambulance unless expressly directed to do so by PVFD Incident Command or PVPD as the result of the potential threat of a riot.
5. The pronouncing crew must remain on scene with the body until released by law enforcement or coroner.



HELICOPTER UTILIZATION



HELICOPTER UTILIZATION

INDICATIONS

1. Need for optimum speed of response and transport (keep in mind helicopter response and loading time)
2. Impaired ground accessibility due to location, traffic, or road conditions
3. Shortage of ground transport units and a need for rapid transit
4. Need to achieve a distribution of patients among area hospitals (mass casualty)

PROCEDURES

1. It is the responsibility of the paramedic to assess the situation and determine the need for air ambulance
2. If an air ambulance is needed, it is the responsibility of the paramedic to contact dispatch and request it
3. It shall be the responsibility of the paramedic or their designee to establish a suitable landing zone. The zone should be:
 - a. 100 square feet in area
 - b. Free of overhead obstructions
 - c. Accessible to aircraft for patient transfer
 - d. A fire department pumper should be requested to wet down the area



INFECTIOUS DISEASE



INFECTIOUS DISEASE

The following information is taken from: ***Assistant Secretary for Preparedness and Response: EMS INFECTIOUS DISEASE PLAYBOOK***

STANDARD PRECAUTIONS:

1. EXAMPLE DISEASES:
 - a. AIDS/HIV, anthrax (cutaneous or pulmonary)
 - b. Botulism
 - c. Cellulitis/abscess
 - d. Dengue fever
 - e. Nonspecific upper respiratory infections
2. PPE:
 - a. Gloves during patient contact
 - b. Goggles/face shield and surgical mask for any airway procedures (intubation, suctioning) or patient with active cough from apparent infectious source
 - c. Impermeable gown for any situation likely to generate splash/liquid exposures
 - d. PPE should be removed in an appropriate doffing area
 - e. Avoid self-contamination
 - f. PPE waste should be placed in a labeled leak-proof container
 - g. Potential exposures should be reported according to existing service protocols
3. PATIENT CARE CONSIDERATIONS:
 - a. Provide a surgical mask for all patients with acute infectious respiratory symptoms who can tolerate it
 - b. Provide tissues to patients for secretion control
 - c. Encourage patient to use hand hygiene and cough etiquette practices
4. TRANSPORT CONSIDERATIONS:
 - a. Standard transportation to appropriate hospital facility
 - b. Turn on patient compartment exhaust fan
5. Complete appropriate AMBULANCE DECONTAMINATION

CONTACT PRECAUTIONS:

1. EXAMPLE DISEASES:
 - a. Excessive wound drainage
 - b. MRSA
 - c. Vancomycin-resistant enterococci (VRE),
 - d. C. difficile
 - e. Norovirus
 - f. Infectious diarrhea
 - g. Head lice/body lice/scabies
2. PPE:
 - a. Ensure strict adherence to standard precautions based on situation (e.g., mask, goggles/face shield for splatter risk or airway interventions)
 - b. Disposable fluid-resistant gown that protects the provider's legs as needed
 - c. Disposable gloves
3. PATIENT CARE CONSIDERATIONS:
 - a. Provide a surgical mask for all patients with acute infectious respiratory symptoms who can tolerate it
 - b. Provide tissues to patients for secretion control

- c. Encourage patient to use hand hygiene and cough etiquette practices
- 4. TRANSPORT CONSIDERATIONS:
 - a. Standard transportation to appropriate hospital facility
 - b. Turn on patient compartment exhaust fan
- 5. Ambulance should be taken out of service for appropriate AMBULANCE DECONTAMINATION

DROPLET PRECAUTIONS:

- 1. EXAMPLE DISEASES:
 - a. Neisseria meningitides
 - b. Mumps
 - c. Mycoplasma
 - d. Streptococcal and many other causes of pneumonia
 - e. Parvovirus
 - f. Pertussis
 - g. Pneumonic plague
 - h. Rhinovirus
 - i. Rubella
 - j. Seasonal influenza
 - k. Streptococcal pharyngitis
- 2. PPE:
 - a. Ensure strict adherence to standard precautions based on situation (e.g., mask, goggles/face shield for splatter risk or airway interventions)
 - b. Disposable surgical mask (N95 respirator not required)
 - c. Disposable gloves
 - d. Eye protection – cleanable goggles or disposable face shield
- 3. PATIENT CARE CONSIDERATIONS:
 - a. Provide a surgical mask for all patients with acute infectious respiratory symptoms who can tolerate it
 - b. Provide tissues to patients for secretion control
 - c. Encourage patient to use hand hygiene and cough etiquette practices
 - d. Personnel not in appropriate PPE: maintain a distance of 6 feet from the patient. Wear gloves
 - e. Minimize use of nebulizers (consider metered dose inhalers)
 - f. Minimize airway interventions that may cause coughing (e.g., suctioning)
- 4. TRANSPORT CONSIDERATIONS:
 - a. Standard transportation
 - b. Consider putting patient compartment exhaust vent on high and isolating the driver compartment if performing aerosol producing procedures (airway suctioning, intubation, aerosolized medication administration)
 - c. Increase ventilation - have air or heat on non-recirculating cycle and/or open windows
 - d. Advise receiving hospital of respiratory symptoms
- 5. Ambulance should be taken out of service for appropriate AMBULANCE DECONTAMINATION

AIRBORNE PRECAUTIONS:

- 1. EXAMPLE DISEASES:
 - a. Measles, TB (suspected or confirmed),
 - b. Varicella (chickenpox)
- 2. PPE:
 - a. PPE:
 - b. Ensure strict adherence to standard precautions based on situation (e.g., mask, goggles/face shield for splatter risk or airway interventions)
 - c. Disposable NIOSH-approved, fit-tested N95 respirator or powered air purifying respirators (PAPRs) with full hood and high efficiency particulate air (HEPA) filter
 - d. Disposable exam gloves

3. PATIENT CARE CONSIDERATIONS:

- a. Ensure strict adherence with standard precautions (e.g., add gown or coverall for significant bodily fluid exposures and follow doffing for contact precautions)
- b. Ask the patient to wear a surgical mask (N95 respirator not required) if they are able to tolerate it
- c. Provide tissues to patients for secretion control
- d. Encourage patient to use hand hygiene and cough etiquette practices
- e. Procedures that can generate small particle aerosols (endotracheal intubation and suctioning)
 - i. Protection of the eyes, nose, and mouth – in addition to gown and gloves – is recommended
 - ii. Use of an N95 respirator is recommended

4. TRANSPORT CONSIDERATIONS:

- a. Notify the receiving hospital of the need for an airborne infection isolation room
- b. Consider having the patient compartment exhaust vent on high and isolating the driver compartment
- c. Consider having the driver compartment ventilation fan set to high without recirculation
- d. If driver/pilot compartment cannot be isolated from the patient compartment, vehicle operator to wear NIOSH-approved, fit-tested N95 respirator
- e. Intubated patients should be ventilated with a bag-valve device or ventilator equipped with a HEPA filter on exhalation port

5. Ambulance should be taken out of service for appropriate AMBULANCE DECONTAMINATION

SPECIAL RESPIRATORY PRECAUTIONS:

1. EXAMPLE DISEASES:

- a. Severe acute respiratory syndrome (SARS)
- b. Novel influenza strains (e.g., H7N9)
- c. Smallpox
- d. Covid

2. PPE:

- a. Ensure strict adherence to standard precautions based on situation (e.g., mask, goggles/face shield for splatter risk or airway interventions)
- b. Disposable NIOSH-approved, fit-tested N95 or equivalent/higher level respirator or PAPR with full hood and HEPA filter
- c. Disposable face shield or disposable or cleanable goggles (if not using hooded PAPR)
- d. Disposable fluid-resistant gown that extends to at least mid-calf or disposable fluid-resistant coveralls
- e. Disposable gloves with extended cuffs (strongly consider double-gloving)
- f. Disposable boot/shoe covers

3. PATIENT CARE CONSIDERATIONS:

- a. CONSIDERATIONS:
- b. Ask the patient to wear a surgical mask or N95 if they are able to tolerate it
- c. Provide tissues to patients for secretion control
- d. Encourage patient to use hand hygiene and cough etiquette practices
- e. Exercise caution when performing aerosol-producing procedures (endotracheal intubation, suctioning, administration of nebulized medication, CPAP/BiPAP, CPR)
 - i. Only perform these procedures if medically necessary and cannot be postponed
 - ii. Patients who are intubated should be ventilated with a bag-valve device or ventilator with a HEPA filter on the exhalation port

4. TRANSPORT CONSIDERATIONS:

- a. Notify the receiving hospital of the need for an isolation room
- b. The patient compartment exhaust vent should be on high and the driver compartment should be isolated from the patient compartment if possible
- c. The driver compartment ventilation fan should be set to high without recirculation
- d. The vehicle operator should wear a NIOSH-approved, fit-tested N95 respirator if cab cannot be isolated
- e. Consider transport by portable isolation unit or ambulance preparation
- f. Have a plan for family members wishing to accompany the patient
- g. Contact the appropriate public health authority for further information or actions

5. Ambulance should be taken out of service for appropriate AMBULANCE DECONTAMINATION

EVD/VHF PRECAUTIONS:

1. EXAMPLE DISEASES:
 - a. Ebola
 - b. Marburg virus
 - c. Crimean-Congo fever
2. PPE:
 - a. Ensure strict adherence to standard precautions based on situation (e.g., mask, goggles/face shield for splatter risk or airway interventions)
 - b. Use NIOSH-approved, fit-tested N95 respirator in combination with impermeable hood that covers head and shoulders and a full-face shield or PAPR with impermeable drape-style hood
 - c. Can add a heavier impermeable apron for high-risk situations
 - d. Disposable gloves with extended cuffs (strongly consider double-gloving)
 - e. Disposable boot/shoe covers
3. PATIENT CARE CONSIDERATIONS:
 - a. Ask the patient to wear a surgical mask or N95 respirator if they are able to tolerate it
 - b. Advise the designated receiving hospital as early as possible about a suspect case
 - c. Anticipate additional stool/vomit to reduce contamination of the HCW and the ambulance (emesis bags, towels, and/or impermeable sheet placed on stretcher, adult undergarments)
 - d. Minimize the number of HCWs who make patient contact
 - e. Dedicated medical equipment (ideally disposable) should be used
 - f. If patient is ambulatory, strongly consider a barrier garment, surgical mask, and gloves if tolerated
 - g. If patient is non-ambulatory, comfortably shroud patient in a barrier sheet, and apply surgical
 - h. Exercise caution when performing aerosol-producing procedures (endotracheal intubation, suctioning, administration of nebulized medication, CPAP/BiPAP, CPR)
 - i. Only perform if medically necessary and cannot be postponed
 - i. Do not perform invasive procedures (IV) unless urgently required for patient care or stabilization
 - j. Handle needles/sharps with caution. Dispose of in a single patient, puncture-proof, sealed container
 - k. Consider giving oral or nasal medicine per protocols rather than injectable
 - l. Use hands-free communications devices (e.g., tactical headsets) inside the PPE ensemble to facilitate communication and avoid contamination of radios
 - m. Charting is either verbal to a partner in a clean area or completed after transport
4. TRANSPORT CONSIDERATIONS:
 - a. If the patient is a highly suspect case and stable, consider specialized ambulance preparation
 - b. Confirmed case or interfacility transport should be performed by EMS personnel with properly prepared ambulances
 - c. For emergency transport, consider applying an impermeable barrier sheet or cocoon to the patient to protect the HCW and environmental surfaces
 - d. The driver's compartment should remain clean! No family members or belongings
 - e. Suspected EVD/VHF cases should be transported to a designated regional receiving center
5. Ambulance should be taken out of service for appropriate AMBULANCE DECONTAMINATION



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS MEDICAL CONTROL FORMAT



MEDICAL CONTROL FORMAT

All patient care and treatment shall follow protocol guidelines. Medical Control shall be contacted when indicated in the individual protocol, or when any problem or question exists. When transporting to a hospital other than the medical control hospital the paramedic shall contact that hospital or have the medical control hospital contact that hospital as soon as possible. This is to ensure quality patient care upon arrival at the destination hospital.

When transporting a patient to a hospital other than the medical control hospital and the patient is stable and no orders are to be requested, the receiving hospital should be contacted directly.

The following information should be transmitted to the medical control hospital in all emergency situations:

1. Paramedic's name and unit
2. Patient's chief complaint and brief history
3. Patient's condition (stable vs unstable)
4. Patient's vital signs
5. Findings of exam and treatment done prior to contact
6. Request for additional orders



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

MEDICAL CONTROL HOSPITAL



MEDICAL CONTROL HOSPITAL

North Kansas City Hospital will serve as the primary medical control hospital for the Pleasant Valley Fire Department's emergency medical services. All references within the Pleasant Valley Fire Department's medical protocols for adults shall be to North Kansas City Hospital.

Children's Mercy Downtown will serve as the primary medical control hospital for all pediatric patients.



NARCOTIC ACCOUNTABILITY



NARCOTIC ACCOUNTABILITY

The purpose of this section is to provide the guidelines for inventory of supply and record keeping procedures for the administration of all scheduled narcotic drugs. The appropriate security procedures for these drugs shall also be described.

1. The following drugs shall be considered scheduled narcotic drugs requiring strict accountability:
 - a. Versed
 - b. Fentanyl
2. Controlled Substance Verification will occur at each shift change.
 - a. On-coming paramedic (or Senior EMT if Medic Unit is BLS for the shift) will verify that the red tag on the outside of the Med Vault is present, intact, and the tag number matches the previous red tag number in the Daily Drug Log
 - i. If there is an issue, refer to the Controlled Substance Verification Procedure
 - b. The on-coming paramedic (or Senior EMT if Medic Unit is BLS for the shift) will then sign their name and radio number in the Daily Drug Log
3. The drugs listed in item 1 above shall be kept in the following manner:
 - a. Limited quantities shall be maintained in a sealed lock box stored in a locking cabinet within the patient compartment of the ambulance. The following shall be considered minimum quantities carried in the ambulance:
 - i. 2 vials of Versed
 - ii. 2 vials of Fentanyl
4. Narcotic drug administration shall be documented in ESO. These records shall be completed at the time of the call in the ePCR by::
 - a. Entering in the "Flow Chart" tab. This includes amount given, route, etc. You will have to enter the old tag number and the new tag number. This is a REQUIRED field and it will not let you lock your report without it.
 - b. If you give multiple doses (i.e. 50 mcg of Fentanyl and another 50mcg 15 minutes later) you will have to enter the tag numbers twice. This is an ESO setting. Do not enter the total dose one time if you administered multiple smaller doses.
 - c. Physician giving order for administration (if not a standing order)
 - d. In the "Signatures" tab, click on "Controlled Substances". In the "Witness to Waste Signature" tab, the paramedic will enter the medication name from the drop down. On the "amount" line, enter the amount given to the patient (i.e. "50 mcg given"). The paramedic then puts their name as the "Personnel Witness Name" and signs in the signature box.
 - e. HOSPITAL STAFF MUST WITNESS DISPOSAL! All unused portions shall be wasted at the hospital prior to clearing. In the "Signatures" tab, click on "Controlled Substances". In the "Witness to Waste Signature" tab, the paramedic will enter the medication name from the drop down. On the "amount" line, enter the amount wasted (i.e. "50 mcg wasted"). When you have the nurse or physician sign as a witness, make sure to click the "other" button and add their First and Last name and Title.
 - f. When you go back to the ambulance, make sure you recount/reseal the Narc Box w the new card on the outside and the old card on the inside.



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

PATIENT DESTINATION CRITERIA



PATIENT DESTINATION CRITERIA

GENERAL ROUTING

1. For all life-threatening emergencies, transport the patient to the nearest appropriate hospital defined as the hospital that requires the least amount of transport time.
2. For all non-life-threatening emergencies, transport the patient to the hospital of the patient's choice.
3. Ambulances should not transport more than one patient in the same ambulance unless the patients have established a previous relationship, such as family or friends and are being transport to the same destination hospital.

TIME CRITICAL DIAGNOSIS ROUTING

TRAUMA

Any trauma patient with any of the following criteria should be routed to a Trauma Center even if it is not the nearest hospital.

1. Physiologic Criteria
 - a. Shock, as defined as BP less than 90mmHg (adults)
 - b. Respiratory distress
 - i. RR > 29 or < 10 (adults) or < 20 (infant less than 1 year)
 - c. Altered mental status with Glasgow Coma Scale of less than 14
2. Mechanism of Injury
 - a. Occupant ejection
 - b. Fall from height > 20 feet (adults), > 10 feet (children)
 - c. Pedestrian/bicycle/motorcycle hit at speed > 20mph
 - d. Death of same car occupant
 - e. Prolonged extrication > 20 minutes
 - f. Vehicle telemetry data consistent with high-risk injury
3. Anatomic
 - a. Penetrating injury to head, neck, torso, or extremities proximal to elbow or knee
 - b. Airway burns
 - c. 20% second degree burns and/or 5% third degree burns
 - d. Flail chest
 - e. Two or more proximal long bone fractures
 - f. Pelvic fracture
 - g. Paralysis
 - h. Amputation proximal to wrist or ankle
 - i. Open and/or depressed skull fracture
 - j. Crushed, degloved, or mangled extremity
 - k. Tonsillectomy and Adenoidectomy bleeding post-surgery
4. For any trauma patients with the following criteria, consider transport to a Trauma Center even if it is not the nearest hospital. Contact medical control for guidance if necessary
 - a. Age < 16 or > 55
 - b. Pregnancy greater than 20 weeks
 - c. Patient with bleeding disorder or patient on anticoagulants
 - d. Patient with any known serious medical disorder (i.e. dialysis)

5. Patients who are less than 16 years old and who meet the physiologic (pediatric), mechanism of injury, or anatomic criteria should be routed to a Missouri state designated pediatric trauma center (Children's Mercy Hospital Downtown)
6. Traumatic cardiopulmonary arrest patients should be taken to the nearest Trauma Center, unless it is "out of service".
7. When there is more than one adult trauma patient, attempt to evenly distribute patients. If this is not feasible, contact central medical control for routing assistance.
8. When more than one patient less than 16 years old meets physiologic or anatomic criteria per the trauma routing protocol, contact central medical control for routing assistance.
9. For the following burn patients, CONSIDER routing directly to the closest Burn Center
 - a. > 20% partial thickness (second degree) burns
 - b. > 5% full thickness (third degree) burns
 - c. Facial burns/inhalation injury
 - d. Circumferential burns
 - e. High voltage electrical burns
 - f. Genital perineum burns
 - g. Burns over joints or exposed bone
 - h. Burns to hands or feet

STEMI ROUTING

1. Patients with identified ST elevation of 1mm or greater in two or more contiguous leads or left bundle branch block should be transported to the closest appropriate STEMI center.

STROKE ROUTING

1. Patients identified with acute stroke symptoms should be transported to the closest appropriate stroke center per the following guidelines:
 - a. Patients with onset of symptoms/last known well less than 3.5hrs AND do not meet any tPA Exclusion Criteria, transport to closest Primary or Comprehensive Stroke Center
 - b. Patients with onset of symptoms/last known well between 3.5hrs and 12hrs OR less than 3.5hrs AND meet tPA Exclusion Criteria, OR woke up with symptoms, transport to closest Comprehensive Stroke Center
 - Thrombolytic (tPA) Exclusion Criteria:
 - Head Trauma at onset
 - History of recent bleeding, current bleeding, surgery or invasive procedure in the past three months
 - Bleeding disorder
 - c. Patients with onset of symptoms/last known well greater than 12hrs may be transported to closest appropriate hospital.

NOTES

Patients Under Arrest:

1. Hospital destination for patients who are under arrest by law enforcement:
 - a. Patients under arrest with a life-threatening emergency will be transported to the closest appropriate hospital as directed by this policy.
 - b. Patients under arrest and considered to have a non-life-threatening emergency have no freedom of movement, or right to designate medical treatment. The patient will be transported to the hospital designated by the arresting officer.

Patients refusing transport to any other facility:

1. Patient care and safety should be the central consideration in all diversion decisions. If a patient requests transport to a hospital that is on “High Volume”, and the patient is made aware of a potential delay in care, then transport to the patient’s hospital of choice:
 - a. Contact the receiving hospital “for information” and inform them that you are bringing the patient per this protocol.
 - b. Do not ask permission of the hospital.
 - c. This should be an informed decision on the patient’s part. The medic should explain that since the hospital is “closed to ambulances” it means that this may delay and/or otherwise impede the patient’s care.

General parent-child considerations:

1. In general, if the parent is supportive, most children should be transported with a parent, unless accompaniment interferes with the care of the child.
2. Children usually respond best to open and honest dialogue.
3. In situations of major trauma or cardiopulmonary arrest, it is usually better to have the parent not ride in the ambulance.



PHYSICIAN ON SCENE



PHYSICIAN ON SCENE

If a licensed physician appears on a scene and desires to assume direction and control of patient care, the physician shall execute a form which declares the physician has assumed responsibility for patient care.

Two scenarios are common:

1. Personal Physician
 - a. Individual is the personal physician of the patient and has previously established a doctor-patient relationship with the patient.
 - b. Most commonly occurs when the patient is still in the doctor's office.
 - c. May also include a "team physician".
2. Intervener Physician
 - a. Physician does not have a prior formal doctor-patient relationship with the patient.
 - b. Most commonly occurs when the doctor happens on scene prior to, concurrent with, or after EMS arrival.

The highest priority is the provision of effective and efficient care and transport of the patient. This policy describes the relationship between the patient, the EMS system, and the physician in these scenarios.

1. Personal Physician:
 - a. If the patient's personal physician is present and desires to continue care, the prehospital provider should defer to the orders of the personal physician as within the paramedic's scope of practice.
 - b. Regardless of a physician executing a physician intervention form; EMS personnel retain the authority to establish medical direction with on-line medical control and assume medical authority of the patient if they believe the care rendered by the personal physician is not the standard of care. EMS personnel will not comply with orders that exceed their scope of practice or are inconsistent with the standard of care.
 - c. The medical direction of EMS personnel shall revert to protocols and/or on-line medical control any time the personal physician is no longer in attendance.
2. Intervener Physician On Scene:
 - a. If an individual identifies themselves as a physician on scene and expresses intent to assume medical care for a patient, EMS personnel will briefly discuss the situation with the physician.
 - b. EMS personnel will ask for proof of licensure to practice medicine in the State of Missouri. The only acceptable evidence of licensure are:
 - i. State of Missouri Medical License card
 - c. If the physician cannot provide this proof of licensure, the paramedic will not transfer medical authority.
 - d. If after establishing proof of licensure listed above, EMS personnel will provide the physician with a "medical intervention form" for review.
 - e. If after reading the document and any further discussion, the physician wishes to take over medical care of the patient the paramedic will perform the following:
 - i. **CONTACT WITH MEDICAL CONTROL IS MANDATORY** to determine whether to relinquish responsibility to the intervener physician.
 - ii. If on-line medical control relinquishes responsibility to the intervener physician, then:
 1. The intervener physician must sign the "medical intervention form".
 2. The intervener physician must accompany the patient to the hospital, sign the PVFD Patient Care Report, and give a report to the receiving hospital physician.
 - iii. If a patient explicitly requests that the intervener physician not provide care then the paramedic maintains medical authority of the patient.
 - iv. If patient care has progressed to time or transport, the paramedic will decline the offer of assistance and transport.

- f. At no time shall patient care be compromised for on-line medical control contact to relinquish responsibility to the intervener physician.
- g. The on-scene paramedic will continue to have ultimate authority for the patient. If the paramedic feels at any time that the intervening physician is either not providing the standard of care or is interfering with the furtherance of patient care, the paramedic will terminate the medical intervention relationship and assume care of the patient per PVFD EMS protocols.



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

PUBLIC SERVICE CALLS



PUBLIC SERVICE CALLS

To provide for adequate and uniform documentation of calls that are public service in nature. These include but are not limited to:

1. Lifting assistance.
2. Assistance to or from a vehicle.
3. Assistance with medical equipment

DOCUMENTATION

1. All public service calls shall be documented.
2. These reports shall be done using the NIFRS (Fire Incident) section of ESO.
3. Any ambulance request that is dispatched with a nature of public service but has a patient that presents with a health complaint requires the Pleasant Valley Fire Department EMS and NFIRS reports to be filled out along with the appropriate accompanying documents.



QUALITY IMPROVEMENT



QUALITY IMPROVEMENT

1. To continually provide suggestions for improvement in the quality of medical care rendered.
2. To improve patient care by adjusting protocols, treatment decisions, and procedures as needed.
3. To ensure proper documentation.

PROCEDURE

1. The EMS Chief or their designee shall review the following categories of calls and forward them to the PEER Review Committee for review.
 - a. All cardiac arrests.
 - b. Any calls marked life threatening.
 - c. Any call involving employee injury or exposure to communicable diseases.
 - d. Refusals that the supervisors find questionable, improperly documented, or outside of standard of care.
 - e. Calls involving any new medication or procedure for the first six months of use.
 - f. Any call that varies from protocol.
 - g. Any call that the attending paramedic desires reviewed.
 - h. Any STEMI activations.
 - i. Any Stroke activations.
 - j. Any Trauma activations.
2. Runs found to be outside of protocol, protocols needing reviewed or revised, or clarification on treatment is needed shall be forwarded to the Medical Director for review.
3. Calls will be reviewed with attending crew on return from Medical Director.
4. Review of audited calls shall be discussed in open forum at the end of CEU meetings.
5. The attending crewmembers will be able to present any details to explain treatment choices, which may vary from protocol. This will allow the attending crew to discuss the call with their peers and enhance educational opportunity for everyone.



REFUSAL OF SERVICE

REFUSAL OF SERVICE

Purpose: The purpose of this policy is to assist EMS personnel with clear rules for managing situations in which the patient or the patient's representative refuses care or transportation by EMS.

Right to Refuse: An adult patient who is conscious, mentally competent, and understands the consequences of his or her decision has the right to refuse care or transportation by EMS. A patient that refuses treatment or transport must be offered treatment or transport and advised of the benefits of treatment or transport and the specific risks of refusing treatment or transport. The offer of treatment or transport and description of the risks of refusal should be documented in the trip sheet and on the Refusal Form.

No Right to Refuse: Conversely, where a patient is a minor or otherwise not competent, or unable to understand the consequences of his or her decision, the patient may not refuse treatment or transport for himself, and EMS may have implied consent to treat or transport. In this situation, EMS must contact Medical Control for directions regarding treatment and transport (Mandatory Medical Control Contact).

Mandatory Medical Control Contact: Medical Control contact is mandatory if the crew believes that the patient may be at significant harm to him or herself by refusing treatment or transport. The crew may contact Medical Control at any time there is a question in regards to the patient condition. Utilization of the EMS radio is mandatory if communication allows. The purpose of mandatory Medical Control Contact is to:

1. Assist the paramedic in persuading a reluctant patient to be transported. Occasionally, a recommendation from a physician may be all that is needed to facilitate transport.
2. Serve to support the paramedic's judgment that the patient is acting against medical advice.

POLICY:

1. **Patient:** Any person who is ill or injured or in need of treatment by medical personnel. This includes any person that has activated the EMS system or for whom the EMS system has been activated, including emergency and non-emergency calls for service, or any person that presents himself to EMS personnel with a medically related complaint such that it could be reasonable inferred that the person is seeking or in need of medical attention.
2. **Not a Patient:** A person who is not ill or injured or in need of treatment by medical personnel. This includes individuals who may have been involved in a situation that either did result or could have resulted in the creation of a patient requiring medical treatment as defined above.
3. **Capacity:** A person 18 years of age or older of sound mind who is able to receive and evaluate information and to communicate a decision. If a patient is able to answer the following questions, the patient may refuse medical treatment:
 - a. Recite three objects to the patient (apple, table, and penny). Ask patient to wait until you say all three words then have them repeat the three words. Tell the patient you will ask them to repeat the words later.
 - b. What year is this?
 - c. What month is this?
 - d. What is the day of the week?
 - e. What were the three objects I asked you to remember?
 - i. Apple
 - ii. Table
 - iii. Penny
4. **Minor:** A person who is less than 18 years of age.
5. **Emancipated Minor:** a minor who is:

- a. Married;
 - b. A parent or legal custodian of a child;
 - c. Enlisted or commissioned in the U.S. Armed Forces;
 - d. Self-supporting and the custodial parent has relinquished the child from parental control by express or implied consent; or
 - e. Declared and adult by a court of competent jurisdiction
 - f. EMS shall use reasonable care and judgment in ascertaining the age of the patient and determining if a minor is emancipated. EMS may treat or transport a minor patient who claims to be emancipated and is refusing treatment or transport if emancipation status cannot be reasonable determined and an emergency exists.
6. **Suicidal Patient:** There is reason to believe that the patient is at risk of self-inflicted physical harm as evidenced by, but not limited to, threats or attempts to commit suicide or to inflict physical harm on himself or herself.
 7. **Durable Power of Attorney for Health Care:** a legal instrument that, if properly drafted and executed, has the effect of delegating legal authority to another in the case of incapacity. If a person at the scene presents a Durable Power of Attorney for Health Care, Medical Control shall be contacted for further advice and direction regarding the validity and effect of the instrument.
 8. **Implied Consent:** consent to medical treatment is implied where an emergency exists if there has been no protest or refusal of consent by a person authorized and empowered to consent or, if so, there has been a substantial change in the condition of the person affected that is material and morbid and there is no one immediately available who is authorized, empowered, willing, and capacitated to consent.
 9. **Emergency:** For purposes of implied consent, “emergency” is defined as a situation where, in competent medical judgment, the proposed medical treatment is immediately or imminently necessary and any delay occasioned by an attempt to obtain consent would reasonably jeopardize the life, health, or limb of the person affected or would reasonably result in disfigurement or impairment of faculties. For purpose of the emergency services statutes, “emergency” is the sudden and, at the time, unexpected onset of a health condition that manifests itself by symptoms of sufficient severity that would lead a prudent layperson, possessing an average knowledge of health and medicine, to believe the absence of immediate medical care could result in: (a) placing a person’s health, or with respect to a pregnant woman, the health of the woman or her unborn child, in significant jeopardy; (b) serious impairment to a bodily function; (c) serious dysfunction of any bodily organ or part; or (d) inadequately controlled pain.

No Right to Refuse

1. **Suicidal Patient:**
 - a. If the patient meets the definition for “suicidal patient”, then the patient cannot refuse medical treatment and transport.
 - b. Assistance from the appropriate law enforcement agency should be sought for those patients exceeding the capabilities of the transporting unit.
 - c. Patients do not have to verbalize suicidal intent to be at risk of harm to themselves. If a patient’s actions, behavior, or verbal comments are consistent with and could reasonable be construed to be an act of suicidal intention, or risk of self-harm, the patient cannot refuse medical treatment and transport.
 - d. If the patient attempts to refuse medical treatment and transport:
 - i. Law enforcement should be contacted for assistance (protective custody).
 - ii. If the patient is not placed in protective custody, Medical Control should be contacted to determine the patient’s disposition.
 - iii. If Medical Control contact is required, the Medical Control physician will be apprised of the patient’s condition, including the history, vital signs, level of consciousness, and mental status, and including all reasons that the patient is determined to be at risk of physical harm to themselves. From this information, the Medical Control physician will determine transport and treatment.
 - iv. If the Medical Control physician determines that the patient is at risk of physical harm and should be transported, but the patient continues to refuse, reasonable restraint (to assure

patient and allied health personnel safety) may be utilized to transport the patient to the hospital.

- v. The appropriate law enforcement agency should be advised that Medical Control physician has determined that the patient is *at risk of physical harm* to themselves and has ordered the patient to be transported for evaluation.
- vi. All actions taken to restrain the patient as well as the Medical Control physician's orders to restrain must be documented on the patient care report.

2. Incompetent Adults (Intellectually Disabled, Mentally Ill, Dementia, Etc.):

- a. If the patient's guardian is on the scene with evidence of guardianship, the patient may not refuse treatment or transport for himself.
- b. If the guardian is not competent, as defined above, Medical Control must be contacted.
- c. If the patient's guardian cannot prove guardianship, Medical Control must be contacted.
- d. If the patient's guardian is not present, Medical Control should be contacted to approve transport. If the patient's condition is urgent, the patient will be transported by standing order prior to guardian contact or Medical Control approval.
- e. If a person at the scene presents a Durable Power of Attorney for Health Care, contact Medical Control for advice and direction regarding the validity and effect of the instrument.

3. Unemancipated Minor (Guardian Not Present):

- a. An unemancipated minor must be treated and transported. Or, a competent adult, as defined above, must be on scene and take responsibility *en loco parentis* for the patient and sign the Refusal Form.
- b. Law enforcement may take these patients into protective custody pending parental notification and response to the scene. In the instances, the law enforcement officer should sign the Refusal Form.

4. Intoxicated Patient:

- a. Intoxicated patients may have serious underlying medical conditions. These patients should be properly assessed to include a blood glucose reading.
- b. If the patient has capacity as defined above, and is turned over to law enforcement, the patient must sign a Refusal Form along with the highest-ranking police officer as a witness.
- c. If the patient has capacity as defined above, but is in such condition that a reasonable person would consider them not capable of being able to adequately care for themselves, the patient should be transported. If the patient continues to refuse, then contact Medical Control for guidance on medical treatment and transport.

5. Special Situations:

- a. **Medication Administration:** If, upon administration of a medication, the patient decides to refuse transport, an assessment of the patient's condition and capacity should be made.
- b. **Law Enforcement-Mandated Transport:**
 - i. If law enforcement does transport the patient, the patient refusal must be signed by the highest-ranking law enforcement agent on the scene.
 - ii. If the transporting paramedic believes the patient is at risk of physical harm if transport is initiated by law enforcement, then he/she should notify the highest-ranking law enforcement officer at the scene that a danger exists.

DOCUMENTATION REQUIREMENTS:

1. Refusal Form

- a. Complete the Refusal Form, including Capacity and Notifications
- b. If the patient refuses to sign, document that verbal review of the information has occurred and, if possible, have a witness (preferably someone related to the patient) sign the Refusal Form.

2. Document all findings during assessment, including two (2) complete sets of vital signs.

3. Document all recommendations given to the patient on the Refusal Form.

4. Document any Medical Control recommendations on the Refusal Form.



REPORT WRITING



REPORT WRITING

Proper documentation is an important part of any EMS call. It can protect both the agency and the individual medic from possible negligence claims. It shall be the policy of this agency that the report form is filled out completely, authorization signatures obtained (if possible), and completed prior to the end of each assigned shift. Listed below are the minimum criteria required for EMS reports.

1. How the call is received from dispatch.
2. Nature of the call on arrival.
3. Patient's chief complaint.
4. How patient was found.
5. H.P.I. & P.M.H.
6. At least two complete sets of vital signs. This is to include blood pressure (not palpated), pulse rate, respiratory rate, and O2 saturation. Document that vitals are being monitored.
7. Number of IV attempts and who initiates the IV (Medic name).
8. Number of intubation attempts and who places the ET tube (Medic name).
9. Time drugs administered and who administers them (Medic name).
10. Procedures done and who completes them (EMT or Medic name).
11. Patient's response to treatments include: pain ratings, and vital signs.
12. If orders are requested, what is requested and who gives the order.
13. If no orders are requested it shall be noted.
14. Attending medic shall sign the report in the appropriate provider signature location.



SCOPE OF PRACTICE



SCOPE OF PRACTICE

DEFINITIONS

1. “Paramedic” means a State of Missouri licensed Paramedic and employee of Pleasant Valley Fire Department or Mutual Aid partner.
2. “EMT” means a State of Missouri licensed Emergency Medical Technician and employee of Pleasant Valley Fire Department or Mutual Aid partner.
3. ALS Ambulance means an ambulance staffed with at least 1 paramedic and 1 EMT
4. Mutual Aid ALS Ambulance means an ambulance from a Mutual Aid partner staffed with at least 1 paramedic and 1 EMT
5. BLS Ambulance means an ambulance staffed with 2 EMTs

AUTHORIZATION

1. Paramedics are authorized to perform the following skills within the Pleasant Valley Fire Department EMS System:
 - a. All patient care activities authorized by the Pleasant Valley Fire Department EMS Protocols.
 - i. If a specific patient care activity is not located within the “Pleasant Valley Fire Department EMS Protocols” document, it is not an authorized activity and therefore should not be performed.
2. EMTs are authorized to perform the following skills within the Pleasant Valley Fire Department EMS System:
 - a. The following interventions **SHALL NOT** be performed by a Pleasant Valley Fire Department EMT:
 - i. Endotracheal intubation.
 - ii. Invasive circulatory procedures such as intravenous cannulation, intraosseous cannulation, and regulating intravenous infusions.
 - iii. Administration of medications EXCEPT as noted below:
 1. Oxygen
 2. Oral glucose
 3. Aspirin
 4. Honey
 5. Patient’s own medications where EMS may assist patient
 - iv. Needle decompression to the chest.
 - v. Defibrillation or cardioversion using a manual defibrillator.
 - vi. Interpretation of electrocardiographs.
 - vii. External cardiac pacing.
 - b. If a specific patient care activity is not located within the “Pleasant Valley Fire Department EMS Protocols” document, it is not an authorized activity and therefore should not be performed.
 - c. When performing patient care activities, EMTs are expected to follow the Pleasant Valley Fire Department EMS Protocols until the arrival of a Pleasant Valley Fire Department Paramedic (paramedic then assumes responsibilities for patient care activities), the arrival of a paramedic with a Mutual Aid ALS Ambulance, or designation of the medical operations officer incident command, except in cases where:
 - i. Following the direction of the paramedic would present a danger to the patient or public OR
 - ii. There are multiple patients on the scene such that the paramedic and EMT are taking care of separate patients, in which case the EMT on the responding ambulance should take responsibility for individual patient care.
 - d. Some manual monitor/defibrillators have an “automatic external defibrillator” (AED) mode. EMTs may operate the AED function of a manual monitor/defibrillator.
 - e. EMTs may obtain a blood sample for, and perform a rapid bedside glucose test as allowed by the State of Missouri and as delineated in the Pleasant Valley Fire Department EMS Protocols. If the blood sample is obtained by the EMT, the blood sample will be obtained with a lancet or similar device.

BLS AMBULANCE RESPONSE

There are times when staffing will require 2 EMTs to staff the ambulance and operate in a BLS capacity. When responding in a BLS capacity, and Mutual Aid ALS Ambulance should be immediately requested when dispatch gives any of the following call natures:

1. Cardiac Arrest
2. Unconscious or Unresponsive Person
3. Altered Mental Status
4. Chest Pain
5. Shortness of Breath, Shortness of Air, or Respiratory Distress
6. Syncope or Passed Out
7. Stroke
8. Seizure
9. Diabetic
10. Obstetrical Emergencies
11. Motor Vehicle Crash on I-35
12. Pedestrian Struck
13. Electrocutation or Lightning Strike
14. Fall > 20Ft (adult) or Fall > 10Ft (pediatric)
15. Shooting
16. Stabbing

A Mutual Aid ALS Ambulance should be requested immediately (if not already enroute) upon BLS Ambulance arrival on scene when the BLS crew is presented with a patient with any of the following chief complaints:

1. Cardiac Arrest
2. Unconscious or Unresponsive Person
3. Altered Mental Status
4. Chest Pain
5. Shortness of Breath, Shortness of Air, or Respiratory Distress
6. Syncope or Passed Out
7. Stroke
8. Seizure
9. Diabetic
10. Obstetrical Emergencies
11. Motor Vehicle Crash on I-35
12. Pedestrian Struck
13. Electrocutation or Lightning Strike
14. Fall > 20Ft (adult) or Fall > 10Ft (pediatric)
15. Gunshot wound to head, neck, chest, abdomen, genitalia, back, upper extremities proximal to elbow, or lower extremities proximal to knee
16. Stab wound to head, neck, chest, abdomen, genitalia, back, upper extremities proximal to elbow, or lower extremities proximal to knee
17. Gross Hemorrhage

BLS AMBULANCE TRANSPORT

In the event the ambulance is staffed BLS, the BLS crew may transport the patient with an EMT attending as long as the patient meets ALL of the following criteria:

1. The patient has not required airway management (OPA, NPA, iGel, ETT) at any time during the current encounter
2. The patient has not required ventilatory support (CPAP or BVM use) at any time during the current encounter
3. The patient has not required chest compressions at any time during the current encounter
4. The patient has not had acute mental status changes or GCS < 14 at any time during the current encounter
5. The patient's vital signs have not been outside of the following parameters at any time during the current encounter
 - a. Pulse Rate 50-120 bpm
 - b. Systolic Blood Pressure 100 – 200mmHg
 - c. Diastolic Blood Pressure < 120mmHg
 - d. Respiratory Rate 10 – 30 per minute
 - e. Pulse Oximeter > 93%
 - f. Glasgow Coma Scale > 13
 - g. Blood Glucose 70 – 350mg/Dl
6. The patient has not required vascular access (IV or IO) at any time during the current encounter
7. The patient has not required 4 Lead or 12 Lead cardiac monitoring, or ETCO2 monitoring at any time during the current encounter
8. The patient has not required any medication found with the PVFD EMS Protocols Medication Formulary (except Oxygen or Oral Glucose) at any time during the current encounter
9. The patient does not require TCD Routing (STEMI, Stroke, or Trauma) per the Patient Destination Criteria Guideline



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

SUSPECTED ABUSE/NEGLECT POLICY



SUSPECTED ABUSE/NEGLECT POLICY

Definition – Includes physical, emotional, or sexual abuse as well as physical or emotional neglect.

Duty to Report

1. State law requires that suspected child abuse or neglect be reported to the Missouri Department of Health and Senior Services. Any Paramedic, EMT, or First Responder having reasonable cause to suspect that a child (any person less than 18 years of age) has been or may be abused or neglected must report or cause a report to be made to the Division of Family Services. Notification should be done by telephoning 1-800-392-3738
2. State law requires that suspected elder abuse or neglect be reported to the Missouri Department of Health and Senior Services. Any Paramedic, EMT, or First Responder having reasonable cause to suspect that a person sixty years of age or older (or handicapped person between the ages of 18 and 59) has been or may be abuse or neglected must report or cause a report to be made to the Department of Health and Senior Services. Notification should be done by telephoning 1-800-392-0210

- a. Notification of suspected elder abuse or neglect may also be made at the State of Missouri website at <https://health.mo.gov/abuse>

Provide necessary medical care according to the appropriate protocol.

Although a preliminary assessment of abuse or neglect is made in the field, appropriate personnel make the final determination upon review of the incident. Accusations of abuse or neglect should not be made in the field.

In a private and confidential environment notify the nurse and/or physician at the receiving hospital of the facts upon which neglect of abuse is suspected.

Notification of the Assistant Chief of EMS of the facts of the neglect or abuse is encouraged.

DOCUMENTATION

The following information shall be included in the attending paramedics report:

1. All facts upon which abuse or neglect is suspected.
2. Identity of nurse and/or physician notified.
3. Whether physician or hospital staff is going to make proper notification.
4. If notification is made by the attending paramedic.



COMBINED GENERAL MEDICAL



The purpose of this protocol is to outline the general approach to the patient with a medical emergency.

SCENE SIZE UP: (BLS or ALS)

- Assure the scene is safe for you, other rescuers, and the patient. It may be appropriate to withdraw from the scene in some situation until a safe environment can be obtained. It may be appropriate to rapidly extricate the patient from a dangerous situation.
- Identify the number of patients and other resources that may be needed
 - Initiate Incident Management System if appropriate
 - Declare and MCI as needed
 - Establish "Medical Command" as needed
 - Call for law enforcement and/or first responder assistance as needed
 - Call for additional EMS units as needed
 - Begin triage as needed
- Consider mechanism of injury

BODY SUBSTANCE ISOLATION: (BLS or ALS)

- Apply universal precautions/body substance isolation as needed

PRIMARY SURVEY

Search for immediate life threats by assessing the patient by using A-B-C process and treating the problems as they are found.

Abnormalities found in the Primary Survey should be addressed with appropriate interventions at time of discovery

It may be appropriate to move directly from the Primary Survey to another protocol

ASSESSMENT: (BLS or ALS)

- Assess AIRWAY:
 - Note patient's ability to speak and any evidence of actual or potential airway obstruction including vomitus, bleeding, dentures, loose teeth, or foreign bodies
 - Secure airway:
 - BLS Maneuvers:
 - Jaw thrust
 - Oral or nasal airway
 - iGel placement
 - Apply capnography (ALS ONLY)
 - ALS Maneuvers:
 - Oral endotracheal intubation
 - Apply capnography and monitor ETCO₂
- Assess BREATHING:
 - Note patient's ability to speak, rate, depth, and quality of respirations, abnormal noises/stridor, retractions, accessory muscle use, nasal flaring, or cyanosis
 - Administer oxygen as needed
 - Assist ventilations as needed
 - BVM ventilation is indicated prior to attempts at iGel placement or endotracheal intubation
- Assess CIRCULATION:
 - Note pulses, level of consciousness, skin abnormalities (color, temperature, capillary refill, moisture)

- ALS ONLY: **ESTABLISH IV/IO ACCESS**
- Assess neurological function (DISABILITY)
 - Note level of consciousness, Glasgow Coma Scale or AVPU Scale, movement of each extremity

IF NO PULSE REFER TO ADULT OR PEDIATRIC CARDIAC ARREST PROTOCOLS AS APPROPRIATE

SECONDARY SURVEY

A systemic history and physical exam, focused on the patient's complaints, searching for problems that may not be immediately life or limb threatening, but may become so if not addressed appropriately

ASSESSMENT: (BLS or ALS)

- Obtain Chief Complaint
- Obtain "SAMPLE" history
- Obtain vital signs (MANUAL Blood Pressure)
- Perform focused physical exam (this evaluation is dependent on the history as well as findings from the primary survey and may be more or less detailed depending on the situation)
- Consider application of cardiac monitor (ALS ONLY)
- Consider application of pulse oximeter
- Consider obtaining rapid blood glucose

Assessment summation: Consider information gathered in primary and secondary survey, determine an impression of the patient's primary problems and proceed to the appropriate treatment protocol

Transport while monitoring vital signs

Patient destination as determined by appropriate protocol

Medical Control contact as determined by appropriate protocol

REFER TO APPROPRIATE PROTOCOL FOR SPECIFIC MEDICAL EMERGENCY

**COMBINED ALLERGIC REACTION**ASSESSMENT: (BLS or ALS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain a brief history
- Apply cardiac monitor (ALS ONLY)
- Apply pulse oximeter

INTERVENTIONS: (BLS or ALS)

- Secure airway
- Administer oxygen and assist ventilation as needed
- ALS Only: **ESTABLISH IV/IO ACCESS**
- Transport while monitoring vital signs

MILD ANAPHYLAXIS:

Patient is alert and appropriate, normotensive, may have wheezing with good air movement and no cyanosis; may have hives

- ALS ONLY:
 - If wheezing: **ADULT/PEDS: ALBUTEROL 2.5MG NEBULIZER MAY REPEAT X2 IF WHEEZING CONTINUES**
 - **ADULT/PEDS:** consider **EPINEPHRINE 1:1000 0.01mg/kg (up to 0.3mg) IM**
 - **THIS MAY BE REPEATED WITH MEDICAL CONTROL APPROVAL**
 - **ADULT/PEDS:** consider **DIPHENHYDRAMINE 1mg/kg (up to 50mg) IV/IO/IM**

ANAPHYLAXIS WITH RESPIRATORY DISTRESS:

Respiratory distress without signs of shock

- ALS ONLY:
 - **ADULT/PEDS: EPINEPHRINE 1:1000 0.01mg/kg (up to 0.3mg) IM**
 - **THIS MAY BE REPEATED WITH MEDICAL CONTROL APPROVAL**
 - **ADULT/PEDS: ALBUTEROL 2.5MG NEBULIZER MAY REPEAT X2 IF WHEEZING CONTINUES**
 - **ADULT/PEDS: DIPHENHYDRAMINE 1mg/kg (up to 50mg) IV/IO/IM**

ANAPHYLACTIC SHOCK:

Hypotension, delayed capillary refill, altered mental status, with or without respiratory distress

- ALS ONLY:
 - **ADULT/PEDS: EPINEPHRINE 1:1000 0.01mg/kg (up to 0.3mg) IM**
 - **MAY REPEAT X1**
 - **CONTACT MEDICAL CONTROL IF ADDITIONAL DOSES ARE NEEDED**
 - **ADULT/PEDS: DIPHENHYDRAMINE 1mg/kg (up to 50mg) IV/IO/IM**
 - **ADULT/PEDS: NS/LR 20mg/kg BOLUS (MAX 1000 mL) IV/IO**
 - **CONSIDER REPEAT IF NEEDED TO MAINTAIN SBP > 90mmHg**
 - **ADULT/PEDS: ALBUTEROL 2.5MG NEBULIZER MAY REPEAT X2 IF WHEEZING CONTINUES**



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

COMBINED ALTERED MENTAL STATUS



ASSESSMENT: (BLS or ALS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history with particular attention to onset and progression of present state. If this information is unobtainable, at least establish when patient was last seen alert and/or appropriate
- Note any signs of trauma
- Check for pill bottles or syringes and bring with patient
- Apply cardiac monitor (ALS ONLY)
- Obtain a rapid blood glucose determination
 - If blood glucose < 70mg/dL, REFER TO HYPOGLYCEMIA PROTOCOL

INTERVENTIONS:

- BLS or ALS:
 - Secure airway as needed
 - Administer oxygen as needed
 - Assist ventilations as needed
- ALS ONLY:
 - **CONSIDER ESTABLISHING IV ACCESS**
 - **CONSIDER NALOXONE**
 - **ADULT: 0.4 – 2mg IV OR UP TO 4mg IN**
 - **PEDS: 0.1mg/kg up to 2mg IV OR 2mg IN**
- Transport while monitoring vital signs



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

COMBINED BEHAVIORAL/PSYCHIATRIC DISORDER



ASSESSMENT: (BLS or ALS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history noting any bizarre or abrupt changes in behavior, suicidal ideation, possible alcohol or drug ingestion, and significant past medical history (diabetes, previous psychiatric disturbances)

PROBABLE PSYCHIATRIC PROBLEM ONLY:

- BLS or ALS
 - Attempt to establish rapport
 - If patient is dangerous to himself or others, have police assist in transport
 - Restrain if absolutely necessary
 - If suicidal, do not leave patient alone and, if possible, remove any dangerous objects (guns, knives, pills, etc)
 - If emergency transport is unnecessary, do as little as possible except to reassure while transporting
 - Consider your own safety and limitations
- ALS ONLY
 - **IF PHYSICAL RESTRAINT IS REQUIRED, STRONGLY CONSIDER CHEMICAL RESTRAINT:**
 - **ADULTS: MIDAZOLAM 0.1mg/kg IM or IN up to 10mg**
 - **PEDS: MIDAZOLAM 0.1mg/kg IM or IN up to 5mg**
 - **CONTACT MEDICAL CONTROL FOR ADDITIONAL PEDS DOSES**
- Transport in position of comfort

AGITATED PATIENTS:

- BLS or ALS
 - Cardiac monitor when possible (ALS ONLY)
 - Pulse oximeter when possible
 - Rapid blood glucose when possible
 - Consider your own safety and limitations
 - If patient is dangerous to themselves or others, have police assist in transport
 - Restrain if absolutely necessary
- ALS ONLY
 - **IF PHYSICAL RESTRAINT IS REQUIRED, STRONGLY CONSIDER CHEMICAL RESTRAINT:**
 - **ADULTS: MIDAZOLAM 0.1mg/kg IM or IN up to 10mg**
 - **PEDS: MIDAZOLAM 0.1mg/kg IM or IN up to 5mg**
 - **CONTACT MEDICAL CONTROL FOR ADDITIONAL PEDS DOSES**
- Consider the possibility of aspiration when deciding on physical restraint position
- As soon as possible administer oxygen
- Monitor ETCO2 if Midazolam has been given
- Transport while frequently monitoring vital signs



COMBINED DYSTONIC REACTION



ASSESSMENT: (BLS or ALS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain a brief history noting any new medication, particularly psychiatric or antiemetic agents, time of last dose, and any known allergic or previous dystonic reactions
- Dystonic reactions are generally characterized by any or all of the following:
 - Deviation of eyes in all directions
 - Forced spasm of the face, jaw, tongue, and pharynx muscles
 - Protrusion of tongue
 - Forced jaw opening
 - Difficulty speaking
 - Facial grimacing
 - Torticollis
 - Lordosis or Scoliosis
 - Tortipelvic crisis
- Mental status is unaffected
- Vital signs are usually normal
- Remaining physical exam findings are normal

INTERVENTIONS:

- BLS or ALS
 - Administer oxygen as needed
 - Assist ventilations as needed
- ALS ONLY
 - **CONSIDER DIPHENHYDRAMINE**
 - **ADULT/PEDS: 1mg/kg IV or IM MAX 50mg**
 - **MAY REPEAT DOSE ONCE WITH MEDICAL CONTROL APPROVAL**

Transport in position of comfort monitoring vitals



COMBINED HYPOGLYCEMIA

ASSESSMENT: (BLS or ALS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain rapid blood glucose
 - IF BLOOD GLUCOSE > 70mg/dL REFER TO APPROPRIATE PROTOCOL
- Obtain brief history to include onset and progression of present state
- Observe for presence of an automated external insulin delivery device (Insulin Pump)
- Apply cardiac monitor (ALS ONLY)
- Apply pulse oximeter

INTERVENTIONS

- BLS or ALS
 - Secure airway as needed
 - Administer oxygen as needed
 - Assist ventilation as needed
 - If blood glucose < 70mg/dL
 - Patient awake and alert, symptomatic, with gag reflex present
 - **ADULT/PEDS: CONSIDER ORAL GLUCOSE**
- ALS ONLY
 - Patient not awake and alert or without gag reflex **OR**
 - Patient remains symptomatic and blood glucose remains < 70mg/dL after Oral Glucose
 - **ESTABLISH IV/IO ACCESS**
 - **CONSIDER OBTAINING 12 LEAD ECG**
 - **ADMINISTER D10%**
 - **ADULTS: 125mL IV/IO**
 - **PEDS: 5mL/kg IV/IO MAX 125mL**
 - **RECHECK BLOOD GLUCOSE AND REPEAT X1 IF PATIENT REMAINS SYMPTOMATIC AND BLOOD GLUCOSE REMAINS < 70mg/dL TO MAX 250mL GIVEN**
 - **ADULTS/PEDS: IF PATIENT REMAINS SYMPTOMATIC AND BLOOD GLUCOSE REMAINS < 70mg/dL CONTACT MEDICAL CONTROL FOR ADDITIONAL DOSES**
- Transport while monitoring vital signs

NOTES:

- Transport is MANDATORY with IO placement
 - The placement and subsequent removal of an IO in the pre-hospital environment is not currently supported by literature
- Any patient that refuses transport should be instructed to contact their physician and consume a meal with complex carbohydrates and protein in addition to general instructions given on all refusals
- Transport is indicated for the following patients:
 - IO placement
 - Patients under age 18
 - Patients with unexplained hypoglycemia
 - Patients taking oral hypoglycemic meds
 - Patients not taking food by mouth
 - Patients who do not have a competent adult to monitor them



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

COMBINED HYPOTENSION/SHOCK



ASSESSMENT: (BLS or ALS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES for pediatric blood pressure and pulse rate parameters
- Obtain brief history to include:
 - Onset
 - Duration
 - Possible etiologies
 - Past history of similar occurrences
 - Fever
 - Blood loss
 - Fluid loss
 - Allergic exposure
 - Trauma
- Apply cardiac monitor (ALS ONLY)
- Apply pulse oximeter
- Obtain rapid blood glucose

IF ANAPHYLACTIC SHOCK REFER TO ALLERGIC REACTION PROTOCOL

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen, or assist ventilations as needed
- ALS ONLY
 - **CONSIDER OBTAINING 12 LEAD ECG (IF STEMI SUSPECTED REFER TO CHEST PAIN/STEMI PROTOCOL)**
 - **ESTABLISH IV/IO ACCESS**
 - **ADULTS: NS/LR 1L IV/IO**
 - **MAY REPEAT X1 (MAX 2L) TO MAINTAIN SBP > 90mmHg**
 - **PEDS: 20mL/kg MAX 1L**
 - **MAY REPEAT X1 (MAX 2L) TO MAINTAIN AGE APPROPRIATE SBP**
 - **CARDIOGENIC SHOCK:**
 - **ADULT/PEDS: NS/LR 10mL/kg MAX 500mL**
 - **ADULTS ONLY:**
 - **CONSIDER EPI 1:100000 (PUSH DOSE) IV/IO**
 - **10-20mcg EVERY 2-3 MINUTES IF SBP < 90mmHg AFTER APPROPRIATE FLUID RESUSCITATION**
- Transport while monitoring vital signs



COMBINED HYPERTHERMIA

ASSESSMENT: (BLS or ALS)

- Obtain vital signs, including temperature (if available), consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history noting length of illness and treatment given
- If suspected environmental etiology, note the type and length of exposure
- Apply cardiac monitor (ALS ONLY)
- Apply pulse oximeter

HYPERTHERMIA WITHOUT ALTERED MENTAL STATUS:

- BLS or ALS
 - Secure airway as needed
 - Administer oxygen as needed
 - Loosen clothing, remove any heavy clothing, and do not cover with blankets
 - CONSIDER cooling the patient by sponging with room temperature water
 - AVOID INDUCING SHIVERING
- ALS ONLY
 - **CONSIDER ESTABLISHING IV ACCESS**

HYPERTHERMIA WITH ALTERED MENTAL STATUS:

- BLS or ALS
 - Obtain rapid blood glucose
 - Secure airway as needed
 - Administer oxygen as needed
 - Assist ventilations as needed
 - Remove patient's clothing and CONSIDER cooling by sponging the patient with room temperature water And, if possible, fan the patient to achieve evaporation
 - Consider Ice Packs placed in the groin, axilla, and neck. Do not place directly on skin. Wrap w a paper towel or place over clothes.
- AVOID INDUCING SHIVERINGALS ONLY
 - **ESTABLISH IV/IO ACCESS**

Transport while monitoring vital signs. Cool while enroute. DO NOT let cooling in the field delay transport

NOTES:

- IF SHOCK PRESENT REFER TO HYPOTENSION/SHOCK PROTOCOL
- Hyperthermia may be the result of environment heat loads, infections, drug ingestions, or chronic disease.
- Elevations in body temperature may induce seizures in children.
- Hyperthermia with an altered mental status is a true emergency regardless of the etiology



COMBINED HYPOTHERMIA

ASSESSMENT: (BLS or ALS)

- Obtain vital signs, including temperature (if available), consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history noting length of exposure and pertinent medical history (drug ingestion, diabetes)
- Note condition of skin for evidence of local injury (frostbite)
- Apply cardiac monitor (ALSO ONLY)
- Apply pulse oximeter

LOCALIZED COLD INJURY (FROSTBITE)

- BLS or ALS
 - Handle injured part very gently. **DO NOT RUB** the injured area. Protect from pressure, trauma, or friction but leave uncovered
 - Do not allow limb to thaw if there is a chance that refreezing will occur before evacuation and transport is complete
 - Remove wet garment, dry and insulate patient to maintain core temperature
 - Active rewarming should be done at the hospital, not in the field
 - Transport while monitoring vitals

SYSTEMIC HYPOTHERMIA

- AWAKE BUT ALTERED LOC
 - BLS or ALS
 - Remove wet garment, dry and insulate patient to maintain core temperature
 - Obtain a rapid blood glucose
 - Administer oxygen as needed
 - Assist ventilations as needed
 - Transport while monitoring vital signs
 - ALS ONLY
 - **ESTABLISH IV/IO ACCESS**
- UNRESPONSIVE
 - BLS of ALS
 - Handle patient as gently as possible. Avoid unnecessary manipulation
 - Remove wet garment, dry and insulate patient to maintain core temperature
 - Secure airway with as little manipulation as possible. Stimulation of the airway may precipitate ventricular fibrillation in the hypothermic patient
 - Administer oxygen as needed
 - Assist ventilations as needed
 - Obtain a rapid blood glucose
 - Transport while monitoring vital signs
 - ALS ONLY
 - **ESTABLISH IV/IO ACCESS**
 - IF CARDIAC ARREST REFER TO ADULT OR PEDIATRIC CARDIAC ARREST PROTOCOLS
 - Pulse and respirations may be very slow and difficult to detect if the patient is severely hypothermic. If no definite pulse and no signs of life begin CPR.



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

COMBINED NON-TRAUMATIC ABDOMINAL PAIN/VOMITING



ASSESSMENT: (BLS or ALS)

- Obtain vital signs with PAIN SCALE, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history
 - Onset
 - Location
 - Duration
 - Radiation of pain
 - Associated signs and symptoms
 - Vomiting
 - Bilious emesis, hematemesis, coffee ground emesis
 - GU symptoms
 - Melena
 - Rectal bleeding
 - Vaginal bleeding
 - Known or suspected pregnancy
 - Recent trauma
- Life threatening causes:
 - MI, AAA, Dissection, Ectopic Pregnancy

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen, or assist ventilations as needed
- ALS ONLY
 - **CONSIDER ESTABLISHING IV ACCESS**
 - **CONSIDER ONDANSETRON (Zofran)**
 - **ADULTS: 4mg IV/IM / ODT (orally disintegrating tablet) ONE TIME ONLY**
 - **PEDS OVER AGE 4 YEARS OLD: 2mg IV/IM ONE TIME ONLY**
- FOR SEVER PAIN CONSIDER REFERRING TO PAIN CONTROL PROTOCOL
- Transport while monitoring vital signs

NOTES:

- ELDERLY PATIENTS
 - More likely to have life-threatening cause of symptoms
 - Shock may be occult, with absent tachycardia in setting of severe hypovolemia
- PEDIATRIC PATIENTS
 - Life threatening causes vary by age
 - Consider:
 - Occult or non-accidental trauma
 - Toxic ingestion
 - Button battery ingestion
 - GI bleed
 - Peritonitis
 - Most pediatric patients without signs of shock, no IV is required and pharmacologic pain management should be limited



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

COMBINED OVERDOSE/POISONING



ASSESSMENT: (BLS or ALS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history
- Bring any containers or package inserts, etc. of suspected ingestion
- Apply cardiac monitor (ALS ONLY)
- Apply pulse oximeter
- Obtain a rapid blood glucose
 - If blood glucose < 70mg/dL, REFER TO HYPOGLYCEMIA PROTOCOL

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen, or assist ventilations as needed
- ALS ONLY
 - **ESTABLISH IV ACCESS**
 - **FOR KNOWN OPIOID/OPIATE/NARCOTIC OVERDOSE WITH RESPIRATORY DEPRESSION:**
 - **NALOXONE**
 - **ADULT: 0.4 – 2mg IV OR UP TO 4mg IN**
 - **PEDS: 0.1mg/kg up to 2mg IV OR 2mg IN**
 - **FOR KNOWN TRICYCLIC ANTIDEPRESSANT (TCA) OVERDOSE WITH WIDE COMPLEX TACHYCARDIA:**
 - **SODIUM BICARBONATE**
 - **ADULTS/PEDS: 1mEq/kg IV MAX 50mEq**
 - **MUST CONTACT MEDICAL CONTROL**
 - **FOR KNOWN CALCIUM CHANNEL BLOCKER OVERDOSE WITH HYPOTENSION**
 - **CALCIUM CHLORIDE 10%**
 - **ADULTS/PEDS: 0.2mL/kg IV (MAX 10mL)**
 - **MUST CONTACT MEDICAL CONTROL**
 - **FOR KNOWN ORGANOPHOSPHATE POISONING**
 - **ATROPINE**
 - **ADULTS: 3mg IV (MUST CONTACT MEDICAL CONTROL)**
 - **PEDS: CONTACT MEDICAL CONTROL**
- Transport while monitoring vital signs



COMBINED PAIN CONTROL

ASSESSMENT: (BLS or ALS)

- Obtain vital signs including PAIN SCALE, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history
 - Mechanism of injury
 - Location of injury
- Apply cardiac monitor (ALS ONLY)

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen, or assist ventilations as needed
- ALS ONLY
 - **ESTABLISH IV ACCESS**
 - **CONSIDER TORADOL FOR MILD TO MODERATE PAIN**
 - **ADULTS: 15mg IV or 30mg IM/IN (15mg ea Nare)**
 - **DO NOT REPEAT**
 - **IF >65 USE CAUTION. CONSIDER HALF DOSE OR ANOTHER PAIN CONTROL METHOD.**
 - **PEDS: 0.5-1mg/kg IV/IM. MAX DOSE 15mg IV/ 30mg IM**
 - **CONSIDER FENTANYL FOR MODERATE TO SEVERE PAIN**
 - **ADULTS: 50-100mcg SLOW IV or IN**
 - **MAY REPEAT EVERY 5 MIN TO TOTAL MAX GIVEN 200mcg**
 - **PEDS: 1mcg/kg SLOW IV or IN MAX 50mcg**
 - **MAY REPEAT X1 MAX 50mcg**
 - **CONSIDER MORPHINE FOR MODERATE TO SEVERE PAIN**
 - **ADULTS: 2-10 mg IV**
 - **No greater than 3 mg/min**
 - **CONTACT MEDICAL CONTROL IF ADDITIONAL DOSES ARE NEEDED**
 - **CONTACT MEDICAL CONTROL IF PATIENT ATTEMPTS TO REFUSE TRANSPORT AFTER RECEIVING PAIN MEDICATION**
- Transport while monitoring vital signs and ETCO2

CONTRAINDICATIONS:

- TRAUMA
 - Head injury
 - Suspected spinal injury
 - Thoraco-abdominal trauma (blunt or penetrating)
 - Burns or facial trauma involving the airway or causing respiratory compromise
- SYSTEMS
 - Respiratory distress or compromise (Asthma, COPD, or CHF)
 - Cardiac dysrhythmia present
 - Altered mental status
 - Pregnancy
- Drug or alcohol use
- Sensitivity or allergy to opiates
- Hypotension (SBP < 100mmHg) or Pulse rate < 60 bpm
- Age < 12 months



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

COMBINED POST RESUSCITATION



ASSESSMENT: ALS ONLY

- ROSC – Return of Spontaneous Circulation – pulse and measurable BP with increasing ETCO₂
- Continue to address specific differentials associated with original dysrhythmia
- Several simultaneous and stepwise interventions must be performed to optimize care and maximize patient outcome
- Focus on prevention of hypoxia, hypo/hypercapnia, and hypotension
 - Survival and neurologic outcome worsen in their presence
- Transport all ROSC patients to a STEMI center

INTERVENTIONS:

- ALS ONLY
 - Optimize oxygenation/ventilation
 - ETCO₂ 35-45mmHg
 - DO NOT HYPERVENTILATE
 - Monitor vital signs and EKG rhythm
 - Obtain 12 Lead ECG
 - SBP < 90mmHg or < Age Appropriate
 - **NS/LR BOLUS**
 - **ADULTS: 500mL IV/IO MAX 2L**
 - **PEDS: 20mL/kg MAY REPEAT X1**
 - **CONSIDER EPI 1:100,000 (PUSH DOSE) IV/IO**
 - **ADULTS ONLY: 10-20mcg EVERY 2-3 MINUTES IF SBP < 90mmHg**
 - IF RECURRENT DYSRHYTHMIA REFER TO APPROPRIATE PROTOCOL
 - **IF SBP > 90mmHg AND iGEL/ETT IN PLACE**
 - **CONSIDER MIDAZOLAM FOR SEDATION AS NEEDED**
 - **ADULTS: 1-2mg IV/IO**
 - **REPEAT AS NEEDED TO MAX GIVEN 5MG**
 - **PEDS: 0.1mg/kg UP TO 2mg**



COMBINED SEIZURE

ASSESSMENT: (BLS or ALS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history including
 - Description of the incident from observers
 - History of medication use
 - Presence/absence of alcohol or substance abuse
 - Recent illness or trauma
- Note any seizure activity
- Apply cardiac monitor (ALS ONLY)
- Apply pulse oximeter
- Obtain rapid blood glucose
 - IF BLOOD GLUCOSE < 70mg/dL REFER TO HYPOGLYCEMIA PROTOCOL

SEIZURE WITH TYPICAL POSTICAL PHASE

- BLS or ALS
 - Secure airway, administer oxygen, or assist ventilations as needed
- ALS ONLY
 - **CONSIDER ESTABLISHING IV ACCESS**
- Transport in lateral decubitus position while monitoring vital signs

REPETITIVE SEIZURES

- BLS or ALS
 - Secure airway, administer oxygen, or assist ventilations as needed
- ALS ONLY
 - **ESTABLISH IV/IO ACCESS**
 - **IF NO KNOWN SEIZURE DISORDER AND 3RD TRIMESTER PREGNANCY OR < 1 MONTH POST-PARTUM AND ACTIVELY SEIZING**
 - **ADULTS ONLY: MAGNESIUM SULFATE 4GM IV/IO OVER 5 MINUTES**
 - **IF NO IV ESTABLISHED:**
 - **MIDAZOLAM**
 - **ADULTS/PEDS: 0.2mg/kg IN MAX 5mg**
 - **MAY REPEAT DOSE X1**
 - **IF IV ESTABLISHED:**
 - **MIDAZOLAM**
 - **ADULTS: 5mg SLOW IV**
 - **MAY REPEAT X1**
 - **PEDS: 0.1mg/kg SLOW IV MAX 5MG**
 - **CONTACT MEDICAL CONTROL FOR ADDITIONAL DOSES IF NO CESSATION OF SEIZURE ACTIVITY**
 - Monitor ETCO2
- Transport while monitoring vital signs

NOTES:

- Seizures may be a sign of hypoxia or cardiac arrest



COMBINED SYNCOPE

ASSESSMENT: (BLS or ALS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history including onset, duration, warning symptoms (light headedness, dizziness, nausea), presence of seizure activity, and precipitating factors
- Apply cardiac monitor (ALS ONLY)
- Apply pulse oximeter
- Obtain rapid blood glucose

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen, or assist ventilations as needed
- ALS ONLY
 - Consider obtaining a 12 Lead ECG in any patient with potential cardiac etiology or previous cardiac history
 - **CONSIDER ESTABLISHING IV ACCESS**
- Transport while monitoring vital signs



ADULT GENERAL AIRWAY

ASSESSMENT: (BLS or ALS)

- Obtain vital signs
- Obtain brief history noting
 - Onset
 - Potential for aspiration of small objects or food
 - Fever or cough
 - Chest pain
 - Past history of prior respiratory or cardiac problems such as Asthma, COPD, or CHF
 - Assess adequacy of airway and ventilation noting
 - Vocalization
 - Rate and depth of respirations
 - Breath sounds
 - Evidence of distress such as accessory muscle use, stridor, cyanosis, or nasal flaring
 - Apply cardiac monitor (ALS ONLY)
 - Apply pulse oximeter

INTERVENTIONS:

- BLS or ALS
 - Is oxygenation/ventilation adequate with appropriate respiratory rate and SpO₂ > 93%
 - IF PRESENT REFER TO APPROPRIATE PROTOCOL
 - BLS Maneuvers:
 - Chin lift/Jaw thrust
 - Nasal/oral airway
 - Is airway patent
 - NO – REFER TO ADULT OBSTRUCTED AIRWAY PROTOCOL
 - YES – administer oxygen as needed to maintain SpO₂ > 93%
 - Prepare to assist ventilations with BVM 6-10 bpm squeezing bag just enough to elicit chest rise and maintain SpO₂ > 93%
 - Consider iGel placement
- ALS ONLY
 - **CONSIDER MONITORING ETCO₂**
 - **CONSIDER CPAP**
 - **CONSIDER ORAL ENDOTRACHEAL INTUBATION if unconscious or in Respiratory Arrest**
 - **ETCO₂ MONITORING IS MANDATORY WITH iGel OR ETT PLACEMENT**
 - If patient is not tolerating iGel or ETT
 - **CONSIDER MIDAZOLAM**
 - **2mg EVERY 5-10 MINUTES UNTIL DESIRED SEDATION ACHIEVED OR SBP < 90mmHg**
- Transport to closest appropriate hospital while monitoring vital signs

NOTES:

- iGel should be left in place if ventilations are adequate
- If oxygenation/ventilations inadequate, CONSIDER Oral Endotracheal Intubation (ALS ONLY)
- DO NOT HYPERVENTILATE – squeeze the BVM just enough for chest rise



ADULT OBSTRUCTED AIRWAY

ASSESSMENT: (BLS or ALS)

- Obtain brief history
 - Events leading up to obstruction
- Assess adequacy of airway and evidence of distress
 - Vocalization
 - Accessory muscle use
 - Stridor
 - Cyanosis
 - Nasal flaring

IS PATIENT RESPONSIVE?

- YES
 - BLS or ALS
 - If patient shows signs of choking
 - Attempt to relieve obstruction with Heimlich Maneuver
 - For visibly pregnant or obese patients perform chest compressions
 - Repeat until obstruction relieved or patient becomes unresponsive
 - When obstruction relieved or if patient is spontaneously breathing or coughing
 - Place in position of comfort
 - Administer oxygen as needed
 - Suction as needed
 - Transport while monitoring vital signs
- NO OR BECOMES UNRESPONSIVE
 - BLS or ALS
 - Position and open airway
 - Remove foreign body if seen
 - Attempt BVM ventilations
 - Begin chest compressions
 - Reposition head and open airway
 - Attempt BVM ventilations
 - BLS: continue previous interventions until obstruction is relieved or ALS is on scene
 - ALS ONLY: **PERFORM DIRECT LARYNGOSCOPY**
 - **REMOVE FOREIGN BODY WITH MAGILL FORCEPS IF VISUALIZED**
 - **ATTEMPT BVM VENTILATIONS AND CONTINUE CHEST COMPRESSIONS IF DIRECT LARYNGOSCOPY UNSUCCESSFUL**
 - **IF UNABLE TO RELIEVE OBSTRUCTION AND ADEQUATELY OXYGENATE/VENTILATE BEGIN EMERGENT TRANSPORT TO CLOSEST FACILITY**

NOTES:

- Anytime obstruction is relieved
 - Place patient in position of comfort or left lateral recumbent if remains unconscious
 - Administer oxygen, monitor SpO2 and vital signs
 - Be prepared for vomiting as is common after an obstruction is relieved
 - Suction as needed



ADULT ASTHMA/COPD



ASSESSMENT: (BLS or ALS)

- Obtain vital signs
- Obtain brief history noting
 - Onset of symptoms
 - Concurrent symptoms
 - Fever, cough, rhinorrhea, tongue/lip swelling, rash, labored breathing, foreign body aspiration
 - Usual trigger of symptoms
 - Cigarette smoke, change in weather, upper respiratory infections
 - Previously intubated
 - Number of ED visits, hospital admissions, and ICU admissions in the previous year
 - Family history of asthma, eczema, or allergies
 - Assess lung sounds
 - Apply cardiac monitor (ALS ONLY)
 - Apply pulse oximeter

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen, or assist ventilation as needed
 - Maintain SpO₂ > 93%
- ALS ONLY
 - **CONSIDER 12 LEAD ECG IF AGE > 35 YEARS OR PATIENT HAS CARDIAC HISTORY**
 - IF STEMI SUSPECTED REFER TO CHEST PAIN/STEMI PROTOCOL
 - **ALBUTEROL 2.5mg NEBULIZED**
 - **REPEAT Albuterol X2 AS NEEDED**
 - OR**
 - **XOPENEX (LEVALBUTEROL) MAY BE USED INSTEAD OF ALBUTEROL FOR LESS STIMULATING EFFECTS. SOLUTION 0.042% (1.25MG IN 3mL) NEBULIZED**
 - **REPEAT XOPENEX AS NEEDED EVERY 10 MIN FOR MAX OF 6 NON-CONTINUOUS DOSES (7.5mg)**
 - **IPRATROPIUM (Atrovent) 0.5mg Nebulized**
 - **DO NOT REPEAT ATROVENT DOSE**
 - **CONSIDER ETCO₂ MONITORING**
 - **ESTABLISH IV ACCESS**
 - If severe distress
 - RR > 30
 - HR > 110
 - Patient speaks less than 3-word sentences
 - **CONSIDER CPAP**
 - **CONSIDER EPINEPHRINE 1:1000 0.3mg IM**
 - **CONSIDER METHYLPREDNISOLONE 125mg**
- Transport monitoring vital signs and respiratory status. Be prepared to assist ventilations as needed

NOTES:

- Epinephrine 1:1000 is CONTRAINDICATED
 - Age 50 or older
 - Cardiac etiology

- Known CAD
- COPD exacerbations are particularly responsive to CPAP and should be considered early

- CPAP CONTRAINDICATIONS:
 - Decreased or altered mental status
 - Facial trauma/burns
 - Vomiting or excessive secretions
 - Cannot maintain own airway
 - SBP < 90mmHg



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

ADULT CHF/PULMONARY EDEMA



ASSESSMENT: (BLS or ALS)

- Obtain vital signs
- Obtain brief history noting
 - Prior respiratory or cardiac problems
 - Prior history of heart failure
- Obtain medications
 - Erectile dysfunction medications in the last 36hrs
 - Such as Viagra, Levitra, or Cialis
- Assess lung sounds
- Apply cardiac monitor (ALS ONLY)
- Apply pulse oximeter
- Consider MI in these patients

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen to maintain SpO₂ > 93%, assist ventilations as needed
- ALS ONLY
 - **OBTAIN 12 LEAD ECG**
 - IF STEMI SUSPECTED REFER TO CHEST PAIN/STEMI PROTOCOL
 - **ESTABLISH IV ACCESS**
 - **NITROGLYCERIN 0.4mg SL EVERY 5 MINUTES IF SBP > 100mmHg**
 - **CONSIDER ETCO₂ MONITORING**
 - **CONSIDER CPAP**
 - **CONSIDER PLACEMENT OF ADVANCED AIRWAY IF PATIENT CONTINUES TO WORSEN**
- Transport while monitoring vital signs and respiratory status
- Be prepared to assist ventilations as needed

NOTES:

- CPAP CONTRAINDICATIONS:
 - Decreased or altered mental status
 - Facial trauma/burns
 - Vomiting or excessive secretions
 - Cannot maintain own airway
 - SBP < 90mmHg
- Avoid nitroglycerin if erectile dysfunction medications have been taken in the past 36 hours



ADULT CHEST PAIN/STEMI

ASSESSMENT: (BLS or ALS)

- Obtain vital signs
- Obtain brief history
 - OPQRST
 - SAMPLE
- Obtain medications and allergies
 - Erectile dysfunction medications in the last 36hrs
 - NTG or ASA for current episode of chest pain
- Assess cardiac and respiratory systems
- Apply cardiac monitor (ALS ONLY)
- Apply oximeter

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen to maintain SpO₂ > 93%, or assist ventilations as needed
- ALS ONLY
 - **OBTAIN 12 LEAD ECG**
 - **ASPIRIN 81mg X4 PO (CHEWED)**
 - If acute STEMI (> 1mm ST segment elevation in 2 or more anatomically continuous leads or NEW LBB)
 - Keep scene time to 15-20 minutes
 - Transport to a STEMI Center
 - Notify receiving hospital as early as possible
 - Transmit 12 Lead ECG to receiving hospital (if possible)
 - **ESTABLISH IV ACCESS**
 - If SBP > 100mmHg
 - **NITROGLYCERIN 0.4mg SL EVERY 5 MINUTES if NOT Inferior (Leads II, III, aVF)**
 - **MAINTAIN SBP > 100mmHg**
 - **CONSIDER FENTANYL**
 - **50 mcg SLOW IVP**
 - **MAY REPEAT EVERY 5 MINUTES TO TOTAL MAX GIVEN 200mcg**
 - **OR**
 - **CONSIDER MORPHINE**
 - **ADULTS: 2-10 mg IV**
 - **No greater than 3 mg/min**
 - **Consider Right-Sided 12-lead if Inferior leads have elevation**

NOTES:

- Nitroglycerin is
 - CONTRAINDICATED in INFERIOR STEMI
 - May be given x1 without IV access ONLY if patient is currently prescribed nitroglycerin
 - Do not give if erectile dysfunction medications have been taken in last 36hrs due to potential for severe hypotension
- Look for ST Depression in reciprocal leads



ADULT BRADYCARDIA

ASSESSMENT: (BLS OR ALS)

- Obtain vital signs
- Obtain brief history of current symptoms, especially current chest pain or shortness of breath
- Assess for SIGNS OF POOR PERFUSION
 - ALTERED MENTAL STATUS
 - CHEST PAIN
 - CHF
 - SHORTNESS OF AIR
 - SYNCOPE
 - HYPOTENSION/SHOCK
- Obtain past medical history
 - Congenital Disease, Previous Dysrhythmias, Coronary Artery Disease, Heart Failure, Heart Murmur
- Current medication looking specifically for
 - Beta Blockers, Calcium Channel Blockers, Clonidine, Digoxin
- Apply Cardiac Monitor (ALS ONLY)
- Apply Pulse Oximeter

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen to maintain SpO₂ > 93%, or assist ventilations as needed
- ALS ONLY
 - IF SUSPECTED CALCIUM CHANNEL BLOCKER OVERDOSE REFER TO OVERDOSE/POISONING PROTOCOL
 - **OBTAIN 12 LEAD ECG**
 - **ESTABLISH IV/IO ACCESS**
 - If signs of poor perfusion exist
 - **ATROPINE 0.5mg IV/IO INITIALLY THEN 1MG IV/IO EVERY 3-5 MINUTES MAX 3MG**
 - **TRANSCUTANEOUS PACING IF NOT RESPONSIVE TO ATROPINE**
 - **CONSIDER EARLY IN 2ND OR 3RD DEGREE AV BLOCK**
 - **CONSIDER MIDAZOLAM 1-2mg IV/IN FOR SEDATION**
 - **REPEAT AS NEEDED TO MAX 5mg**
 - **IF SIGNS OF POOR PERFUSION PERSISTS:**
 - **CONSIDER NS/LR 10ml/kg MAX 500mL OR**
 - **EPI 1:100000 (PUSH DOSE) 10-20mcg IV/IO EVERY 2-3 MINUTES**
- Transport monitoring vital signs

NOTES:

- Atropine in the presence of MI may worsen heart damage
- Atropine is ineffective in cardiac transplant patients
- DO NOT DELAY pacing patients with poor perfusion and presence of MI or 2nd or 3rd degree AV Block



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

ADULT TACHYCARDIA: NARROW COMPLEX



ASSESSMENT: (BLS OR ALS)

- Obtain vital signs
- Obtain brief history of current symptoms, especially current chest pain or shortness of breath
- Assess for SIGNS OF POOR PERFUSION
 - ALTERED MENTAL STATUS
 - CHEST PAIN
 - CHF
 - SHORTNESS OF AIR
 - SYNCOPE
 - HYPOTENSION/SHOCK
- Obtain past medical history
 - Congenital Disease, Previous Dysrhythmias, Coronary Artery Disease, Heart Failure, Heart Murmur
- Current medication looking specifically for
 - Beta Blockers, Calcium Channel Blockers, Clonidine, Digoxin
- Apply Cardiac Monitor (ALS ONLY)
- Apply Pulse Oximeter

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen to maintain SpO₂ > 93%, or assist ventilations as needed
- ALS ONLY
 - **ESTABLISH IV/IO ACCESS**
 - SIGNS OF POOR PERFUSION PRESENT
 - **SYNCHRONIZED CARDIOVERSION**
 - **120J**
 - **REPEAT @ 150J (IF NEEDED)**
 - **REPEAT @ 200J (IF NEEDED)**
 - **CONSIDER MIDAZOLAM**
 - **1-2mg IV/IN FOR SEDATION**
 - **REPEAT AS NEEDED TO MAX 5mg GIVEN**
 - NO SIGNS OF POOR PERFUSION PRESENT
 - **OBTAIN 12 LEAD ECG**
 - **CONSIDER VALSALVA**
 - **CONSIDER ADENOSINE**
 - **12mg IV/IO RAPID PUSH THEN RAPID FLUSH**
 - **IF NO CONVERSION: REPEAT X2 MAX**
- Transport monitoring vital signs

IF CARDIAC ARREST OCCURS REFER TO ADULT CARDIAC ARREST VF/PULSELESS VT OR ASYSTOLE/PEA PROTOCOLS

NOTES:

- Adenosine is not known to be effective in Atrial Fibrillation or Atrial Flutter
- If patient is asymptomatic, consider close observation and/or fluid bolus rather than immediate treatment with medication



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

ADULT TACHYCARDIA: WIDE COMPLEX



ASSESSMENT: (BLS or ALS)

- Obtain vital signs
- Obtain brief history of current symptoms, especially current chest pain or shortness of breath
- Assess for SIGNS OF POOR PERFUSION
 - ALTERED MENTAL STATUS
 - CHEST PAIN
 - CHF
 - SHORTNESS OF AIR
 - SYNCOPE
 - HYPOTENSION/SHOCK
- Obtain past medical history
 - Congenital Disease, Previous Dysrhythmias, Coronary Artery Disease, Heart Failure, Heart Murmur
- Current medication looking specifically for
 - Beta Blockers, Calcium Channel Blockers, Clonidine, Digoxin
- Apply Cardiac Monitor (ALS ONLY)
- Apply Pulse Oximeter

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen to maintain SpO₂ > 93%, or assist ventilations as needed
- ALS ONLY
 - **ESTABLISH IV/IO ACCESS**
 - SIGNS OF POOR PERFUSION PRESENT
 - **SYNCHRONIZED CARディオVERSION**
 - **120J**
 - **REPEAT @ 150J (IF NEEDED)**
 - **REPEAT @ 200J (IF NEEDED)**
 - **CONSIDER MIDAZOLAM 1-2mg IV/IN FOR SEDATION**
 - **REPEAT AS NEEDED TO MAX 5mg GIVEN**
 - **OBTAIN 12 LEAD ECG AFTER CONVERSION**
 - NO SIGNS OF POOR PERFUSION PRESENT
 - **OBTAIN 12 LEAD ECG**
 - REGULAR VT WITH PULSE OR SVT WITH ABERRANCY
 - **AMIODARONE 150mg IV/IO OVER 10 MIN (mix w 100mL saline bag if no premix bag in stock)**
 - IRREGULAR: TORSADES DE POINTES
 - **MAGNESIUM SULFATE 2GM IV/IO OVER 2 MINUTES (mix w 100mL saline bag if no premix bag in stock)**
- Transport monitoring vital signs

IF CARDIAC ARREST OCCURS REFER TO ADULT CARDIAC ARREST VF/PULSELESS VT OR ASYSTOLE/PEA PROTOCOLS

NOTES:

- Torsades de Pointes may not synchronize for cardioversion. IF SIGNS OF POOR PERFUSION are present, proceed with Unsynchronized Cardioversion (defibrillation) at the above listed doses.



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

ADULT CARDIAC ARREST: VF/PULSELESS VT



ASSESSMENT: (BLS or ALS)

- Events leading up to cardiac arrest
- Estimated down time
- Past medical history
 - Existence of terminal illness
 - Left ventricular assist device
- Medications
- Maternal arrest – manual displace uterus to patient's left side, defibrillation is safe at all energy levels
- Persistent VF/Pulseless VT should be transported to a STEMI center

INTERVENTIONS:

- BLS or ALS
 - Initiate manual continuous chest compressions (CCC)
 - Push hard, push fast (100-110/min)
 - Change compressors every 2 minutes
 - Deploy mechanical CPR device after 3 cycles (6 minutes) of manual chest compressions and sufficient manpower present to high-quality manual continuous chest compressions during set up
 - Return immediately to manual chest compressions if issue with mechanical CPR device develops
 - Initiate BVM ventilations at 6/min until iGel ready
 - DO NOT HYPERVENTILATE – squeeze the BVM just enough for chest rise
 - BLS on scene awaiting ALS arrival:
 - Attach AED and follow prompts
 - ALS on SCENE
 - Attach cardiac monitor/defibrillator
 - Establish iGel (BLS or ALS) or **ET TUBE (ALS ONLY)** and continue ventilations at 6/min
 - **ATTACH CAPNOGRAPHY AND MONITOR ETCO2 (ALS ONLY)**
 - ETCO2 < 20mmHg could be an indication of inadequate chest compressions
 - Pulse checks should take no more than 5-10 seconds
- ALS ONLY
 - Check rhythm, check pulse after each 2 min CCC cycle and treat accordingly
 - **DEFIBRILLATE 200J**
 - **RESUME CCC X 2 MIN (ALWAYS RESUME CCC AFTER DEFIBRILLATION)**
 - **ESTABLISH IV/IO ACCESS**
 - **EPINEPHRINE 1:10000 1mg IV/IO EVERY 3-5 MINUTES**
 - **DEFIBRILLATE 200J**
 - **RESUME CCC X 2 MIN**
 - **AMIODARONE 300mg IV/IO**
 - **REPEAT 150mg IV/IO X1 AFTER 5 MINUTES**
 - **IF TORSADES DE POINTES PRESENT: MAGNESIUM SULFATE 2GM IV/IO OVER 2 MIN**
 - **FOR RENAL FAILURE/DIALYSIS PATIENTS:**
 - **CALCIUM CHLORIDE 1GM IV/IO, IV FLUSH at least 20-30 mL between Ca+ and Bicarb**
 - **SODIUM BICARBONATE 1mEq/kg IV/IO**

REFER TO ADULT CARDIAC ARREST: ASYSTOLE/PEA PROTOCOL ANYTIME ASYSTOLE/PEA PRESENT

IF ROSC ACHIEVED REFER TO POST RESUSCITATION CARE PROTOCOL



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

ADULT CARDIAC ARREST: ASYSTOLE/PEA



ASSESSMENT: (BLS or ALS)

- Events leading up to cardiac arrest
- Estimated down time
- Past medical history
 - Existence of terminal illness
 - Left ventricular assist device
- Medications
- Maternal arrest – manual displace uterus to patient's left side, defibrillation is safe at all energy levels
- Persistent VF/Pulseless VT should be transported to a STEMI center

INTERVENTIONS:

- BLS or ALS
 - Initiate manual continuous chest compressions (CCC)
 - Push hard, push fast (100-110/min)
 - Change compressors every 2 minutes
 - Deploy mechanical CPR device after 3 cycles (6 minutes) of manual chest compressions and sufficient manpower present to high-quality manual continuous chest compressions during set up
 - Return immediately to manual chest compressions if issue with mechanical CPR device develops
 - Initiate BVM ventilations at 6/min until iGel ready
 - BLS on scene awaiting ALS arrival:
 - Attach AED and follow prompts
 - ALS on SCENE
 - Attach cardiac monitor/defibrillator
 - Establish iGel (BLS or ALS) or **ET TUBE** (ALS ONLY) and continue ventilations at 6/min
 - **ATTACH CAPNOGRAPHY AND MONITOR ETCO2** (ALS ONLY)
 - Pulse checks should take no more than 5-10 seconds
- ALS ONLY
 - Check rhythm, check pulse after each 2 min CCC cycle and treat accordingly
 - **ESTABLISH IV/IO ACCESS**
 - **EPINEPHRINE 1:10000 1mg IV/IO EVERY 3-5 MINUTES**
 - **CONSIDER REVERSIBLE CAUSES**
 - **FOR RENAL FAILURE/DIALYSIS PATIENTS:**
 - **CALCIUM CHLORIDE 1GM IV/IO, IV FLUSH at least 20-30 mL between Ca+ and Bicarb**
 - **SODIUM BICARBONATE 1mEq/kg IV/IO**

REFER TO ADULT CARDIAC ARREST: VF/PULSELESS VT PROTOCOL ANY TIME VF/PULSELESS VT PRESENT

IF ROSC ACHIEVED REFER TO POST RESUSCITATION CARE PROTOCOL

IF CONSIDERING TERMINATION REFER TO TERMINATION OF RESUSCITATION PROTOCOL



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

ADULT TERMINATION OF RESUSCITATION



ADULT TERMINATION OF RESUSCITATION

Numerous studies have proven that a certain population of cardiac arrest patients will not survive hospital discharge if they fail pre-hospital therapy. Patients that meet the below criteria will be eligible for termination of resuscitative efforts in the field and do not require transport to an Emergency Department. If there is any doubt as to whether to transport or not, it is generally safer to transport per the existing cardiac arrest protocol.

INCLUSION CRITERIA

1. Primary cardiac arrest.
2. Age >21 years of age.
3. Either asystole or PEA as presenting rhythm.
4. Asystole must be the terminal rhythm.
5. Advanced airway placement (ETT or SGA)
6. IV or IO access.
7. > 20 minutes of appropriate therapy.

EXCLUSION CRITERIA

1. Non-primary cardiac arrest (cause of arrest possibly attributed to other pathology). Including but not limited to the following:
 - a. Hypothermia
 - b. Drug overdose (possible acute overdose).
 - c. Primary respiratory arrest (i.e. status asthmaticus).
 - d. Carbon monoxide poisoning
 - e. Traumatic arrest.
 - f. Drowning
2. Ventricular fibrillation or ventricular tachycardia at any time during resuscitation.
3. Any return of spontaneous circulation during resuscitation.
4. Any neurologic activity.
5. Family desires transport (either voiced or perceived).
6. Other, such as unstable scene or disagreement among family members.
7. EMS responder witnessed cardiac arrest.

PROCEDURE

1. Follow appropriate cardiac arrest protocol to include:
 - a. Successful early advanced airway placement.
 - b. Successful early IV drug therapy.
 - c. Minimum of 20 minutes of appropriate therapy.
2. If the patient fits the inclusion criteria and does not fit any exclusion criteria and the above procedures have been accomplished the paramedic may initiate the termination of resuscitation.
PARAMEDIC MUST CONTACT MEDICAL CONTROL FOR ORDERS.
3. Physician and paramedic must both agree on termination.
4. The appropriate law enforcement agency must be notified.
5. Rhythm strips must accompany report and report must contain the name of the medical control physician.



ADULT STROKE



ASSESSMENT: (BLS or ALS)

- Obtain vital signs
- Obtain brief history at scene
 - Determine Time of Onset/Last Known Well
 - Determine progression of symptoms
- Medical history elements of importance noting
 - Medications (especially anticoagulants or antihypertensives)
 - History of hypertension, MI, prior CVA, known bleeding problems, or cardiac dysrhythmias
 - History of DVT or PE
- Administer the Cincinnati Pre-Hospital Stroke Scale
- Apply cardiac monitor (ALS ONLY)
- Apply pulse oximeter
- Obtain rapid blood glucose
 - IF BLOOD GLUCOSE > 70mg/dL REFER TO HYPOGLYCEMIA PROTOCOL

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen to maintain SpO2 > 93%, or assist ventilations as needed
- ALS ONLY
 - **ESTABLISH IV ACCESS**
 - **CONSIDER OBTAINING 12 LEAD ECG**

Transport to appropriate Stroke Facility per Stroke Destination Guidelines while frequently monitoring vital sign



CHILDBIRTH

ASSESSMENT: (BLS or ALS)

- Obtain vital signs
- Obtain brief history
 - Due date, Pre-Natal care, History of multiple births, Onset of labor, Timing of contractions, History of ruptured membranes including fluid color, Previous perinatal complications
- Assess pelvic area for
 - Crowning, Abnormal presentation, Significant vaginal bleeding

INTERVENTIONS:

DELIVERY NOT IMMINENT:

- BLS or ALS
 - Monitor contractions including strength and frequency
 - Secure airway, administer oxygen, or assist ventilations as needed
 - Transport in position of comfort as soon as possible while monitoring vital signs
 - If the patient is hypotensive, place in left lateral decubitus position
 - Notify receiving hospital of impending arrival
 - Be prepared to stop transfer to deliver infant if indicated
- ALS ONLY
 - **CONSIDER ESTABLISHING IV ACCESS**

DELIVERY IMMINENT:

FOR BREECH PRESENTATION, PROLAPSED CORD, OR NUCHAL CORD, REFER TO COMPLICATED CHILDBIRTH PROTOCOL

FOR OTHERWISE NORMAL APPEARING DELIVER:

- ALS ONLY
 - **ESTABLISH IV ACCESS (IF TIME ALLOWS)**
- BLS or ALS
 - Secure airway, administer oxygen, or assist ventilations as needed
 - Do not attempt to impair or delay delivery
 - Support and control delivery of head as it emerges
 - Protect perineum with gentle hand pressure
 - Suction mouth then nose as soon as head is delivered
 - Guide head and neck downward to deliver anterior shoulder
 - Rest should deliver passively – Be sure to have a good grip on baby!!
 - Dry baby and place in warm blanket
 - Double clamp cord 6" from baby after cord stops pulsating and cut cord between clamps
 - Document APGAR at 1 and 5 minutes
 - IF BABY IS APNEIC, CYANOTIC, OR HR < 100, REFER TO NEONATAL RESUSCITATION PROTOCOL
 - Gentle uterine massage can decrease bleeding after delivery
 - IF MOM DEVELOPS PROFUSE HEMORRHAGE OR SHOCK, REFER TO HYPOTENSION/SHOCK PROTOCOL
 - Transport while monitoring vital signs. Do not wait for placental delivery. If placenta delivers spontaneously, bring to the hospital



COMPLICATED CHILDBIRTH

ASSESSMENT: (BLS or ALS)

- Obtain vital signs
- Obtain brief history
 - Due date, Pre-Natal care, History of multiple births, Onset of labor, Timing of contractions, History of ruptured membranes including fluid color, Previous perinatal complications
- Assess pelvic area for
- Crowning, Abnormal presentation, Significant vaginal bleeding
- Consider need for additional resources

INTERVENTIONS:

BREECH DELIVER:

- BLS or ALS
 - Discourage pushing from mother
 - Never attempt to pull infant from vagina by presenting part
 - If legs are delivered, gently elevate trunk and legs to aid delivery of head
 - If head does not deliver within 30 seconds, reach 2 fingers into vagina to locate infant's mouth and press vaginal wall away to access an airway
 - Maintain this position until baby delivers or relieved by physician at hospital
 - CONSIDER rapid transport to closest appropriate facility
 - Do not wait for placenta delivery
 - If the placenta delivers spontaneously, bring to the hospital

PROLAPSED CORD:

- BLS or ALS
 - Discourage mother from pushing
 - Elevate mother's hips
 - Placed gloved hand in mother's vagina and elevate presenting fetal part off of cord until relieved by physician
 - Keep cord moist and warm
 - CONSIDER rapid transport to closest appropriate facility
 - Do not wait for placenta delivery
 - If the placenta delivers spontaneously, bring to the hospital

NUCHAL CORD:

- BLS or ALS
 - Reduce by using 2 gloved fingers to slip under cord and gently remove from around neck
 - If unable to reduce, clamp cord in 2 places and cut cord between the 2 clamps
 - CONSIDER rapid transport to closest appropriate facility
 - Do not wait for placenta delivery
 - If the placenta delivers spontaneously, bring to the hospital



NEONATAL RESUSCITATION

ASSESSMENT: (BLS or ALS)

- Assess airway, breathing, and circulation
- The need for resuscitation can be guided by heart rate, respiratory rate and effort, tone, and color
- Do not be overly concerned about exact APGAR calculation
- If BLS crew consider requesting Mutual Aid ALS Ambulance if not already enroute/on scene

INTERVENTIONS:

- BLS or ALS
 - Suction the mouth then the nose with the bulb syringe or low suction. Do not suction for more than 10 seconds
 - Warm, dry, and stimulate the neonate
 - If heart rate > 100 and no gasping for air or apnea:
 - Routine care
 - Provide warmth
 - Clear airway if necessary
 - Begin ongoing assessment
 - If heart rate < 100 OR baby is gasping for air or is apneic:
 - Assist ventilations @ 40-60 bpm
 - If possible, use 2 person BVM technique
 - Take care not to overinflate the lungs
 - Reassess after 60 seconds
 - After 60 seconds of ventilations, if HR < 60:
 - Begin chest compressions
 - 2-thumb encircling hands technique
 - 3:1 ratio of compressions to ventilations
 - Reassess after 60 seconds
 - After 60 seconds of chest compressions, if HR remains < 60:
 - BLS
 - Continue 3:1 ratios of compressions to ventilations and reassess every 60 seconds
 - Continue until baby improves or relieved by ALS
 - ALS ONLY
 - **ESTABLISH IV/IO ACCESS**
 - **EPINEPHRINE 1:10000 0.01mg/kg IV/IO EVERY 3-5 MINUTES**



PEDIATRIC GENERAL AIRWAY

ASSESSMENT: (BLS or ALS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history noting
 - Onset
 - Potential for aspiration of small objects or food
 - Fever or cough
 - Chest pain
 - Past history of prior respiratory or cardiac problems such as Asthma, COPD, or CHF
 - Assess adequacy of airway and ventilation noting
 - Vocalization
 - Rate and depth of respirations
 - Breath sounds
 - Evidence of distress such as accessory muscle use, stridor, cyanosis, or nasal flaring
 - Apply cardiac monitor (ALS ONLY)
 - Apply pulse oximeter

INTERVENTIONS:

- BLS or ALS
 - Is oxygenation/ventilation adequate with appropriate respiratory rate and SpO₂ > 93%
 - IF PRESENT REFER TO APPROPRIATE PROTOCOL
 - BLS Maneuvers:
 - Chin lift/Jaw thrust
 - Nasal/oral airway
 - Is airway patent
 - NO – REFER TO PEDIATRIC OBSTRUCTED AIRWAY PROTOCOL
 - YES – administer oxygen as needed to maintain SpO₂ > 93%
 - Prepare to assist ventilations with BVM 10-12 bpm squeezing bag just enough to elicit chest rise and maintain SpO₂ > 93%
 - Consider iGel placement
- ALS ONLY
 - **CONSIDER ORAL ENDOTRACHEAL INTUBATION if unconscious or in respiratory arrest**
 - **ETCO₂ MONITORING IS MANDATORY WITH iGel OR ETT PLACEMENT**
- Transport to closest appropriate hospital while monitoring vital signs

NOTES:

- Oral endotracheal intubation in patients younger than 8 years old is very controversial. 2 person BVM ventilation is preferred in patients less than 8 years of age.
- Proper positioning and preparation are necessary before an attempt at advanced airway placement is made



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

PEDIATRIC OBSTRUCTED AIRWAY



ASSESSMENT: (BLS or ALS)

- Obtain brief history
 - Events leading up to obstruction
- If battery was ingested without airway obstruction, refer to Battery Ingestion protocol
 - Assess adequacy of airway and evidence of distress
 - Vocalization
 - Accessory muscle use
 - Stridor
 - Cyanosis
 - Nasal flaring

IS PATIENT RESPONSIVE?

- YES
 - BLS or ALS
 - If patient shows signs of choking:
 - Age < 1 year: Give 5 back blows followed by 5 chest compressions
 - Repeat until obstruction relieved or patient becomes unresponsive
 - Age > 1 year: Give 5 abdominal thrusts
 - Repeat until obstruction relieved or patient becomes unresponsive
 - When obstruction relieved or if patient is spontaneously breathing or coughing
 - Place in position of comfort
 - Administer oxygen as needed
 - Suction as needed
 - Transport while monitoring vital signs
- NO OR BECOMES UNRESPONSIVE
 - BLS or ALS
 - Position and open airway
 - Remove foreign body if seen
 - Attempt BVM ventilations
 - Age < 1 year: Give 5 back blows followed by 5 chest compressions
 - Age > 1 year: Give chest compressions
 - Reposition head and open airway
 - Attempt ventilations
 - BLS: continue previous interventions until obstruction is relieved or ALS is on scene
 - ALS ONLY: **PERFORM DIRECT LARYNGOSCOPY**
 - **REMOVE FOREIGN BODY WITH MAGILL FORCEPS IF VISUALIZED**
 - **ATTEMPT BVM VENTILATIONS AND CONTINUE CHEST COMPRESSIONS IF DIRECT LARYNGOSCOPY UNSUCCESSFUL**
 - **IF UNABLE TO RELIEVE OBSTRUCTION AND ADEQUATELY OXYGENATE/VENTILATE BEGIN EMERGENT TRANSPORT TO CLOSEST FACILITY**

NOTES:

- Anytime obstruction is relieved
 - Place patient in position of comfort or left lateral recumbent if remains unconscious
 - Administer oxygen, monitor SpO2 and vital signs

- Be prepared for vomiting as is common after an obstruction is relieved
- Suction as needed



Battery Ingestion

Indications:

- Battery ingestion
- Suspected button battery ingestion is less than 12 hours
- Patient is 12 months or older and can swallow without difficulty

Assessment: (ALS or BLS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history noting
 - Type of battery and time of ingestion

Intervention:

- 10 mL (2 teaspoons) PO every 10 minutes, no max dose.

Contraindications:

- Stop administering honey if the patient vomits or shows signs of respiratory distress.
- Patient is younger than 12 months
- Inability to swallow or protect airway or an airway obstruction

Notes:

- Immediate action and emergency medical care are mandatory following battery ingestion. The honey protocol is a first-aid measure designed to mitigate damage while the patient is en route to the emergency room; it is not a cure or a substitute for urgent medical care.
- Honey is not safe for infants younger than one year due to the risk of infant botulism. Administration is typically recommended only if the ingestion occurred within the last 12 hours.
- Do not induce vomiting. This action can potentially worsen the injury by causing the battery to move.
- When a button battery becomes lodged in the esophagus, it can cause a caustic, alkaline burn due to the production of hydroxide ions, which can lead to severe injury rapidly. The properties of honey may help to neutralize the alkaline environment and coat the battery, potentially slowing the progression of injury.



PEDIATRIC WHEEZING

ASSESSMENT: (BLS or ALS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history noting
 - Onset of symptoms
 - Concurrent symptoms
 - Fever, cough, rhinorrhea, tongue/lip swelling, rash, labored breathing, foreign body aspiration
 - Usual trigger of symptoms
 - Cigarette smoke, change in weather, upper respiratory infections
 - Previously intubated
 - Number of ED visits, hospital admissions, and ICU admissions in the previous year
 - Family history of asthma, eczema, or allergies
 - Assess lung sounds
 - Apply cardiac monitor (ALS ONLY)
 - Apply pulse oximeter

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen, or assist ventilation as needed
 - Maintain SpO₂ > 93%
- ALS ONLY
 - **MONITOR ETCO₂ (IF AVAILABLE)**
 - **CONSIDER XOPENEX OVER ALBUTEROL FOR PEDIATRICS DUE TO ITS LESS STIMULATING EFFECTS.**
 - **LEVALBUTEROL (XOPENEX) SOLUTION 0.042% (1.25mg in 3 mL) NEBULIZED**
 - **REPEAT AS NEEDED EVERY 10 MIN FOR MAX 3 NON-CONTINUOUS DOSES (3.75MG).**
 - **OR**
 - **ALBUTEROL 2.5mg NEBULIZED**
 - **REPEAT X2 AS NEEDED**
 - **If over 8 yrs old, consider IPRATROPIUM (Atrovent) 0.5mg NEBULIZED. DO NOT REPEAT.**
 - **CONSIDER ESTABLISHING IV ACCESS**
 - **CONSIDER METHYLPREDNISOLONE 2mg/kg IV OVER 2 MIN**
 - If severe distress: Speaks less than 3-word sentences, Severe retractions or, Silent chest
 - **MONITOR ETCO₂**
 - **CONSIDER EPINEPHRINE 1:1000 0.01mg/kg IM MAX 0.3mg**
 - **CONSIDER NS/LR BOLUS 20mL/kg**
- Transport monitoring vital signs and respiratory status. Be prepared to assist ventilations as needed

NOTES:

- Consider cause:
 - If patient 2 years old or younger, most likely bronchiolitis
 - If patient older than 2 years, more likely reactive airway disease
 - Bronchiolitis:
 - Viral illness characterized by fever, secretions, and respiratory distress typically seen November through April
 - Most important interventions are supplemental oxygen and suction secretions
 - Bronchodilators may not work
 - Croup:

- Most common cause of stridor in children
- Most often seen in children < 9 years old

- Child will have stridor, barking cough, and URI symptoms often with a sudden, nocturnal onset
- Agitation can worsen stridor and respiratory distress

- Epiglottitis:
 - Extremely rare
 - Allow patient to assume position of comfort, do not agitate
 - Do not manipulate airway – best done in the hospital



PEDIATRIC BRADYCARDIA

ASSESSMENT: (BLS OR ALS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history of current symptoms, especially current chest pain or shortness of breath
- Assess for SIGNS OF POOR PERFUSION
 - ALTERED MENTAL STATUS
 - CHEST PAIN
 - CHF
 - SHORTNESS OF AIR
 - SYNCOPE
 - HYPOTENSION/SHOCK
- Obtain past medical history
 - Congenital Disease, Previous Dysrhythmias, Coronary Artery Disease, Heart Failure, Heart Murmur
- Current medication looking specifically for
 - Beta Blockers, Calcium Channel Blockers, Clonidine, Digoxin
- Apply Cardiac Monitor (ALS ONLY)
- Apply Pulse Oximeter

INTERVENTIONS:

- AGE < 8 YEARS
 - BLS or ALS
 - Secure airway, administer oxygen to maintain SpO₂ > 93%, or assist ventilations as needed
 - Do chest compressions if, despite adequate oxygenation and ventilation:
 - HR < 80 beats/min in infants (age < 1 year)
 - HR < 60 beats/min in child (age 1-8 years)
 - ALS ONLY
 - **ESTABLISH IV/IO ACCESS**
 - **EPINEPHRINE 1:10000 0.1mg/kg IV/IO**
 - **MAY REPEAT EVERY 3-5 MINUTES TO DESIRED RATE EFFECT**
 - **CONSIDER ATROPINE 0.02mg/kg IV/IO MINIMUM DOSE 0.1mg**
 - **MAY REPEAT X1 AFTER 3-5 MINUTES MAX 0.5mg**
 - **APPLY EXTERNAL PACER**
- AGE > 8 YEARS
 - ALS ONLY
 - **ESTABLISH IV/IO ACCESS**
 - **ATROPINE 0.02mg/kg IV/IO MINIMUM DOSE 0.1mg EVERY 3-5 MINUTES MAX 1mg**
 - **TRANSCUTANEOUS PACING IF NOT RESPONSIVE TO ATROPINE**
 - **CONSIDER EARLY IN 2ND OR 3RD DEGREE AV BLOCK**
 - **IF POOR PERFUSION PERSISTS:**
 - **CONSIDER NS/LR 10mL/kg (MAX 500mL) TO MAINTAIN AGE APPROPRIATE SBP**



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

PEDIATRIC TACHYCARDIA: NARROW COMPLEX



ASSESSMENT: (BLS OR ALS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history of current symptoms, especially current chest pain or shortness of breath
- Assess for SIGNS OF POOR PERFUSION
 - ALTERED MENTAL STATUS
 - CHEST PAIN
 - CHF
 - SHORTNESS OF AIR
 - SYNCOPE
 - HYPOTENSION/SHOCK
- Obtain past medical history
 - Congenital Disease, Previous Dysrhythmias, Coronary Artery Disease, Heart Failure, Heart Murmur
- Current medication looking specifically for
 - Beta Blockers, Calcium Channel Blockers, Clonidine, Digoxin
- Apply Cardiac Monitor (ALS ONLY)
- Apply Pulse Oximeter

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen to maintain SpO₂ > 93%, or assist ventilations as needed
- ALS ONLY
 - **ESTABLISH IV/IO ACCESS**
 - SIGNS OF POOR PERFUSION PRESENT
 - **SYNCHRONIZED CARDIOVERSION**
 - **1ST ATTEMPT: 0.5J/kg (120J MAX)**
 - **2ND ATTEMPT: 1J/kg (150J MAX)**
 - **3RD ATTEMPT: 2J/kg (200J MAX)**
 - **4TH ATTEMPT: 3-4J/kg (200J MAX)**
 - NO SIGNS OF POOR PERFUSION PRESENT
 - **ADENOSINE 0.1mg/kg MAX 6mg IV**
 - **ADENOSINE 0.2mg/kg MAX 12mg IV**
- Transport monitoring vital signs

IF CARDIAC ARREST OCCURS REFER TO PEDIATRIC CARDIA ARREST VF/PULSELESS VT OR ASYSTOLE/PEA PROTOCOLS

NOTES:

- If patient is asymptomatic, consider close observation and/or fluid bolus rather than immediate treatment with medication



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

PEDIATRIC TACHYCARDIA: WIDE COMPLEX



ASSESSMENT: (BLS OR ALS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history of current symptoms, especially current chest pain or shortness of breath
- Assess for SIGNS OF POOR PERFUSION
 - ALTERED MENTAL STATUS
 - CHEST PAIN
 - CHF
 - SHORTNESS OF AIR
 - SYNCOPE
 - HYPOTENSION/SHOCK
- Obtain past medical history
 - Congenital Disease, Previous Dysrhythmias, Coronary Artery Disease, Heart Failure, Heart Murmur
- Current medication looking specifically for
 - Beta Blockers, Calcium Channel Blockers, Clonidine, Digoxin
- Apply Cardiac Monitor (ALS ONLY)
- Apply Pulse Oximeter

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen to maintain SpO₂ > 93%, or assist ventilations as needed
- ALS ONLY
 - **ESTABLISH IV/IO ACCESS**
 - SIGNS OF POOR PERFUSION PRESENT
 - **SYNCHRONIZED CARDIOVERSION**
 - **1ST ATTEMPT: 0.5J/kg (120J MAX)**
 - **2ND ATTEMPT: 1J/kg (150J MAX)**
 - **3RD ATTEMPT: 2J/kg (200J MAX)**
 - **4TH ATTEMPT: 3-4J/kg (200J MAX)**
 - NO SIGNS OF POOR PERFUSION PRESENT
 - **AMIODARONE 5mg/kg IV/IO MAX 300mg OVER 20 MINUTES**
- Transport monitoring vital signs

IF CARDIAC ARREST OCCURS REFER TO PEDIATRIC CARDIA ARREST VF/PULSELESS VT OR ASYSTOLE/PEA PROTOCOLS

NOTES:

- If patient is asymptomatic, consider close observation and/or fluid bolus rather than immediate treatment with medication



PEDIATRIC CARDIAC ARREST: VF/PULSELESS VT



ASSESSMENT: (BLS or ALS)

- Events leading up to cardiac arrest
- Estimated down time
- Past medical history
 - Prematurity
 - Congenital heart disease
- Medications

INTERVENTIONS:

- BLS or ALS
 - Initiate continuous chest compressions (CCC)
 - Push hard, push fast (100-120/min)
 - Change compressors every 2 minutes
 - Initiate BVM ventilations with 2 rescuer technique at 10-12/min until iGel ready
 - DO NOT HYPERVENTILATE – squeeze the BVM just enough for chest rise
 - BLS on scene awaiting ALS arrival:
 - Attach AED (Age > 1 year) and follow prompts
 - ALS on SCENE
 - Attach cardiac monitor/defibrillator
 - Establish iGel (BLS or ALS) and continue ventilations at 10-12/min
 - ALS ONLY
 - **CONSIDER ORAL ENDOTRACHEAL INTUBATION**
 - **ATTACH CAPNOGRAPHY AND MONITOR ETCO2 WITH ANY ADVANCED AIRWAY**
 - DO NOT interrupt chest compressions for advanced airway placement
 - Pulse checks should take no more than 5-10 seconds
- ALS ONLY
 - Check rhythm, check pulse after each 2 min CCC cycle and treat accordingly
 - **DEFIBRILLATE 4J/kg MAX 200J**
 - **RESUME CCC X 2 MINUTES (ALWAYS RESUME CCC AFTER DEFIBRILLATION)**
 - **ESTABLISH IV/IO ACCESS**
 - **EPINEPHRINE 1:10000 0.01mg/kg IV/IO EVERY 3-5 MINUTES**
 - **DEFIBRILLATE 4J/kg MAX 200J**
 - **RESUME CCC X2 MINUTES**
 - **AMIODARONE 5mg/kg IV/IO MAX 300MG**
 - **MAY REPEAT 2.5mg/kg IV/IO MAX 150mg X1 AFTER 5 MIN**

REFER TO PEDIATRIC CARDIAC ARREST: ASYSTOLE/PEA PROTOCOL ANY TIME ASYSTOLE/PEA PRESENT

IF ROSC ACHIEVED REFER TO POST RESUSCITATION CARE PROTOCOL



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

PEDIATRIC CARDIAC ARREST: ASYSTOLE/PEA



ASSESSMENT: (BLS or ALS)

- Events leading up to cardiac arrest
- Estimated down time
- Past medical history
 - Prematurity
 - Congenital heart disease
- Medications

INTERVENTIONS:

- BLS or ALS
 - Initiate continuous chest compressions (CCC)
 - Push hard, push fast (100-120/min)
 - Change compressors every 2 minutes
 - Initiate BVM ventilations with 2 rescuer technique at 10-12/min until iGel ready
 - DO NOT HYPERVENTILATE – squeeze the BVM just enough for chest rise
 - BLS on scene awaiting ALS arrival:
 - Attach AED (Age > 1 year) and follow prompts
 - ALS on SCENE
 - Attach cardiac monitor/defibrillator
 - Establish iGel (BLS or ALS) and continue ventilations at 10-12/min
 - ALS ONLY
 - **CONSIDER ORAL ENDOTRACHEAL INTUBATION**
 - **ATTACH CAPNOGRAPHY AND MONITOR ETCO2 WITH ANY ADVANCED AIRWAY**
 - DO NOT interrupt chest compressions for advanced airway placement
 - Pulse checks should take no more than 5-10 seconds
- ALS ONLY
 - Check rhythm, check pulse after each 2 min CCC cycle and treat accordingly
 - **ESTABLISH IV/IO ACCESS**
 - **EPINEPHRINE 1:10000 0.01mg/kg IV/IO EVERY 3-5 MINUTES**
 - **CONSIDER REVERSABLE CAUSES**

REFER TO PEDIATRIC CARDIAC ARREST: VF/PULSELESS VT PROTOCOL ANY TIME VF/PULSELESS VT PRESENT

IF ROSC ACHIEVED REFER TO POST RESUSCITATION CARE PROTOCOL



PEDIATRIC BRUE



BRUE: BRIEF RESOLVED UNEXPLAINED EVENT:

- An event in an infant less than 1 year old reported by a bystander as sudden, brief (less than 1 minute), completely resolved upon EMS arrival that includes one or more of the following
 - Absent, decreased, or irregular breathing
 - Color change (central cyanosis or pallor)
 - Marked change in muscle tone (hyper or hypotonia)
 - Altered level of responsiveness

ASSESSMENT: (BLS or ALS)

- Obtain vital signs, , consult PEDIATRIC VITAL SIGN RANGES in appendix for age appropriate vitals
 - Also obtain: Temperature, SpO2, and Blood Glucose
- Obtain history to include:
 - Observer's impression of the infant's color, respirations, and muscle tone
 - Was the infant apneic, cyanotic, or limp during event
 - Was there seizure-like activity
 - Was any resuscitation attempted or required, or did event resolve spontaneously
 - How long did the event last
- Detailed past medical history
 - Recent trauma, infection (fever or cough)
 - History of GERD
 - History of Congenital Heart Disease
 - History of seizures
 - Medication history
- Complete head to toe assessment

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen to maintain SpO2 > 93%, or assist ventilations as needed
 - Any pediatric patient with BRUE should be transport while monitoring vital signs

NOTES:

- **CONTACT MEDICAL CONTROL IF PARENTS REQUEST TO REFUSE TRANSPORT OF THESE PATIENTS**
- The following is not BRUE and could indicate serious illness:
 - Abnormal vital signs for age (including fever)
 - Vomiting
 - Signs of trauma (non-accidental trauma)
 - Noisy breathing
 - There is an identifiable cause of the event
 - History or exam is concerning for child abuse or neglect



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

COMBINED GENERAL TRAUMA



The purpose of this protocol is to outline the general approach to the patient with a traumatic injury.

SCENE SIZE UP: (BLS or ALS)

- Assure the scene is safe for you, other rescuers, and the patient. It may be appropriate to withdraw from the scene in some situation until a safe environment can be obtained. It may be appropriate to rapidly extricate the patient from a dangerous situation.
- Identify the number of patients and other resources that may be needed
 - Initiate Incident Management System if appropriate
 - Declare an MCI as needed
 - Establish "Medical Command" as needed
 - Call for law enforcement and/or first responder assistance as needed
 - Call for additional EMS units as needed
 - Begin triage as needed
- Consider mechanism of injury

BODY SUBSTANCE ISOLATION: (BLS or ALS)

- Apply universal precautions/body substance isolation as needed

PRIMARY SURVEY

Search for immediate life threats by assessing the patient by using C-A-B process (circulation survey and controlling external hemorrhage, airway, and breathing) and treating the problems as they are found.

Abnormalities found in the Primary Survey should be addressed with appropriate interventions at time of discovery.

It may be appropriate to move directly from the Primary Survey to another protocol

ASSESSMENT: (BLS or ALS)

- Assess CIRCULATION:
 - Note pulses, level of consciousness, skin abnormalities (color, temperature, capillary refill, moisture)
 - ALS ONLY: **ESTABLISH IV/IO ACCESS (DO NOT DELAY TRANSPORT FOR IV/IO ATTEMPTS)**
- Assess AIRWAY WITH SIMULTANEOUS CERVICAL SPINE MOTION RESTRICTION:
 - Note patient's ability to speak and any evidence of actual or potential airway obstruction including vomitus, bleeding, dentures, loose teeth, or foreign bodies
 - Secure airway:
 - BLS Maneuvers:
 - Jaw thrust
 - Oral or nasal airway
 - iGel placement
 - Apply capnography (ALS ONLY)
 - ALS Maneuvers:
 - Oral endotracheal intubation
 - Apply capnography and monitor ETCO2
- Assess BREATHING:
 - Note patient's ability to speak, rate, depth, and quality of respirations, abnormal noises/stridor, retractions, accessory muscle use, nasal flaring, or cyanosis
 - Administer oxygen as needed
 - Assist ventilations as needed

- BVM ventilation is indicated prior to attempts at iGel placement or endotracheal intubation

- Assess neurological function (DISABILITY)
 - Note level of consciousness, Glasgow Coma Scale or AVPU Scale, movement of each extremity
- If major bleeding is present, control with sterile dressing and direct pressure. If direct pressure fails:
 - Consider approved tourniquet to control extremity hemorrhage
- DO NOT remove impaled objects unless airway management is compromised

IF NO PULSE REFER TO COMBINED TRAUMATIC CARDIAC ARREST PROTOCOL

SECONDARY SURVEY

A systemic history and physical exam, focused on the patient's complaints, searching for problems that may not be immediately life or limb threatening, but may become so if not addressed appropriately

ASSESSMENT: (BLS or ALS)

- Obtain Chief Complaint
- Obtain "SAMPLE" history
- Obtain vital signs (MANUAL Blood Pressure)
- Perform focused physical exam (this evaluation is dependent on the history as well as findings from the primary survey and may be more or less detailed depending on the situation)
- Consider application of cardiac monitor (ALS ONLY)
- Consider application of pulse oximeter
- Consider obtaining rapid blood glucose

Assessment summation: Consider information gathered in primary and secondary survey, determine an impression of the patient's primary problems and proceed to the appropriate treatment protocol

Assessment summation: Consider information gathered in primary and secondary survey, determine an impression of the patient's primary problems and proceed to the appropriate treatment protocol

Transport while monitoring vital signs

Patient destination as determined by appropriate protocol

Medical Control contact as determined by appropriate protocol

REFER TO APPROPRIATE PROTOCOL FOR SPECIFIC BODY AREA INJURY

For pain control, REFER TO COMBINED PAIN CONTROL PROTOCOL



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

COMBINED ABDOMINAL TRAUMA



ASSESSMENT: (BLS or ALS)

- Obtain vitals, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history of events leading to injury
- Abdominal exam should note presence of:
 - Bruising
 - Bleeding
 - Distension
 - Open wounds
 - Evisceration

INTERVENTIONS:

- BLS or ALS
 - Consider rapid transport to a trauma center
 - If evisceration is present, cover the exposed viscera with sterile saline/water-soaked pads, then cover w Saran wrap.
 - Do not attempt to replace the exposed viscera in the abdominal cavity
- ALS ONLY
 - **CONSIDER ESTABLISHING IV ACCESS**
 - **CONSIDER IV NS/LR**
 - **ADULTS: 1000mL IV REPEAT X1 TO A MAX OF 2L TO MAINTAIN SBP > 90mmHg**
 - **PEDS: 20mL/kg MAX 1L TO MAINTAIN AGE APPROPRIATE SBP**



COMBINED BURN

ASSESSMENT: (BLS or ALS)

- Obtain brief history noting type of burn
 - Thermal
 - Electrical
 - Chemical
 - Scald
- Assess the Percentage Total Body Surface Area (%TBSA) noting depth and extent
 - Patient's palm represents approximately 1% of their TBSA
 - Use "Rule of 9's" to estimate TBSA for large burns
 - Use "Superficial", "Partial Thickness" and "Full Thickness" to describe burn thickness
- Be alert for associated injuries
- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Apply cardiac monitor (ALS ONLY)
- Apply pulse oximeter
- Prevent hypothermia regardless of burn size

INTERVENTIONS:

- BLS or ALS
 - STOP THE BURN
 - Remove clothing if not stuck
 - Remove jewelry or any constricting items
 - Dry Powder – brush off
 - Wet substance – copiously irrigate
 - Leave blisters intact
 - Presence of: Respiratory Distress, Hoarseness, Stridor, Singed Nasal hair, Oral Burns, or Carbonaceous Sputum
 - Secure airway, administer oxygen, or assist ventilations as needed
 - CRITICAL BURNS
 - > 20% Partial Thickness, > 5% Full Thickness, Facial burns/Inhalation injury, Circumferential burns, High voltage electrical burns, Genital/Perineum burns, Burns of joints or exposed bones, Burns to hands or feet
 - ALS ONLY
 - **ESTABLISH IV/IO ACCESS**
 - **NS/LR**
 - **ADULTS: 500mL/hr**
 - **PEDS: 20mL/kg MAX 500mL**
 - NON-CRITICAL BURNS
 - < 20% Partial Thickness or does not meet critical burn criteria
 - Gently wrap in dry kerlex
 - CONSIDER moistened gauze if < 10% TBSA burn
 - Avoid hypothermia
 - NEVER apply ice directly
 - % TBSA estimations are unreliable in high voltage electrical burns
 - For pain control REFER TO COMBINED PAIN CONTROL PROTOCOL
 - Transport as soon as possible



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

COMBINED CARBON MONOXIDE EXPOSURE



ASSESSMENT: (BLS or ALS)

- Remove from area of exposure
- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history
 - Length of exposure
- Assess for symptoms
 - Headache
 - Dizziness
 - Nausea
 - Vomiting
 - Weakness
 - Numbness
 - Blurry or double vision
 - Difficulty breathing
 - Chest pain or tightness

INTERVENTIONS:

- BLS or ALS
 - If symptoms present
 - Oxygen 15 LPM via NRB
 - Apply cardiac monitor (ALS ONLY)
 - Apply pulse oximeter
 - Transport to hospital monitoring vitals

NOTES:

- Carbon Monoxide exposure can occur in any closed space or fire ground where there are elevated CO levels are measured
- Pulse oximetry can be unreliable
- Pregnant women may be asymptomatic but the fetus could be in distress
 - Should be transported to hospital

**COMBINED CHEST TRAUMA**ASSESSMENT: (BLS or ALS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Exam chest noting:
 - Open wounds
 - Flail segments
 - Tracheal deviation
 - Unequal breath sounds
 - Subcutaneous emphysema
 - Adequacy of ventilations
- With penetrating chest injury, consider the potential for remote injury

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen to maintain SpO₂ > 93%, or assist ventilations as needed
 - Flail chest with respiratory distress
 - CONSIDER positive pressure ventilations
 - Cover and open chest wound with non-porous material secured on three sides to act as “flap”
 - Use Chest Seal device if available
- ALS ONLY
 - **REFER TO NEEDLE THORACOSTOMY PROCEDURE IF TENSION PNEUMOTHORAX PRESENT**
 - **CONSIDER ESTABLISHING IV/IO ACCESS**
 - **CONSIDER IV FLUIDS:**
 - **ADULTS: NS/LR 1000mL REPEAT X1 MAX 2L TO MAINTAIN SBP > 90mmHg**
 - **PEDS: NS/LR 20mL/kg MAX 1L TO MAINTAIN AGE APPROPRIATE SBP**
 - **APPLY CARDIAC MONITOR**
 - **MONITOR FOR DYSRHYTHMIAS IN BLUNT TRAUMA**
- Transport to closest trauma center if patient meets trauma routing criteria while frequently monitoring vital signs



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

COMBINED DROWNING/NEAR DROWNING



ASSESSMENT: (BLS or ALS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain history including circumstances surrounding incident:
 - Diving accident and/or suspected trauma
 - Total time of submersion
 - Water temperature (if known)
 - Water contamination

INTERVENTIONS:

- IF NO PULSE PRESENT REFER TO THE APPROPRIATE CARDIAC ARREST PROTOCOL
- REFER TO SPINAL MOTION RESTRICTION PROCEDURE IF DIVING ACCIDENT OR TRAUMA SUSPECTED
- BLS or ALS
 - Remove wet clothes
 - Dry, warm, and insulate the patient
 - Secure airway, administer oxygen to maintain SpO₂ > 93%, or assist ventilations as needed
- ALS ONLY
 - **CONSIDER ESTABLISHING IV ACCESS**
- Transport while monitoring vital signs

NOTES:

- Near drowning patients may or may not be conscious. Regardless of consciousness, these patients should be transported due to increased risk of delayed death
- Drowning/submersion is commonly associated with hypothermia
- Good outcomes even after prolonged hypothermic arrest are possible



COMBINED EXPOSURE TO TASER TYPE WEAPONS



This protocol is intended to provide guidelines for care of patients following the use of TASER type weapons by law enforcement. For situations involving level of consciousness, significant trauma, and/or other medical problems, follow the applicable protocol(s).

There have been reports of death involving the use of a TASER weapon on combative patients. These deaths may be associated with agitated delirium, hyperdynamic (adrenergic) drugs particularly cocaine, methamphetamine and phencyclidine, and hyperthermia as major co-morbid factors. Therefore, it is imperative that these patients receive a thorough assessment for these risk factors, and are not restrained in an improper position. If a patient remains combative, follow the Patient Restraint Procedure.

ASSESSMENT: (BLS or ALS)

- Assure the scene is secure. Use of this type of weapon to subdue a violent person implies they were a risk to themselves or others. Do not enter scene until advised to do so by law enforcement personnel
- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Apply pulse oximeter
- Apply cardiac monitor (ALS ONLY)
- Perform 12 Lead ECG if patient has any cardiac or respiratory symptoms (ALS ONLY)

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen, or assist ventilations as needed to maintain SpO2 > 93%
 - Stabilize dart(s) in place and transport to ED if the dart(s) is/are embedded in the:
 - Eyelid/globe of eye
 - Face/neck
 - Genitalia
 - Darts in other locations may be carefully removed
 - Control minor bleeding
 - Irrigate area
 - Apply appropriate dressing
- ALS ONLY
 - **ESTABLISH IV ACCESS AS INDICATED BY OTHER SYMPTOMS OR APPROPRIATE PROTOCOLS**

NOTES:

- Dart removal
 - Place one hand on the patient in the area where the dart is embedded and stabilize the skin surrounding the puncture site. Firmly grasp the dart with the other hand
 - In one fluid motion, pull the dart straight out from the puncture site
 - Assure the dart is intact and no portion of the dart remains inside the patient's skin
 - Removed darts should be handled like contaminated sharps



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

COMBINED EXTREMITY TRAUMA



ASSESSMENT: (BLS or ALS)

- Assess the injured extremity noting neurovascular adequacy (color, temperature), deformity, open wounds, and distal sensation and movement

INTERVENTIONS:

- FRACTURES/DISLOCATIONS/SPRAINS
 - BLS or ALS
 - Apply sterile moistened dressing to all open wounds
 - Apply the appropriate splint
 - Chemical cold packs may be applied after splinting
- AMPUTATIONS
 - BLS or ALS
 - Apply pressure with sterile gauze pads and elevate to control external bleeding
 - If direct pressure fails to control bleeding, CONSIDER combat tourniquet if able to apply in appropriate area
 - Amputated part should be rinsed with sterile water or saline, wrapped in a dry dressing, placed in a water tight container, and placed on a chemical cold pack during transport
 - Amputated parts should never be soaked or placed in water or saline, placed directly on ice or ice packs, cooled with dry ice
 - Partial amputations should be dressed and splinted in alignment with the extremity. Avoid torsion
 - To assure viability of the amputated part, transport should proceed as promptly as possible – CONSIDER transport to a trauma center
- PENETRATING EXTREMITY WOUNDS
 - BLS or ALS
 - Apply pressure with sterile gauze pads and elevate to control external bleeding
 - If direct pressure fails to control bleeding, CONSIDER combat tourniquet if able to apply in appropriate location
 - Apply the appropriate splint
- ENTRAPMENT/CRUSH INJURY
 - BLS or ALS
 - Assess for length of entrapment
 - Apply pressure with sterile gauze pads and elevate to control external bleeding
 - If direct pressure fails to control bleeding, CONSIDER combat tourniquet if able to apply in appropriate location
 - Apply the appropriate splint
 - ALS ONLY
 - **ESTABLISH IV/IO ACCESS**
 - **NS/LR**
 - **ADULTS: 1000mL REPEAT X1 MAX 2L TO MAINTAIN SBP > 90mmHg**
 - **PEDS: 20mL/kg MAX 1L TO MAINTAIN AGE APPROPRIATE SBP**
 - **APPLY CARDIAC MONITOR AND OBTAIN 12 LEAD EKG**
 - **IF SIGNS OF HYPERKALEMIA PRESENT**
 - **CONTACT MEDICAL CONTROL TO CONSIDER:**
 - **ADULTS/PEDS: CALCIUM CHLORIDE 1GM IV/IO (flush saline between Ca+ & Bicarb)**
 - **ADULTS/PEDS: SODIUM BICARB 1mEq/kg IV/IO**
- For pain control, REFER TO COMBINED PAIN CONTROL PROTOCOL



COMBINED HEAD TRAUMA



ASSESSMENT: (BLS or ALS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history noting mechanism of injury, use of safety devices, and level of consciousness
- Neurologic exam should note:
 - Level of consciousness (Glasgow Coma Scale, AVPU Scale)
 - Pupil size and reactivity
 - Presence of posturing or paralysis
- Be alert for associated injuries. Assume cervical spine injury is present in all patients with significant head trauma

INTERVENTIONS:

- BLS or ALS
 - Secure airway, administer oxygen, or assist ventilations as needed to maintain SpO2 > 93%
- ALS ONLY
 - **CONSIDER ESTABLISHING IV/IO ACCESS**
 - **NS/LR**
 - **ADULTS: 1000mL REPEAT X1 MAX 2L TO MAINTAIN SBP > 120mmHg**
 - **PEDS: 20mL/kg MAX 1L TO MAINTAIN AGE APPROPRIATE SBP**

IF PATIENT ARRESTS REFER TO TRAUMATIC CARDIAC ARREST PROTOCOL

NOTES:

- Prehospital intubation is not a priority if airway can be adequately maintained with basic maneuvers



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

COMBINED OPHTHALMOLOGIC EMERGENCIES



ASSESSMENT: (BLS or ALS)

- Obtain vital signs, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history
 - Possible chemical exposure or trauma
 - Loss or change in vision
 - Eye pain
 - Eye redness
- Assess and treat the ABCs and other life-threatening emergencies. Do not allow an ophthalmologic problem distract you from a life-threatening emergency
- Do a brief examination without touching the eyes
- Document the presence of gross abnormalities noting
 - Foreign bodies
 - Pupil irregularity
 - Pupil reactivity
 - Hyphema (blood behind the cornea)
 - Subconjunctival hemorrhage
- If possible, determine gross visual acuity. Ability to read print, count fingers, or determine light from dark are important findings. Determine this for each eye independently

EYE TRAUMA WITH THE POSSIBILITY OF A RUPTURED GLOBE OR ABNORMAL PUPIL

- BLS or ALS
 - Prevent further injury. Use a cup to protect the eye. Tape the cup onto bony prominences about the orbit and do not apply any pressure to the eyeball (globe). Gauze is not needed under the cup
 - If an object remains impaled in the eye, do not remove it. Use gauze and/or a cup to stabilize the object. Do not put pressure on the globe
 - Minimize intraocular pressure by elevating the head of the bed, if possible. Try to have the patient avoid sneezing, straining, or blowing nose
 - Transport while monitoring vital signs

CHEMICAL EXPOSURES

- BLS or ALS
 - CONSIDER need for HAZMAT/Decon
 - The first priority in all cases is immediate, thorough, and continuous irrigation with water, NS, or LR until emergency department personnel assume care. Tap water, bottle water, IV fluids, etc can provide this
 - Attempt to identify the offending substance causing the injury and take in any containers
 - Transport while continuously irrigating the effected eye(s)

**COMBINED SPINAL TRAUMA**ASSESSMENT: (BLS or ALS)

- Obtain vitals, consult PEDIATRIC VITAL SIGN RANGES in appendix for age-appropriate vitals
- Obtain brief history
- INDICATIONS FOR C-COLLAR PLACEMENT
 - GCS < 15
 - Evidence of intoxication
 - Painful or distracting injury
 - Pain during unassisted range of motion
 - Signs of Spinal Cord Injury:
 - Sensory loss
 - Weakness
 - Paralysis
 - Numbness, tingling, burning
 - Severe pain

INTERVENTIONS:

- BLS or ALS
 - CONSIDER transport to a trauma center
 - Indications for c-collar placement present
 - REFER TO SPINAL MOTION RESTRICTION PROCEDURE
 - Secure airway, administer oxygen, or assist ventilations as needed to maintain SpO₂ > 93%
- ALS ONLY
 - **CONSIDER ESTABLISHING IV ACCESS**
- Transport while monitoring vital signs

NOTES:

- Exercise caution for injury and consider c-collar if patient is:
 - Over age 65
 - Under age 5
 - Or history of
 - Osteoporosis
 - Bone disease
 - Vertebral disease
- Helmet removal
 - Football helmet
 - Preferred left in place
 - If concerns for airway obstruction
 - Remove facemask OR
 - Remove helmet and shoulder pads at same time while maintaining manual stabilization
 - Motorcycle or other type helmet
 - Remove while maintaining manual stabilization
- Spinal shock
 - Can have low blood pressure and bradycardia masking hemorrhagic shock/hypovolemia
- REFER TO HYPOTENSION/SHOCK PROTOCOL IF HYPOTENSION PRESENT



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

COMBINED TRAUMATIC CARDIAC ARREST



ASSESSMENT: (BLS or ALS)

- Assess for C-A-B (Circulation, Airway, Breathing)
 - If obvious signs of death or presence of non-survivable injuries
 - REFER TO DOA PROTOCOL
- Event leading up to arrest, mechanism of injury
- Estimated down time

INTERVENTIONS:

- BLS or ALS
 - DO NOT delay transport to perform interventions
 - Keep scene time 10 minutes or less
 - Initiate manual continuous chest compressions (CCC)
 - Push hard, push fast (100-110/min)
 - Change compressors every 2 minutes
 - Initiate BVM ventilations
 - **ADULTS:** 6-10 breaths per minute
 - **PEDS:** 10-12 breaths per minutes
 - BLS on scene awaiting ALS arrival:
 - Attach AED and follow prompts
 - ALS on SCENE
 - Attach cardiac monitor/defibrillator
 - Establish iGel (BLS or ALS) or **ET TUBE** (ALS ONLY) and continue ventilations at 6/min
 - **ATTACH CAPNOGRAPHY AND MONITOR ETCO2** (ALS ONLY)
 - ETCO2 < 20mmHg could be an indication of inadequate chest compressions
 - Pulse checks should take no more than 5-10 seconds
- ALS ONLY
 - **ESTABLISH IV/IO ACCESS**
 - **CONSIDER NS/LR**
 - **ADULTS:** 1000mL REPEAT X1 MAX 2L TO MAINTAIN SBP > 90mmHg
 - **PEDS:** 20mL/kg MAX 1L TO MAINTAIN AGE APPROPRIATE SBP



12-LEAD ECG MONITORING

INTRODUCTION: The principal purpose of obtaining a 12-lead ECG in the prehospital setting is to identify patients having an acute ST-elevation myocardial infarction (STEMI) and to relay this information to the receiving hospitals allowing them to appropriately prepare for patient arrival. This has been shown to decrease “door to drug” and “door to invasive intervention” time, both of which improve patient survival.

INDICATIONS:

- I. A prehospital 12-lead ECG should specifically be considered during the assessment of patients being treated as per the following protocols:
 - A. Suspected myocardial infarction:
 1. Chest pain
 - B. Respiratory Distress (possible congestive heart failure/pulmonary edema)
 1. Asthma
 2. COPD
 3. CHF
 4. Pulmonary Edema
 - C. Dysrhythmia diagnosis:
 1. Cardiac Dysrhythmia (may be particularly helpful in pediatric patients)
 - D. Stroke
 - E. Return of Spontaneous Circulation (ROSC)
 - F. Altered Mental Status
 - G. Hypotension/Shock
 - H. Syncope

CONTRAINDICATIONS: Do not delay the treatment and/or transport of unstable patients to obtain a 12-lead EKG.

PRECAUTIONS:

1. Do not prolong scene time to obtain 12-lead ECG (should take less than 5 minutes)
2. A normal (or nonspecific) 12-lead ECG does not rule out myocardial infarction or ischemia, therefore do not base prehospital treatment on the results of the ECG
3. If a patient on whom you have obtained a 12-lead ECG is refusing transport you should strongly consider contacting medical control for advice

TECHNIQUE:

1. The patient should be in the supine position. If the patient cannot tolerate that position, place them in the semi-reclining or sitting position.
2. Prep the skin and shave hair as needed.
3. Apply electrodes as described
 - A. Do not use nipples as reference points as locations vary widely.
 - B. If the patient has pendulous breasts, place electrodes under breasts.

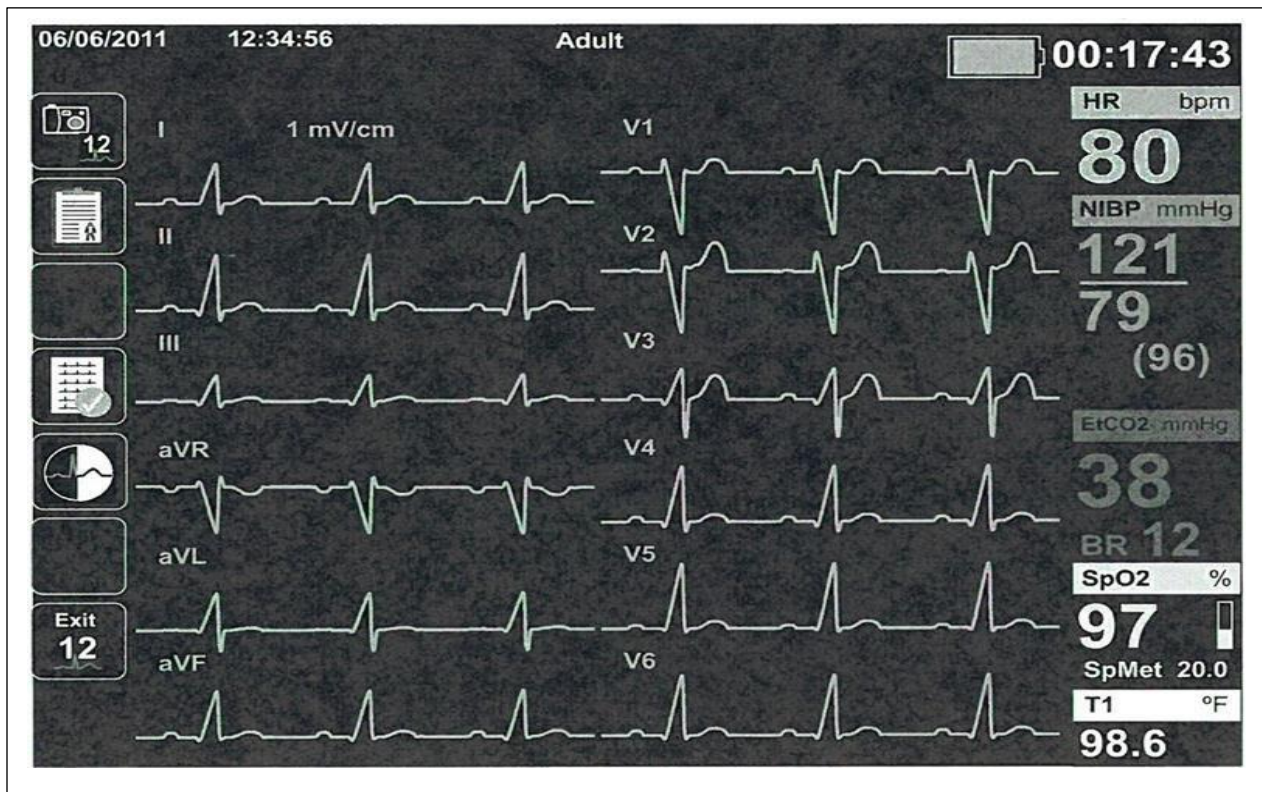
Limb Leads:

Right arm (RA) – Right wrist
Right leg (RL) – Right ankle
Left arm (LA) – Left wrist
Left leg (LL) – Left ankle

Precordial (Chest) Leads:

V1 – 4th intercostal space to the right of the sternum
V2 – 4th intercostal space to the left of the sternum
V3 – Directly between leads V 2 and V 4
V4 – 5th intercostal space at midclavicular line
V5 – Level with V4 at left anterior axillary line
V6 – Level with V5 at left midaxillary line

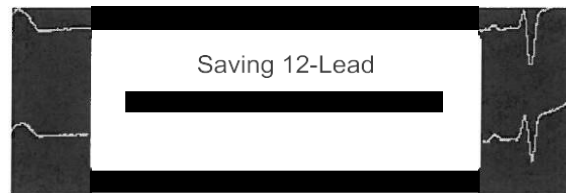
4. Attach 12-lead cable to electrodes and machine – avoid patient, cable or vehicle movement and 60 cycle interference.
5. Enter appropriate patient information.
 - A. Age and Sex (both necessary for accurate 12SLTM interpretation).
 - B. First 3 letters of patients last name (for identification purposes).
6. Obtain 12-lead EKG.
 - A. 12-Lead Acquisition – the X-Series unit begins pre-acquisition of 12- lead data when you attach the electrodes to the patient, as follows:
 - i. Attach electrodes to the patient lead wires.
 - ii. Attach lead wires and electrodes to the patient.
 - iii. Attach the V-lead cable to the 12-lead ECG cable. (when v- leads are not in use, ensure the v-leads protective cap is plugged into the v-lead connector)
 - iv. Attach the 12-lead cable to the ECG connector on the left side of the X-Series product. Arrange the 12-lead cable such that it is neat and not dangling or looped, and assure that it is not pulling on individual electrodes.
 - v. To observe the 12-Lead waveform traces, press **12** . The screen displays all twelve waveform traces, with the size displayed above the waveform traces.



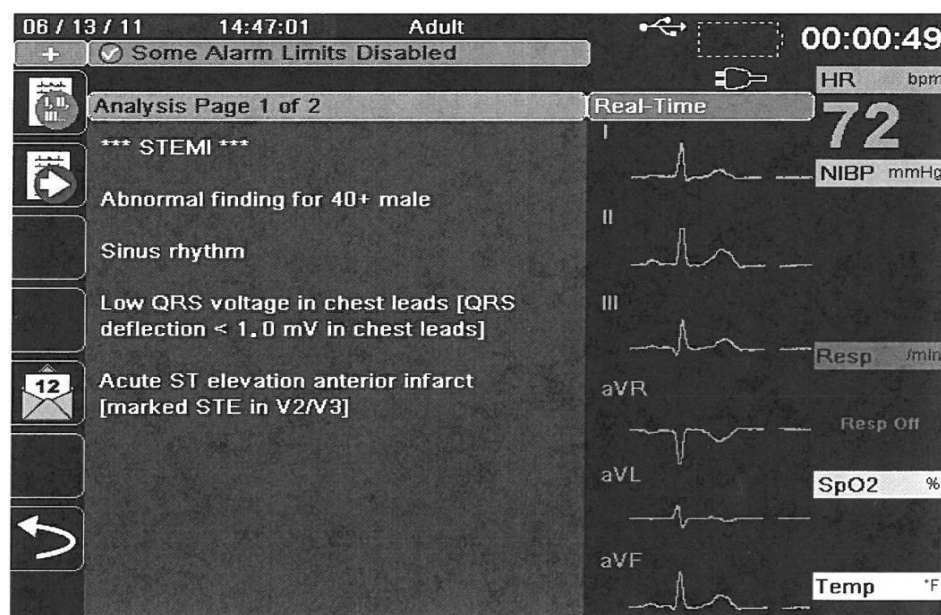
- vi. After observing the patient's ECG and determining that all 12-lead traces display correctly, you can initiate 12- Lead Interpretive Analysis.
 - 1. *Note: 12-Lead Interpretive Analysis works for only Adult patients.
- vii. To begin 12-Lead Interpretive Analysis, press the Acquire quick access key . The X Series unit displays the Acquiring 12-Lead Status bar as it collects 10 seconds of 12-Lead ECG data:



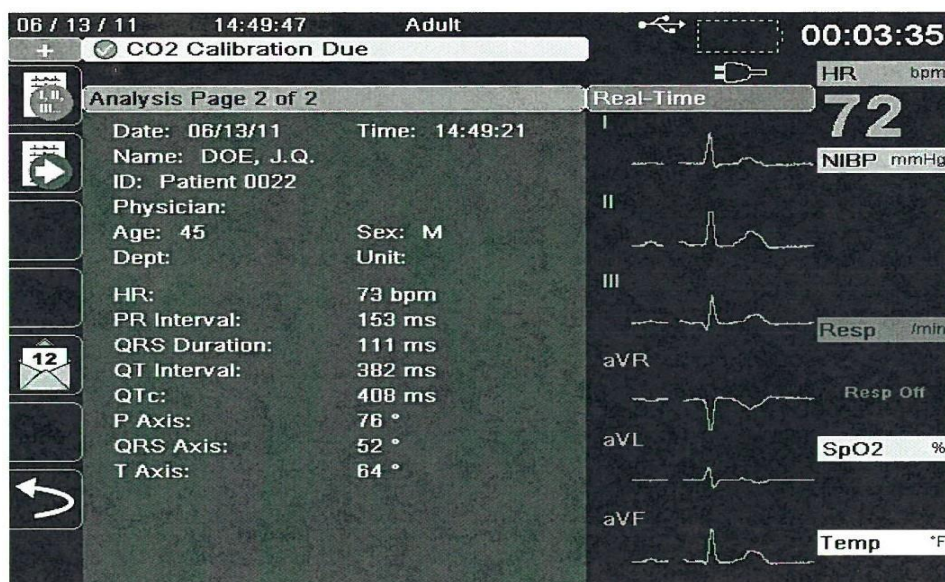
- viii. After acquiring the ECG data, the unit saves the data and displays the Saving 12-Lead Status Bar in the following manner:



- ix. After saving the data, the unit performs the post-acquisition Interpretive Analysis and displays the first page of 12-Lead Interpretive Analysis information. In the example below, the interpretive statement, *****STEMI*****, indicates the occurrence of ST-Elevation Myocardial Infarction. The interpretive statements that the X Series unit displays are produced by the Audicor software of Inovise Medical.



1. *Analysis Page 2 displays the identifying information that has been entered for the patient and additional analysis statements:



7. If the 12-lead ECG shows evidence of an acute myocardial infarction, then contact the destination emergency department as early in the call as possible to allow them the time to prepare appropriately for the patient. Review 12SL analysis for confirmation.
 - A. "Evidence of myocardial infarction" is defined as:
 - i. ST segment elevation of ≥ 1 mm in two contiguous leads
 - ii. The 12SLTM interpretation reports "acute" or "possible acute" myocardial infarction.
8. Transmission of 12-lead ECG (when time permits)
 - A. After acquisition, press envelope button on the left side of the display
 - B. Select desired destinations from list
 - C. Transmit ECG data

COMPLICATIONS: Delay in treatment or transport to obtain a 12-lead ECG may result in patient deterioration

DOCUMENTATION REQUIREMENTS:

1. Print 1 copy of the 12-lead ECG for the receiving facility.
2. Upload ECG from Zoll to ePCR. If unsuccessful, print a copy for submission with the run ticket.
3. Run ticket documentation should include the 12SLTM interpretation as well as the paramedic interpretation. Paramedic interpretation should include the heart rate and rhythm as well as the presence or absence of evidence of myocardial infarction

NOTES:

Reference: Zoll X-Series 12-Lead Monitoring insert, February 2012



Chest Seal



CHEST SEAL (HYFIN VENT) APPLICATION – BLS/ALS

INDICATIONS

- Penetrating chest trauma
- Open pneumothorax (“sucking chest wound”)
- Frothing or bubbling at opening of wound

PROCEDURE

1. Apply personal protective equipment: gloves.
2. Remove clothing.
3. Open package using external red tabs.
4. Wipe any dirt or fluid from skin with gauze.
5. Grip red tab and peel the transparent backing from the chest seal.
6. Center vent over the wound.
7. Firmly press onto the skin to ensure a good seal.
8. Smooth out all edges flat against the skin.
9. Assess front and back of patient carefully for additional wounds and apply second seal as indicated.
10. Monitor patient for development of tension pneumothorax if blood accumulates in all three vented channels of the chest seal.





PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

COMBAT APPLICATION TOURNIQUET



INDICATIONS: Extremity wounds where direct pressure with a sterile dressing alone does not control active bleeding

CONTRAINDICATIONS: Do not place around the neck

PRECAUTIONS: Do not place directly over a wound

TECHNIQUE:

1. Expose the wound and place the wounded extremity through the loop of the Omni-Tape band. The CAT will normally be 2-4 inches above the injury.
2. Pull the free end of the Omni-Tape band tight and secure it back onto itself using the Velcro.
3. Do not secure the band past the windlass clip.
4. Twist the windlass rod until bright red bleeding stops
5. Secure the rod by inserting it into the windlass clip.
6. Small extremities will allow you to extend the Omni-Tape band over the top of the windlass rod and clip.
7. Pull the Velcro windlass strap tight over the windlass clip and Omni-Tape band.
8. Mark the patient's head with the letter "T" and annotate the date/time/location of the CAT on the field medical Card. Patient is now ready for transport unless further medical assessment is required.
9. Never remove or cover a tourniquet and only use on a leg or arm.

COMPLICATIONS: N/A

DOCUMENTATION REQUIREMENTS:

1. Indications for placement
2. Time of tourniquet placement
3. Reassessment after placement



CONFIRMATION OF ADVANCED AIRWAY PLACEMENT

INDICATIONS: All endotracheal intubations and iGel Supraglottic Airway placements must have objective confirmation of correct placement.

CONTRAINDICATIONS: N/A

PRECAUTIONS: No one clinical sign is an absolute indicator of proper endotracheal tube or iGel placement.

TECHNIQUE:

1. Placement Confirmation Using Continuous Waveform ETCO₂: See Zoll X-Series Continuous Waveform ETCO₂ Monitoring Procedure
2. Clinical Signs of Placement:
 - a. Direct visualization
 - b. Equal breath sounds (high in mid-axillary line); if breath sounds are decreased on left, consider right mainstem intubation, pull endotracheal tube back and recheck breath sounds
 - c. No gastric sounds with ventilation
 - d. Cuff inflation of endotracheal tube palpated in supra-sternal notch
 - e. Chest wall movement
 - f. "Fogging" of endotracheal tube or iGel
 - g. Airflow felt through endotracheal tube or iGel
 - h. Compliance of bag with ventilation
 - i. Observation of pink membranes and "non-deterioration"
 - j. Gastric contents in endotracheal tube imply in esophagus
 - k. Vocalization implies endotracheal tube in esophagus
 - l. Palpation of endotracheal tube passing the cords via Sellick's Maneuver

COMPLICATIONS: N/A

DOCUMENTATION REQUIREMENTS:

1. Time of intubation or iGel placement
2. Size of endotracheal tube or iGel
3. Presence and quality of breath sounds
4. Absence of air movement in the epigastric area
5. Application of Continuous Waveform ETCO₂
6. Method used to secure ET tube or iGel
7. Confirmation of placement after moving patient
8. Record ETCO₂ readings in the appropriate section of the ePCR
9. Any complications

NOTES:

1. Head movement has been shown to dislodge correctly placed and secured endotracheal tubes. Secure the head with a c-collar and head block device to restrict movement.
2. Endotracheal tube and iGel placement should be confirmed after each patient movement.
3. When removing the patient from the unit, disconnect the BVM from the endotracheal tube or iGel until the patient is out and the wheels are lowered.
4. None of the clinical signs are absolute; there are documented cases of unrecognized esophageal intubation using each of these techniques.
5. "When in doubt, take it out" and assure oxygenation and ventilation.
6. If an initial attempt at intubation results in esophageal intubation, the endotracheal tube may be left in the esophagus to provide a landmark for another attempt if this does not impede BVM ventilation.

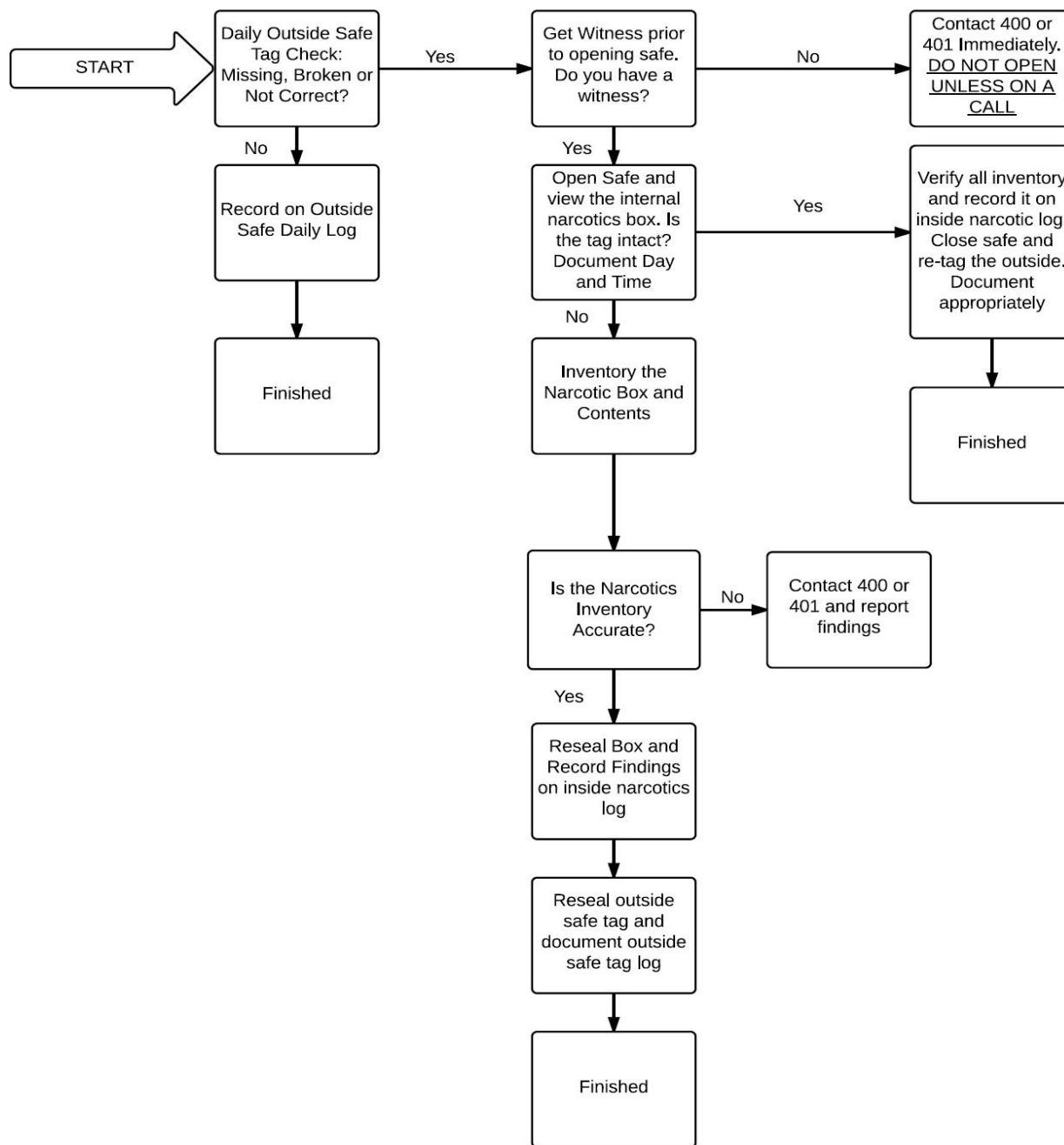


PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

CONTROLLED SUBSTANCE VERIFICATION



Pleasant Valley Fire Department Daily Controlled Substance Verification Procedure





CPAP



INDICATIONS:

CPAP is indicated in all patients in whom inadequate ventilation is suspected. This could be as a result of pulmonary edema, pneumonia, asthma, COPD, etc. The patient must have an adequate mental status and a spontaneous respiratory drive in order to permit the CPAP device to function properly.

CONTRAINDICATIONS:

1. Need for immediate intubation
2. Decreased or Altered Mental Status.
3. Facial features or deformities that prevent an adequate mask seal.
4. Vomiting or excessive respiratory secretions.
5. Inability to maintain own airway.
6. SBP < 90mmHg

PRECAUTIONS:

1. Requires patient cooperation. (The major complication is the inability to tolerate the mask; in which case the mask should be removed and an alternate airway should be used.)
2. Emesis may result in aspiration
3. Adequate supply of oxygen is required.

TECHNIQUE:

1. Ensure adequate oxygen supply to CPAP device. Limit the time spent on portable oxygen.
2. Explain the procedure to the patient.
3. Place the delivery mask over the mouth and nose. To improve compliance and tolerance, manually hold the mask for 3-5 minutes while continuously reassuring the patient. Oxygen should be flowing through the device at this point.
4. Secure the mask with provided straps starting with the lower straps until minimal air leak occurs.
5. Encourage the patient to allow forced ventilation to occur. Observe closely for signs of complications.
6. Evaluate the response of the patient, assessing general appearance, breath sounds, and oxygen saturation.
7. Frequently reassess the patient, being vigilant for worsening respiratory failure or arrest.
8. Add nebulizer as needed.

DOCUMENTATION:

1. Document time, vital signs, FIO2 and clinical response on EPCR.
2. When adding medications, document medication, time, vital signs, and clinical response on epcr.



END TIDAL CO₂ MONITORING

INTRODUCTION:

1. CO₂ is a product of cellular metabolism that is removed from the body by ventilation, thus end-tidal air has a high pCO₂. Inspired air has essentially no CO₂. The CO₂ monitor is a device which measures the pCO₂ of expired air and displays a numerical value of measured end-tidal CO₂. The unit can display a capnogram that demonstrates the change in pCO₂ during inspiration and expiration by a waveform on a monitor and will calculate and display respiratory rate based on the measured time interval between detection peaks of the CO₂ waveform.
2. In the cardiac arrest patient, CO₂ may not be carried to the lungs because of poor perfusion. Therefore, the expired pCO₂ may be very low.

INDICATIONS:

1. The Zoll X-Series EtCO₂ option is indicated for the continuous noninvasive monitoring of end tidal carbon dioxide (EtCO₂) and respiration rate in intubated patients.
2. The X-Series EtCO₂ option supports two methods for continuous measurement of end tidal carbon dioxide and respiration rate. The method used will be the Oridion Microstream Filterline CO₂ Sensor attached to an airway adapter that connects to an endotracheal tube, mask or disposable mouthpiece.
3. Any patient that has had an iGel (or other) supraglottic airway placed.
4. Any patient that has had an endotracheal tube placed.
5. Any patient that is receiving BVM ventilations via tracheostomy tube.

CONTRAINDICATION:

1. Mouth to adjunct ventilations
2. Patients less than 15 kg (30 lbs) in weight

PRECAUTIONS:

1. The Zoll X-Series is able to monitor ETCO₂ for adult, pediatric, and neonatal patients.
2. The CO₂ monitor is an adjunct used to help confirm correct advanced airway placement and continuously monitor placement. It does not replace clinical evaluation.
3. Always ensure the integrity of the patient breathing circuit after insertion of the airway adapter by verifying proper ventilation by auscultation and by verifying a proper CO₂ waveform (capnogram) on the monitor display.
4. Failure to recognize deterioration in patient status with changes in ETCO₂ levels or loss of capnogram.

TECHNIQUE:

- I. Initial Verification of Endotracheal Tube Placement:
 - a. Turn on the monitor and ensure that the CAPNOSTAT Sensor is plugged into the monitor.
 - b. Attach CAPNOSTAT Sensor to the adapter and ensure adaptor is placed between endotracheal tube and BVM or ventilator.
 - c. Once setup is complete, press CO₂ soft button on the left side of the unit. The numeric CO₂ display appears on the screen and displays the message "initializing". The CO₂ display gives the current EtCO₂ value and after a delay of approximately 1 minute, the patients respiratory rate (in bpm), identified as BR. Check to see if displayed properly by displaying of wave form.
 - d. Continually monitor indicator and waveform which changes with ventilation, throughout transport.
1. Patients who do not require intubation but do require the use of nebulizer, the detector may be connected to the nebulizer and the same procedure followed to monitor EtCO₂ and to observe waveform capnography.
2. Continuing Verification of Endotracheal Tube Placement:
 - a. If the EtCO₂ monitor changes to very low or no indicated EtCO₂ and/or loss of capnogram, then either:

- i. The endotracheal tube has become dislodged from the trachea.
OR
 - ii. The patient has deteriorated and now has poor pulmonary perfusion.
OR
 - iii. There is a problem with the sampling circuit of the device.
3. Ensure that the Oridion CO2 detector is correctly plugged into the Zoll monitor and that the cable is intact, that the detection device is placed properly and attached to the airway adapter, and the airway adapter is properly placed in the airway circuit and CO2 button has been pressed.

DOCUMENTATION REQUIREMENTS:

- 1. Time of intubation or iGel placement
- 2. Size of endotracheal tube or iGel
- 3. Presence and quality of breath sounds
- 4. Absence of air movement in the epigastric area
- 5. Application of Continuous Waveform ETCO2
- 6. Method used to secure ET tube or iGel
- 7. Confirmation of placement after moving patient
- 8. Record ETCO2 readings in the appropriate section of the ePCR
- 9. Any complications

NOTES:

- 1. Not effective for detecting right mainstem or pharyngeal intubation
- 2. If there is no ETCO2 level detected and no capnogram, then recheck equipment and/or use other means to verify placement.



iGEL



INDICATIONS:

1. An unconscious patient without a gag reflex who is apneic or is demonstrating inadequate respiratory effort.
2. Inability to adequately ventilate a patient with a Bag Valve Mask with or without airway adjuncts thus requiring a more advanced airway

CONTRAINDICATIONS:

1. Trismus, limited mouth opening, pharyngo-perilaryngeal abscess, trauma or mass

PRECAUTIONS:

1. Do not use excessive force to insert the device

TECHNIQUE:

1. Prepare and position the patient. Head extended into sniffing position unless concern for C-Spine injury.
2. Select appropriate i-gel airway. Have suction ready.
3. Apply a generous amount of water-based lubricant.
4. Visualize oropharynx and remove any gross debris.
5. Insert i-gel following the anatomy using steady pressure until a definitive resistance is felt.
6. Ensure the i-gel is fully seated, locating the solid black line at approximately tooth level.
7. Secure the iGel.
8. Apply Waveform Capnography (EtCO₂).
9. Auscultate for bilaterally equal breath sounds and absence of sounds over the epigastrium. If you are unsure of placement, remove i-gel and ventilate patient with bag-valve mask.

COMPLICATIONS:

1. Potential for bleeding in the airway after prolonged use while chest compressions are provided

DOCUMENTATION:

1. Record Waveform Capnography (EtCO₂) readings into the ePCR
2. Document i-gel size, time, and attempts/success in the ePCR.
3. Document positive or negative breath sounds before and after each movement of the patient.



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

INTRANASAL DRUG ADMINISTRATION



INDICATIONS:

1. Emergent need for medication administration
2. No IV has been established
3. Seizures (Midazolam)
4. Altered mental status suspected narcotic overdose (Naloxone)
5. Analgesic medication administration (Fentanyl)
6. Chemical restraint for combative patient (Midazolam)

CONTRAINDICATIONS:

1. Excessive nasal discharge
2. Bleeding from nares
3. Mucosal destruction
4. Current patient use of nasal vasoconstrictors (neosynephrine)

PRECAUTIONS: Total volume delivered should not exceed 0.5mL per nare per administration

TECHNIQUE:

1. Draw up proper dosage (see appropriate protocol)
2. Expel air from syringe
3. Attach the MAD device via luer lock
4. If volume of dose to be delivered exceeds 0.5mL, divide volume equally and administer into both nares
5. Place MAD device just inside nare and briskly compress the syringe plunger to deliver medication onto nasal mucosa
6. Repeat in other nare if dose volume requires as indicated above

COMPLICATIONS:

1. Gently pushing the plunger will not result in adequate atomization
2. Fluid may escape from nares
3. Not 100% effective

DOCUMENTATION REQUIREMENTS:

1. Dosage used
2. Response to intervention

NOTES: Does not replace standard therapy (IV, O2, and monitoring)



Intraosseous (IO)

Indications:

1. Patients where rapid, regular IV access is unavailable with any of the following:
 - a. Cardiac arrest
 - b. Multisystem trauma with severe hypovolemia and/or a significantly burned patient with no IV access.

Severe dehydration with vascular collapse and/or loss of consciousness.

- c. Respiratory failure / Respiratory arrest.
2. Any other immediately life-threatening, peri-arrest clinical condition in which IV access is unobtainable.

CONTRAINDICATIONS:

1. Suspected Fracture at or proximal to proposed intraosseous insertion site.
2. History of Osteogenesis Imperfecta
3. Current or recent infection at proposed intraosseous insertion site.
4. Previous intraosseous insertion or joint replacement at the selected site.
5. Previous orthopedic procedures (knee placements)

PRECAUTIONS:

1. Transport is required for any patient that receives intraosseous (IO) placement and does not meet Termination of Resuscitation criteria.
 - a. The placement and subsequent removal of IO needles in the pre-hospital environment is NOT currently supported by literature.
2. Needles are one time use only.
3. Never attempt a second insertion in the same leg.

TECHNIQUE:

1. Don personal protective equipment (gloves, eye protection, etc.).
2. Identify appropriate insertion site.
 - a. Adults:
 - i. Proximal Humerus: On an adducted arm with a flexed elbow and an internally rotated humerus, approximately 1 to 2 cm above the surgical neck, on the most prominent aspect of the greater tubercle.
 - ii. Proximal Tibia: On an extended leg, approximately 2 cm medial to the tibial tuberosity, or approximately 3 cm (two finger widths) below the patella and approximately 2 cm medial, along the flat aspect of the tibia.
 - b. Pediatrics:
 - i. Proximal Tibia: On an extended leg, approximately 1 cm medial to the tibial tuberosity, or just below the patella (approximately 1 cm or one finger width) and slightly medial (approximately 1 cm or one finger width), along the flat aspect of the tibia.
3. Prep the selected site with a providone-iodine or chlorhexidine solution.
4. Appropriate needle angle:
 - a. Proximal Humerus: Aim the needle tip at a 45-degree angle to the anterior plane and posteromedial.
 - b. Proximal Tibia (Adult) and Proximal Tibia (Pediatric): Aim the needle tip at a 90-degree angle to the center of the bone.
5. Push needle tip through the skin until tip rests firmly against the bone. The 5 mm mark from the hub must be visible above the skin for confirmation of
6. adequate needle set length.
7. Power the driver until a “pop” or “give” is felt indicating loss of resistance. Do not advance the needle any further! [Note: Avoid recoil, do NOT pull back on the driver when releasing the trigger.]
9. Remove the stylet and place in an approved sharps container.
10. Stabilize and secure the needle with the dressing included in the packaging.
11. Verify needle placement.

- a. Needle feels secure.
 - b. Attach a syringe filled with at least 5 ml NS
 - c. Aspirate bone marrow
12. Inject at least 5-10 ml of NS in an adult or 2-5 cc of NS in a pediatric patient to clear the lumen of the needle and open the intramedullary space.
13. Attach the IV line and adjust flow rate. A pressure bag may assist with achieving desired flows.
14. Re-verify placement prior to each infusion and assess frequently for complications, including extravasation which can lead to compartment syndrome. Following the administration of any intraosseous medications, flush the line with IV fluid.



LUCAS DEVICE



INDICATIONS:

1. Patient in Cardiac Arrest
 - a. DO NOT DELAY INITIATION OF MANUAL CHEST COMPRESSIONS
 - b. Deploy LUCAS device when available and man power is sufficient
 - c. Keep any pauses in manual chest compressions to a minimum

CONTRAINDICATIONS:

1. It is not possible to position the LUCAS device safely or correctly on the patient's chest
2. Too small patient: if the LUCAS device alerts with 3 fast signals when lowering the Suction Cup and you cannot enter the PAUSE mode or ACTIVE mode.
3. Too large patient: If you cannot lock the Upper Part of the LUCAS device to the Back Plate without compressing the patient's chest.

PRECAUTIONS:

1. Only place device when sufficient manpower is present to maintain manual chest compressions while device is being set up and placed
2. If any issues arise with device, IMMEDIATELY return to manual chest compressions

TECHNIQUE:

1. Power on the LUCAS device
2. Place the BACK PLATE
 - a. Assure minimal interruptions to manual chest compressions
 - b. Place under the patient immediately below the armpits
3. Attach the UPPER PART
4. Push down SUCTION CUP
 - a. Lower edge of suction cup should be immediately above the end of the sternum
 - b. Adjust position if needed
5. Lock position and start compressions
6. Attach stabilization strap
 - a. Failure to do so may result in patient injury due to movement of the device from its operating position
7. Follow appropriate cardiac arrest protocols.

COMPLICATIONS:

1. Possible rib fractures



NEEDLE THORACOSTOMY

INDICATIONS:

1. Tension pneumothorax is a clinical diagnosis which should be considered when the following signs/symptoms are present:
 - a. Progressing severe respiratory distress
 - b. Progressing shock
 - c. Decreased or absent breath sounds on the involved side
 - d. Jugular venous distension
 - e. Tympany to percussion on the involved side
 - f. Tracheal deviation away from the involved side
2. 2 of the above signs/symptoms MUST be present or approval from a Medical Control should be obtained prior to performing the procedure.
3. Tension pneumothorax is most common in the patient with:
 - a. Chest trauma
 - b. The intubated patient with high airway pressures causing rupture of the bronchioles or alveoli

CONTRADICTIONS:

1. Clinical signs suggestive of diagnosis, but patient is not in shock and/or respiratory distress.

PRECAUTIONS: N/A

TECHNIQUE:

1. Locate the second intercostal space in the mid clavicular line on the involved side of the chest.
2. Cleanse with betadine or alcohol as time permits.
3. Insert a large gauge "over the needle" catheter (use 18 gauge if < 2 months age) through the open end of a glove fingertip, puncture, and advance needle to hub of catheter to establish the flutter valve mechanism.
4. Insert the needle/catheter with attached flutter valve into the chest just over the top of the third rib.
5. As you enter the pleural space air and/or blood will escape.
6. Advance the catheter and remove the needle.
7. Secure in place.
8. The catheter has a tendency to kink. If re-accumulation of air in the pleural space is occurring, proceed with repeat needle thoracostomy.

COMPLICATIONS:

1. Creation of pneumothorax.
2. Damage to lung or viscera.
3. Bleeding (intercostal vessels are below each rib, therefore always go above the rib).
4. Infection.

DOCUMENTATION REQUIREMENTS:

1. Indications for procedure
2. Response to intervention



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

ORAL ENDOTRACHEAL INTUBATION



INDICATIONS:

1. An unconscious patient without a gag reflex who is apneic or is demonstrating inadequate respiratory effort.
2. Inability to adequately ventilate a patient with a Bag Valve Mask with or without airway adjuncts thus requiring a more advanced airway.

CONTRAINDICATIONS:

1. Presence of gag reflex

PRECAUTIONS:

1. Oxygenate patient prior to procedure
2. Do not use teeth as a fulcrum

TECHNIQUE:

1. Prepare, position and oxygenate the patient with 100% Oxygen.
2. Select proper laryngoscope, ET tube and stylette, if used. Have suction, Waveform Capnography (EtCO₂), and back-up Endotracheal Tube Introducer (Gum Elastic Bougie) ready.
3. Using laryngoscope, visualize vocal cords. (Use Sellick maneuver/BURP to assist you if needed).
4. Limit each intubation attempt to no more than 30 seconds with adequate re-oxygenation utilizing a BVM between attempts.
5. Visualize ET tube passing through vocal cords.
6. Inflate the cuff with 10mL of air.
7. Confirm and document tube placement using Continuous Waveform Capnography (EtCO₂).
8. Auscultate for bilaterally equal breath sounds and absence of sounds over the epigastrium. If you are unsure of placement, remove tube and ventilate patient with bag-valve mask.
9. Secure the tube to the patient.
10. After 3 ventilations, ETCO₂ should be >20 or comparable to pre-intubation values. If < 20, check for adequate circulation, equipment, and ventilatory rate. If ETCO₂ still < 20 without physiologic explanation, remove the ET Tube and ventilate by BVM.

COMPLICATIONS:

1. Hypoxia during procedure
2. Oral trauma from procedure

DOCUMENTATION:

1. Record Waveform Capnography (EtCO₂) readings in the ePCR.
2. Document ETT size, time, attempts/success, and placement location by the centimeter marks either at the patient's teeth or lips in the ePCR.
3. Document positive or negative breath sounds before and after each movement of the patient.



Pediatric Patient Transport



Pedi Mate +



The Pedi Mate + Pediatric Restraint System quickly adapts an ambulance cot for the safe transport of children ranging in size from 10-100 lb (4.5-43.5 kg).

- **Fully-adjustable**, five-point harness system securely holds patients, providing safe restraint for transport
- **Rolls compactly** for convenient storage and easy deployment
- **Vinyl construction** is nontoxic and easy-to-clean
- **Three restraint straps** easily attach to any cot

How to video: <https://www.youtube.com/watch?v=2D9jQLQrxvc>

Notes: ***The number one thing to avoid is allowing a parent or other caregiver to hold a baby or child during transport.***

References: <https://emlrc.org/wp-content/uploads/FL-PEDReady-safe-transport-Educational-Presentation-2023.pdf>

If unable to use padimate, refer to next page for proper use of car seats.



Pediatric Patient Transport

Car Seat on Stretcher



Rear-facing only car seats/infant car seats CANNOT be installed on ambulance cots or in rear-facing captain's chairs

Recommendations for Using Convertible Child Safety Seat, Car Bed, and Securing Child on Cot in Emergency Ground Ambulances



Convertible Child Safety Seat (CSS)

Child Size: 5 to 40 lbs

Installation Recommendations:

- Install with rear-facing and forward-facing belt paths.
- Choose seat with 5-point internal harness.
- Position seat facing rear of ambulance.
- Elevate cot backrest to fully upright position.
- Adjust restraint recline mechanism to fit snugly against cot seat back.





PATIENT RESTRAINT



The purpose of this policy is to set forth principles and procedures for the prehospital patient restraint (PPR) of persons at risk for harm to self or others, including emergency medical services personnel.

The safety of EMS personnel is paramount for the consideration, initiation, and application of PPR. Also, the protection of patients from self-harm or harm to others while maintaining patient dignity to the greatest extent possible while utilizing the least restrictive method of restraint appropriate to the situation.

This policy applies to all patients whom have demonstrated violence toward self or others, or there is reason to believe that combative or violent behavior is imminent, through actions or verbalizations by the patient or reasonable layperson bystanders.

INDICATIONS:

1. All patients who are demonstrating physical harm to self or others including, but not limited to, evidence of suicidal or homicidal behavior, by act or verbalization.
2. All patients demonstrating decreased ability to care for self as evidence by, but not limited to, cognitive impairment or suspected intoxication leading to exposure of self or others to potentially limb or life-threatening situations.
3. Patient with altered mental status leading to decreased capacity to understand situation and circumstances of their current medical condition resulting in aggressive or combative behavior towards emergency personnel.
4. Patients in custody of law enforcement or under involuntary detention.

CONTRAINDICATIONS:

1. Do not restrain the patient in the prone position

PRECAUTIONS:

1. Always assure access to patient's airway should airway management be needed
2. Always assure access to extremities should vascular access be needed

TECHNIQUE:

1. When at all possible, the patient should be approached in a calm, non-threatening and supporting manner. Verbal de-escalation efforts should be attempted unless there is imminent danger to patient or bystanders. If the EMS personnel believe that their personal safety is at risk, law enforcement should be contacted for assistance.
2. When imminent danger to the patient or others is identified, a unified, multiple emergency personnel approach should be made to gain control of the patient. Placement of physical restraints, including approved hood with spit shield, approved soft wrist and ankle restraints should be considered first. Progressive restraint beginning with wrists, then ankles, to immobilization on long spine board should be determined to provide the least amount of restraint necessary to accomplish protection of crew and patient.
3. Hard extremity restraints including handcuffs and shackles should be used only when other methods fail or unusable due to circumstances and efforts to replace with approved restraints made.
4. If handcuffs or shackles are used, they should only be placed by appropriate law enforcement personnel and when law enforcement will travel with the patient or in immediate vicinity of the ambulance. EMS personnel should always have available, the means to immediately remove restraints.
5. Patients under arrest shall be thoroughly searched by law enforcement prior to being placed into any ambulance. Law enforcement agencies retain the responsibility for safe transport of patients under arrest or involuntary detention.
6. Patients should be restrained in a supine position and immobilized on a long spine board whenever possible. It is never appropriate to "sandwich" patient between boards, or to "hog-tie" in any fashion a patient for transport by EMS. Any restraint placed should not compromise normal respiratory efforts or circulation.

7. Vital signs and patient monitoring should be accomplished as soon as possible after placement of appropriate restraints. Potential causes of the behavior, including hypoxemia and hypoglycemia, should be considered and efforts to identify and manage as soon as possible.
8. **ONCE PHYSICAL RESTRAINTS ARE IN PLACE, CHEMICAL RESTRAINT SHOULD BE STRONGLY CONSIDERED AND ADMINISTERED IF THE PATIENT CONTINUES TO EXHIBIT BEHAVIOR IN RESTRAINTS THAT COULD HARM SELF OR OTHERS. CHEMICAL RESTRAINT SHOULD ONLY BE UTILIZED WHEN THE PATIENT CAN BE ADEQUATELY AND CONTINUOUSLY MONITORED BY THE PARAMEDIC. REFER TO COMBINED BEHAVIORAL/PSYCHIATRIC DISORDER PROTOCOL**
9. Once restrained, the patient should be isolated and placed into the ambulance as soon as possible and transported to the closest appropriate facility as per the Determination of Hospital Destination Policy.
10. Physical restraints should be reassessed at a minimum of every ten minutes during transport for the first hour, and every thirty minutes thereafter. The restrained extremities should be assessed for neurovascular function and documented.
11. Always considered consultation with on-line medical control for questions or assistance in determining the most appropriate restraint method for any situation. Patients with medical conditions that appear to compromise their ability to consent for or decline care may be restrained as necessary and transported without law enforcement authority in situations in which a life-threatening emergency exists or potentially exists. Medical control should be contacted in all instances where patients are transported against their consent unless under arrest or duly exercised involuntary detention.

COMPLICATIONS:

1. Injury to responders or patient during procedure

DOCUMENTATION REQUIREMENTS:

1. Detailed description of scene, situation, and patient behaviors
2. Note assistance by other emergency providers
3. Complete vital signs including pulse oximetry initially and at hospital arrival at a minimum
4. EKG rhythm analysis
5. Detailed description of patient injuries and any bystander injuries
6. Description of restraint method(s) and re-examination(s)



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

EPINEPHRINE 1:100000 (PUSH DOSE)



INDICATIONS: For use in the management of bradycardia, hypotension/shock, and post-resuscitation care

CONTRAINDICATIONS: Known hypersensitivity

PRECAUTIONS: In patients taking digitalis, epinephrine may exacerbate ventricular ectopy

TECHNIQUE:

1. Take a 10mL saline flush and eject 1mL of normal saline
2. Into this syringe draw up 1mL of Epinephrine from the 1:10000 preload.
3. This now gives you 100mcg Epinephrine 1:100000 in 10mL syringe, 10mcg/mL

COMPLICATIONS: N/A

DOCUMENTATION REQUIREMENTS:

1. Time medication given
2. Paramedic giving medication
3. Dose given
4. Route given



SEDATION



INDICATIONS:

1. For patients about to undergo a painful procedure (cardioversion)

CONTRAINDICATIONS:

1. Known sensitivity or allergy to benzodiazepines
2. Shock

PRECAUTIONS:

1. ETCO2 should be monitored any time benzodiazepines are given

TECHNIQUE

1. Administer oxygen as appropriate
2. Establish IV access (ALS ONLY)
3. Place patient on pulse oximeter and obtain baseline saturation
4. Place patient on cardiac monitor (ALS ONLY)
5. **CONSIDER VERSED** (ALS ONLY)
 - a. **ADULTS ONLY: 2mg SLOW IV or IN. MAY REPEAT EVERY 5 MINUTES TO TOTAL MAX 5mg GIVEN**
6. Monitor ETCO2

COMPLICATIONS:

1. Respiratory depression
2. Hypotension



SPINAL MOTION RESTRICTION

BACKGROUND: Spinal Motion Restriction utilizes a cervical collar, the cot mattress, and adequately and appropriately securing the patient in order to limit the excessive motion of the spinal column during patient transport. The goal is to minimize secondary injury to spine in patients who have, or may have, an unstable spinal injury, and most importantly, to minimize patient morbidity from the use of immobilization devices.

INDICATIONS:

Spinal Motion Restriction is indicated in trauma patients who sustain a mechanism of injury sufficient to cause a neck or back injury.

1. GCS < 15
2. Evidence of intoxication
3. Painful or distracting injury
4. Pain during unassisted range of motion
5. Signs of spinal cord injury such as:
 - a. Sensory loss
 - b. Weakness
 - c. Paralysis
 - d. Numbness, tingling, or burning
 - e. Pain, deformity, or tenderness to the neck or back
6. Exercise caution for injury and consider c-collar placement:
 - a. Patients over 60 years old
 - b. Patients under 5 years old
 - c. Any patient with a history of osteoporosis, bone disease, or vertebral disease

CONTRAINDICATIONS:

1. Some patients, due to size or age, will not be able to be managed through in-line stabilization and standard cervical collars. Never force a patient from a non-neutral position. Manual stabilization or improvised padding may be required during transport
2. Patients with penetrating injury to the neck should not be placed in a cervical collar or other spinal precautions regardless of whether they are exhibiting neurologic symptoms or not. Doing so can lead to delayed identification of injury or airway compromise, and has been associated with increased mortality

PRECAUTIONS:

1. Assess the scene to determine the mechanism of injury
2. Patients may be safely immobilized by being placed supine on gurney with cervical collar and straps and will not generally require a spine board
3. Reserve long spine board use for the movement of patients whose injuries limit ambulation. Remove from the long spine board as soon as is practical
4. If a patient absolutely needs transport on a spinal board, long boards should be padded
5. These patients (except in cardiac arrest) should NOT be transported to the hospital while still secured to a long spine board. Dangers of transporting a patient on a long spine board include but are not limited to:
 - a. Potential airway compromise or aspiration in immobilized patient with nausea/vomiting, or with facial/oral bleeding
 - b. Excessively tight immobilization straps can limit chest excursion and cause hypoventilation
 - c. Prolonged immobilization on spine board can lead to ischemic pressure injuries to skin
 - d. Prolonged immobilization on spine board can be very uncomfortable for patient
6. Children are abdominal breathers, so immobilization straps should go across chest and pelvis and not across the abdomen, when possible

7. Children have disproportionately larger heads. When securing pediatric patients to a spine board, the board should have a recess for the head, or the body should be elevated approximately 1-2 cm to accommodate the larger head size and avoid neck flexion when immobilized
8. In an uncooperative patient, avoid interventions that may promote increased spinal movement

TECHNIQUE:

1. Gather appropriate devices
2. Assess neurological exam and distal pulses pre-procedure
3. While maintaining in-line stabilization in a neutral/middle position, place the patient in an appropriately sized cervical collar (stabilization should not involve traction or tension)
4. If patient is clinically stable, alert, and without neurological deficits:
 - a. If currently ambulating, position patient onto the stretcher and place supine
 - b. If patient is sitting in vehicle or chair and feels able to, they may self-extricate to the stretcher
 - c. If patient is sitting in vehicle or chair and feels unable to self-extricate, use a long spine board to extricate from vehicle or chair to cot, removing the patient from the long spine board via log-roll after placement on the stretcher.
 - d. If patient is lying on the ground or other surface, use a long spine board or scoop stretcher to move the patient to the stretcher, removing the patient from the long spine board via log-roll after placement on the stretcher
5. Ensure that the patient is adequately secured to the stretcher, ensuring minimal movement
6. Preferred position for transport is supine
7. Manual in-line stabilization must be used during any transfer
8. Assess neurological exam and distal pulses post-procedure
9. All stretcher-to-stretcher transfers should be accomplished using manual in-line stabilization.

DOCUMENTATION:

1. History and physical exam findings indicating the use of SMR
2. Patient complaint of neck or spine pain
3. Spinal tenderness
4. Mental status/GCS
5. Neurologic examination
6. Evidence of intoxication
7. Documentation of multiple trauma
8. Documentation of mechanism of injury
9. Document patient capacity with: Any and all barriers to patient care
10. Patient age
11. Patients under age and not emancipated: Guardian name, contact, and relationship
12. How the patient was moved to the ambulance cot
13. Neurologic exam before and after movement and/or intervention
14. Removal from spine board

NOTES:

From NASEMSO Model Clinical Guidelines



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

TRANSCUTANEOUS CARDIAC PACING



INDICATIONS:

1. Signs of poor perfusion in the presence of bradycardia
 - a. Altered mental status
 - b. Chest Pain
 - c. CHF
 - d. Shortness of Air
 - e. Syncope
 - f. Hypotension/Shock
2. Bradycardic/asystolic cardiac arrest
 - a. The outcome of prolonged bradycardic/asystolic cardiac arrest is dismal even with pacing. Indiscriminate pacing of this rhythm is unwarranted.
 - b. Pacing of bradycardic/asystole of short duration, especially post countershock bradycardic/asystole is more likely to be useful.

CONTRAINDICATIONS: Prophylactic pacing – the pacer electrodes may be placed in the stable bradycardic patient but should not be used unless the patient deteriorates.

PRECAUTIONS: Transcutaneous pacing is not the initial step in treatment of either unstable bradycardia or bradycardic/systolic cardiac arrest. Appropriate BLS and ALS maneuvers, as mandated by protocol, should be carried out first.

TECHNIQUE:

1. Connect ECG electrodes to patient in standard positions (pacer will not operate without ECG monitor intact).
2. Clean & dry chest. Remove excess hair if necessary to obtain electrode to skin contact.
3. Connect pacing electrodes to matching pacing cable.
4. Peel off protective covering and attach electrodes to patient's chest as described:
 - a. Anterior – Posterior (Preferred due to better capture rates and will not interfere with defibrillation.)
 - i. Place negative electrode on left anterior chest halfway between the xiphoid process and left nipple, with the upper edge of the electrode below the nipple line.
 - ii. Place positive electrode on left posterior chest beneath the scapula and lateral to the spine.
 - b. Anterior – Anterior
 - i. Place negative electrode on left chest mid-axillary over fourth interspace.
 - ii. Place positive electrode on right chest, sub-clavicular area.
5. Push “pacer” button.
6. If no intrinsic beats then skip this step. Ensure that each QRS complex is being sensed. If not, then adjust the QRS size and change between leads I, II, III to get the best size.
7. Select pacing rate
 - a. Adults and children ≥ 12 years, begin at 70 bpm.
 - b. Children ≥ 1 year and < 12 years, begin at 90 bpm.
 - c. Infants < 1 year, begin at 100 bpm.
8. Activate pacer by pushing pacer button. The pacer will display a drop-down menu to permit selection of rate, mA and to initiate and terminate pacing.
9. Increase current by 10 mA increments until electrical capture occurs. (Note: The pacer adjusts upwards by 10 mA and downwards by 5 mA.)

10. Determine that mechanical capture has occurred. If so, consider using tapeto secure electrodes to the patches to prevent displacement.

11. To terminate pacing, select STOP from the pacer screen drop down box.

COMPLICATIONS:

1. Patient discomfort
 - a. Pacing may be painful to the awake patient
 - b. Sedation may be useful (simple reassurance may also be useful) (ALS ONLY)

DOCUMENTATION REQUIREMENTS:

1. Monitor information uploaded to EPCR.
2. Documentation of indications for and response to pacing (including mechanical capture)

NOTES:

1. Electrical capture
 - a. Usually evidenced by wide QRS
 - b. In some patients, capture is less obvious and only manifested by minor changes in QRS configuration
2. Mechanical capture
 - a. Manifested by signs of improving cardiac output (i.e. increased level of consciousness or BP)
 - b. Do not confuse skeletal muscle contractions for carotid or femoral pulse
3. CPR may be safely accomplished during this procedure
4. Defibrillation/cardioversion may be done without removing pacing electrodes. Charging the defibrillator automatically prompts to turn off the pacer. When pacing is restarted, the pacer returns to the last used setting.
5. To obtain adequate tissue perfusion, rates may need to be higher than the initial setting.
6. If pacing is not successful, consider further ALS interventions as indicated.
7. While pacing, if you initiate the defibrillation sequence by selecting an energy level, a pop-up box will display a selection to stop pacing.
8. After defibrillation, when reinitiating pacing, the machine returns to the last used settings.



ADENOCARD (ADENOSINE)



Indications: PSVT

Contraindications: Second or third-degree heart block
Known hypersensitivity to the drug
Sick-sinus syndrome or symptomatic bradycardia Atrial fibrillation or flutter

Side Effects: Dyspnea,
Chest pressure/pain Dizziness
Facial flushing
Arrhythmias including blocks

Precautions: Patients with asthma
Children

Dose: **ADULTS:** 12mg rapid IV/IO push.
If no conversion: May repeat x1
PEDS: 0.1mg/kg MAX 6mg IV/IO
If no conversion: 0.2mg/kg MAX 12mg IV/IO
CONTACT MEDICAL CONTROL PRIOR TO GIVING TO PEDIATRIC PATIENTS

Route: IV/IO; Should be administered into the medication port closest to the patient and followed by flushing the line.

Protocol Used in: Adult Tachycardia: Narrow Complex
Pediatric Tachycardia: Narrow Complex



ALBUTEROL



Indications: Wheezing

Contraindications: Known sensitivity
Heart rate >150

Side Effects: Increased heart rate
Increased blood pressure
Tremors, nervousness
May exacerbate congestive heart failure

Dose: **ADULTS:** 2.5 mg nebulized. May repeat x2 as needed
PEDS: 2.5 mg nebulized. May repeat x2 as needed

Route: Nebulized

Protocol Used In: Adult Asthma/COPD
Combined Allergic Reaction
Pediatric Wheezing



AMIODARONE



Indications:	VF Pulseless VT Wide complex Tachydysrhythmias
Contraindications:	Known hypersensitivity to the drug or any of its components including iodine Sinus bradycardia Second or third-degree heart block
Precautions:	Exacerbation of presenting arrhythmia can occur with concurrent use of other antiarrhythmic medications (whatever antiarrhythmic you start with, stay with) CHF Children
Side Effects:	Prolonged QT interval Hypotension Bradycardia
Doses:	VF/Pulseless VT: <u>ADULTS:</u> 300mg IV/IO repeat 150mg IV/IO X1 after 5 minutes <u>PEDS:</u> 5mg/kg IV/IO MAX 300mg repeat 2.5mg/kg IV/IO MAX 150mg X1 after 5 minutes Wide complex tach: <u>ADULTS:</u> 150mg IV/IO over 10 minutes <u>PEDS:</u> 5mg/kg IV/IO MAX 300mg over 20 minutes <u>CONTACT MEDICAL CONTROL PRIOR TO GIVING TO PEDIATRIC PATIENTS</u>
Route:	IV, IO
Protocol Used In:	Adult Cardiac Arrest: VF/Pulseless VT Adult Tachycardia: Wide Complex Pediatric Cardiac Arrest: VF/Pulseless VT Pediatric Tachycardia: Wide Complex



ASPIRIN



Indications:	Chest pain suggestive of AMI
Contraindications:	Known hypersensitivity to the drug Patients with bleeding disorders Children <12 with flu-like symptoms Peptic ulcer
Precautions:	GI bleeding or upset stomach Patients with asthma
Side Effects:	Heartburn Nausea Wheezing GI bleeding Reye's syndrome in children
Dose:	<u>ADULTS:</u> 81mg x4 PO
Route:	oral/must be chewed
Protocol Used In:	Adult Chest Pain/STEMI



ATROPINE



Indications:	Bradycardia Organophosphate poisoning
Contraindications:	Tachycardia/Tachydysrhythmias
Side Effects:	Palpitations, Tachycardia, PVCs, HTN, Pupil dilation, Nausea/vomiting
Doses:	Unstable bradycardia: <u>ADULTS:</u> 0.5 mg initially, then 1 mg IV every 3-5 minutes MAX 3mg. <u>PEDS:</u> 0.02mg/kg IV/IO minimum dose 0.1mg every 3-5 minutes MAX 1mg Organophosphate poisoning: <u>ADULTS:</u> 3 mg IV (<u>MUST CONTACT MEDICAL CONTROL</u>) <u>PEDS:</u> <u>CONTACT MEDICAL CONTROL</u>
Route:	IV, IO, IM (only for organophosphate poisoning)
Protocol Used In:	Combined Overdose/Poisoning Adult Bradycardia Pediatric Bradycardia



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

ATROVENT (Ipratropium Bromide)



PHARMACOLOGY AND ACTIONS:

Atrovent is chemically related to atropine, but has minimal systemic absorption in the inhaled form. Atrovent causes bronchodilation without the anticholinergic side effects of atropine.

INDICATIONS:

Bronchospasm in patients with emphysema or chronic bronchitis.

CONTRAINDICATIONS: N/A

PRECAUTIONS:

Do not neglect more basic maneuvers such as oxygen, airway control, and proper ventilation. Generally considered safe but is controversial in patients with asthma.

ADMINISTRATION:

For specific doses, refer to the applicable protocol.
Acceptable routes of administration include nebulizer only.

PROTOCOLS USED IN:

- A. Adult COPD/Asthma
- B. Pediatric Wheezing



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

BENADRYL (DIPHENHYDRAMINE)



Indications:	Allergic Reactions Anaphylaxis with respiratory distress Anaphylactic shock Dystonic reaction
Contraindications:	Acute asthma attack
Precautions:	Patient with asthma
Side Effects:	Drowsiness, Headache, Palpitations, Dizziness, Dilated pupils Thickening of bronchial secretions
Dose:	<u>ADULT/PEDS:</u> 1 mg/kg, up to a total dose of 50 mg
Routes:	IV, IO, IM
Protocol Used In:	Combined Allergic Reaction Combined Dystonic Reaction



CALCIUM CHLORIDE

Indications:	Calcium channel blocker overdoses with hypotension Known or suspected hyperkalemia Known or suspected hypocalcemia
Contraindications:	Hypercalcemia VF Digitalis toxicity
Precautions:	Patients on digitalis Children Renal disease and respiratory disease Make sure IV is completely patent because it will cause severe tissue necrosis. Calcium chloride will precipitate when mixed with sodium bicarbonate. Flush 20-30 mL between medications or use separate lines.
Side Effects:	Pain or burning at IV site Necrosis of tissue Shortened QT Heart block Arrhythmias
Dose:	<u>ADULTS/PEDS:</u> 0.2 ml/kg IV MAX 10 ml <u>MUST CONTACT MEDICAL CONTROL</u>
Route:	IV
Protocol Used In:	Combined Overdose/Poisoning; Adult Cardiac Arrest- RENAL FAILURE/DIALYSIS

**D10**

Indications:	Hypoglycemia
Contraindications:	Hyperglycemia Delirium tremens Cranial/spinal hemorrhage CHF
Precautions:	Will cause severe tissue necrosis
Side Effects:	Local venous irritation Pulmonary edema Hyperglycemia
Dose:	<u>ADULTS:</u> 125mL IV/IO. <u>PEDS:</u> 5mL/kg IV/IO MAX 125mL Recheck blood glucose and repeat x1 if patient remains symptomatic and blood glucose remains < 70mg/dL. <u>If additional doses needed beyond 250mL contact Medical Control</u>
Route:	IV, IO (If IO established, transport is MANDATORY)
Protocol Used in:	Combined Hypoglycemia

**EPINEPHRINE 1:1000**

Indications:	Asthma Mild anaphylaxis Anaphylaxis with respiratory distress Anaphylactic shock Children with unstable bradydysrhythmias
Contraindications:	N/A
Precautions:	Epinephrine should be used with caution in patients with suspected myocardial ischemia, history of coronary artery disease, or age greater than 50. In patients greater than 50 years of age, the role epinephrine in asthma is controversial. In patients taking digitalis, epinephrine may exacerbate ventricular ectopy.
Side Effects:	Tremors Anxiety Palpitations Tachycardia HTN Dysrhythmias Dyspnea
Doses:	Asthma: <u>ADULTS:</u> 0.3mg IM <u>PEDS:</u> 0.01mg/kg IM MAX 0.3mg Mild anaphylaxis/Anaphylaxis with respiratory distress: <u>ADULT/PEDS:</u> 0.01 mg/kg IM, up to 0.3 mg IM <u>THIS MAY BE REPEATED WITH MEDICAL CONTROL APPROVAL</u> Anaphylactic shock: <u>ADULT/PEDS:</u> 0.01mg/kg up to 0.3mg IM may repeat x1 <u>CONTACT MEDICAL CONTROL IF ADDITIONAL DOSES ARE NEEDED</u> Unstable Bradycardia in children < 8yrs old: <u>PEDS:</u> 0.02mg/kg IV/IO minimum dose 0.1mg every 3-5 minutes MAX 1mg
Route:	IV, IO, IM
Protocol Used In:	Combined Allergic Reaction Adult Asthma/COPD Pediatric Bradycardia Pediatric Wheezing



EPINEPHRINE 1:10,000



Indications:	Asystole/PEA Persistent or recurrent VF/VT
Contraindications:	None in pulseless arrest situations
Precautions:	None in pulseless arrest situations
Doses:	<u>ADULTS:</u> 1 mg IV/IO every 3-5 minutes <u>PEDS:</u> 0.01mg/kg IV/IO every 3-5 minutes
Route:	IV, IO
Protocol Used In:	Adult Cardiac Arrest: Asystole/PEA Adult Cardiac Arrest: VF/Pulseless VT Pediatric Cardiac Arrest: Asystole/PEA Pediatric Cardiac Arrest: VF/Pulseless VT



EPINEPHRINE 1:100,000



Indications:	Used to treat unremitting hypotension after appropriate fluid resuscitation in post-cardiac arrest, cardiogenic shock, or other medical etiology. Used in bradycardia if poor perfusion persists secondary to transcutaneous pacing and appropriate fluid resuscitation.
Contraindications:	Cardiac Arrest Trauma etiology Pediatric patients
Precautions:	Use caution in patients with suspected myocardial infarction or history of coronary artery disease. Epinephrine may exacerbate ventricular ectopy.
Doses:	<u>ADULTS ONLY:</u> 10-20mcg every 2-3 minutes if SBP < 90mmHg
Route:	IV/IO
Protocol Used In:	Combined Hypotension/Shock Combined Post Resuscitation Adult Bradycardia



FENTANYL



Indications:	The relief of moderate to severe acute pain
Contraindications:	- Trauma - Head injury - Paralysis or new neurologic finding - Thoraco-abdominal trauma (blunt or penetrating) - Burns involving the airway or causing respiratory compromise - Systems - Respiratory distress or compromise (Asthma, COPD, or CHF) - Cardiac dysrhythmia present - Altered mental status - Pregnancy - Drug or alcohol use - Sensitivity or allergy to opiates - Hypotension or suspected shock (SBP < 100mgHG) or Pulse rate < 60bpm - Age < 12 months
Precautions:	Should only be given while monitoring pulse oximetry and end tidal CO2
Side Effects:	- Muscle Rigidity Must be given slow or it can cause muscle rigidity (stone chest or rigid chest syndrome) - Bradycardia - Respiratory Depression - Laryngospasm - Hypotension
Dose:	<u>ADULTS:</u> 50-100mcg slow IV or IN. May repeat every 5 minutes to a total MAX given of 200mcg. <u>PEDS:</u> 1mcg/kg slow IV or IN. May repeat x1. <u>Must call Medical Control if more than 200mcg are needed.</u>
Route:	IV or IN
Protocol Used In:	Combined Pain Control



KETOROLAC (TORADOL)



Indications: The relief of mild to moderate acute pain.

Contraindications: Allergy to NSAIDs
NSAID use in last 4 hours
Severe renal disease or kidney transplant or one kidney
Bleeding disorders
Pregnancy or nursing mother

Precautions: This is an NSAID.

Side Effects: Ringing in ears
Headache
Dizziness
Bloating/gas
N/V/D
Diaphoresis
Blurred Vision

Dose: **ADULTS:** 15mg IV, 30mg IM/IN. Do not repeat.

PEDS: 0.5mg/kg to max dose 15IV, 30mg IM/IN. Must be over 10 years old. Do not repeat.

Route: IV or IN/IM

Protocol Used In: Combined Pain Control



Levalbuterol (Xopenex)

Indications:	Bronchospasm
Contraindications:	Hypersensitivity to Levalbuterol or albuterol. Severe Tachycardia (relative contraindication)
Adverse Reactions:	Tachycardia Palpitations Tremors Anxiety Dysrhythmias Headaches Vomiting
Dosage:	Adults: 1.25mg in 3mL by nebulizer Repeat as needed every 10 minutes. Max up to 6 non-continuous doses (7.5mg) Pediatrics: 1.25mg in 3mL by nebulizer. Repeat as needed every 10 minutes. Max up to 3 non-continuous doses (3.75mg)
Note:	Can combine w Atrovent.



MAGNESIUM SULFATE



Indications:	Torsades de pointes Eclampsia
Contraindications:	Renal disease Acute surgical abdomen Nausea/Vomiting
Side Effects:	Sharply lowered blood pressure Respiratory depression
Precautions:	Known to cause bradycardia Known to cause hypotension Known to cause respiratory depression
Doses:	Torsades: <u>ADULTS ONLY:</u> 2gm IV/IO over 2 minutes Eclampsia: <u>ADULTS ONLY:</u> 4gm IV/IO over 5 minutes.
Routes:	IV, IO
Protocol Used In:	Combined Seizure Adult Cardiac Arrest: VF/Pulseless VT Adult Tachycardia: Wide Complex



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

Methylprednisolone (Solu-Medrol)



PHARMACOLOGY AND ACTIONS:

- A. Methylprednisolone is a synthetic steroid that suppresses acute and chronic inflammation.
- B. Potentiates vascular smooth muscle relaxation by beta-adrenergic agonists and may alter airway Hyperactivity

INDICATIONS:

- A. Severe asthma
- B. COPD
- C. Suspected Addisonian crisis (cardiovascular collapse in patient at risk for adrenal insufficiency)

CONTRAINDICATIONS:

- A. Evidence of active GI bleed

PRECAUTIONS:

- A. Most adverse reactions are a result of long-term therapy and include gastrointestinal bleeding, hypertension, and hyperglycemia

ADMINISTRATION:

- A. For specific doses, refer to the applicable protocol
- B. Only acceptable routes of administration include intravenous (IV), and intraosseous (IO)

SPECIAL CONSIDERATIONS:

- A. Must be reconstituted and used immediately
- B. The effect of methylprednisolone is generally delayed for several hours
- C. Methylprednisolone is not considered a first line drug. Be sure to attend to the patient's primary treatment priorities (i.e. airway, ventilation, beta-agonist nebulization) first.
- D. If primary treatment priorities have been completed and there is time while in route to hospital, then methylprednisolone can be administered.
- E. Don not delay transport to administer this drug

PROTOCOLS USED IN:

- A. Adult Asthma/COPD
- B. Pediatric Wheezing



Morphine



CLASS:

- narcotic analgesic
- opiate receptor agonist

ACTIONS:

- The actions are a result of its effects on the CNS
- Depresses the vasomotor center therefore increasing venous capacitance and decreasing PVR
- Pain and anxiety are relieved by a central effect that raises the pain threshold, producing euphoria sedation.
- Depresses the respiratory and cough centers in the brain.

INDICATIONS:

- Suspected cardiac-related chest pain unrelieved by nitroglycerin
- Cardiogenic pulmonary edema (CHF).
- Systolic BP (SBP) greater than 90 mmHg.
- Pain associated with burns and extremity trauma.

PHARMACOKINETICS:

- Onset: Immediate
- Duration: 2 to 7 hours

DOSAGE/ADMINISTRATION:

Adult:

- 2-10 mg IV/SL
- No greater than 3 mg/minute

SIDE EFFECTS:

- CNS: sedation, somnolence, euphoria
- EENT: facial flushing
- CV: hypotension, bradycardia
- RESP: respiratory depression
- GI: nausea, vomiting



NARCAN (NALOXONE)



Indications:	Narcotic overdoses Coma or confusion of unknown origin
Contraindications:	Known sensitivity
Precautions:	May cause withdrawal effects in patients dependent on narcotics.
Side Effects:	Abrupt reversal may cause nausea, vomiting, sweating, tachycardia, increased blood pressure, seizures
Dose:	<u>ADULTS:</u> 0.4 – 2mg IV or up to 4mg IN <u>PEDS:</u> 0.1mg/kg up to 2mg IV or up to 4mg IN
Routes:	IV, IO, IN Do not give IM. Absorption rates are unpredictable.
Protocol Used In:	Combined Altered Mental Status Combined Overdose/Poisoning



NITROGLYCERIN

Indications:	Chest pain CHF
Contraindications:	Systolic BP <100 Pulse rate < 50 and >140 Presence of Inferior STEMI Use of Erectile Dysfunction medications over the last 36 hours Known sensitivity
Precautions:	Monitor vital signs to maintain SBP > 100mmHg
Side Effects:	Headache Postural hypotension Flushing Dizziness
Dose:	Chest Pain: <u>ADULTS ONLY:</u> 0.4 mg SL every five minutes until chest pain is relieved or systolic blood pressure becomes < 100mmHg. CHF: <u>ADULTS ONLY:</u> 0.4 mg SL every five minutes until distress is relieved or systolic blood pressure becomes < 100mmHg.
Route:	SL
Protocol Used In:	Adult Chest Pain/STEMI Adult CHF/Pulmonary Edema



SODIUM BICARBONATE



Indications:	Tricyclic antidepressant overdose with wide complex tachycardia
Contraindications:	Respiratory Alkalosis Metabolic Alkalosis
Precautions:	May cause alkalosis or tetany Calcium chloride will precipitate when mixed with sodium bicarbonate. Flush 20-30 mL between medications or use separate lines.
Dose:	<u>ADULTS/PEDS:</u> 1 meq/kg to a total of 50 meq <u>Must contact medical control prior to administering for tricyclic antidepressant overdoses.</u>
Route:	IV/ IO
Protocol Used In:	Combined Overdose/Poisoning; Adult Cardiac Arrest- RENAL FAILURE/DIALYSIS

**VERSED (MIDAZOLAM)**

Indications:	Seizures Chemical restraint Sedation
Contraindications:	Known sensitivity to the drug or benzodiazepines Shock
Precautions:	Monitor BP Monitor SpO2 Monitor ETCO2
Side Effects:	Respiratory depression Apnea Hypotension Nausea/vomiting
Dose:	Chemical restraint: <u>ADULTS:</u> 0.1mg/kg IM or IN up to 10mg <u>PEDS:</u> 0.1mg/kg IM or IN up to 5mg <u>CONTACT MEDICAL CONTROL FOR ADDITIONAL DOSES</u> Seizures: <u>NOI</u>ESTABLISHED: <u>ADULTS/PEDS:</u> 0.2mg/kg IN MAX 5mg may repeat x1 <u>IV</u> ESTABLISHED: <u>ADULTS:</u> 5mg slow IV may repeat x1 <u>PEDS:</u> 0.1mg/kg slow IV MAX 5mg <u>CONTACT MEDICAL CONTROL IF ADDITIONAL DOSES NEEDED.</u> Sedation: <u>ADULTS ONLY:</u> 2mg slow IV or IN. May repeat every 5 minutes to total MAX 5mg given.
Route:	IV, IN, IM (only use IM route on combative patients)
Protocol Used In:	Combined Behavioral/Psychiatric Disorders Combined Seizures Sedation Procedure



ZOFRAN (ONDANSETRON)



Indications:	Nausea/Vomiting
Contraindications:	Known sensitivity Severe Liver Disease Prolonged QT syndrome Cardiac dysrhythmias
Precautions:	May prolong QT interval resulting in cardiac dysrhythmia
Dose:	<u>ADULTS:</u> 4 mg IV/IM <u>PEDS OVER 4 YEARS OLD:</u> 2 mg IV/IM
Route:	IV, IM, ODT (orally disintegrating tablet)
Protocol Pages:	Combined Non-Traumatic Abdominal Pain/Vomiting

**12 LEAD ECG INTERPRETATION**

I Lateral	aVR	V1 Septal	V4 Anterior
II Inferior	aVL Lateral	V2 Septal	V5 Lateral
III Inferior	aVF Inferior	V3 Anterior	V6 Lateral
SITE		FACING	RECIPROCAL
SEPTAL		V1, V2	NONE
ANTERIOR		V3, V4	NONE
ANTEROSEPTAL		V1, V2, V3, V4	NONE
LATERAL		I, aVL, V5, V6	II, III, aVF
ANTEROLATERAL		I, aVL, V3, V4, V5, V6	II, III, aVF
INFERIOR		II, III, aVF	I, aVL
POSTERIOR		NONE	V1, V2, V3, V4



APGAR SCORE

THE APGAR SCORE

MNEMONIC	0 POINTS	1 POINTS	2 POINTS
A pppearance	blue or pale	blue extremities pink body	body and extremities pink, no cyanosis
P ulse	absent	< 100 beats per minute	>100 beats per minute
G rimace	no response to stimulation, floppy	grimace on suction or aggressive stimulation	cry on stimulation
A ctivity	none	some flexion of arms and legs	active flexion against resistance
R espirations	absent	weak, irregular and slow	strong crying

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PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

CINCINNATI PREHOSPITAL STROKE SCALE



Interpretation: if any of these 3 signs is abnormal, the probability of a stroke is 72%

 Arm Drift <i>The patient closes eyes and extends both arms straight out, with palms up for 10 seconds</i> • Normal – both arms move the same or both arms do not move at all (other findings, such as pronator drift, may be helpful) • Abnormal – one arm does not move or one arm drifts downward	Facial Droop <i>The patient shows teeth or smile</i> • Normal – both sides of the face move equally • Abnormal – one side of the face does not move as well as the other side  Abnormal Speech <i>The patient repeats “you can’t teach an old dog new tricks”</i> • Normal – patient uses correct words with no slurring • Abnormal – patient slurs words, uses the wrong words, or is unable to speak
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**GLASGOW COMA SCALE****ADULT GLASGOW COMA SCALE**

Eye Opening Response	
• Spontaneous – open with blinking at baseline	4
• To verbal stimuli, command, speech	3
• To pain only (not applied to face)	2
• No response	1
Verbal Response	
• Oriented	5
• Confused conversation, but able to answer questions	4
• Inappropriate words	3
• Incomprehensible speech	2
• No response	1
Motor Response	
• Obeys commands for movement	6
• Purposeful movement to painful stimulus	5
• Withdraws in response to pain	4
• Flexion: decorticate posturing	3
• Extension: decerebrate posturing	2
• No response	1
Total	

PEDIATRIC GLASGOW COMA SCALE (PGCS)

	> 1 Year		< 1 Year	Score
EYE OPENING	Spontaneously		Spontaneously	4
	To verbal command		To shout	3
	To pain		To pain	2
	No response		No response	1
MOTOR RESPONSE	Obeys		Spontaneous	6
	Localizes pain		Localizes pain	5
	Flexion-withdrawal		Flexion-withdrawal	4
	Flexion-abnormal (decorticate rigidity)		Flexion-abnormal (decorticate rigidity)	3
	Extension (decerebrate rigidity)		Extension (decerebrate rigidity)	2
	No response		No response	1
	> 5 Years	2-5 Years	0-23 months	
VERBAL RESPONSE	Oriented	Appropriate words/phrases	Smiles/coos appropriately	5
	Disoriented/confused	Inappropriate words	Cries and is consolable	4
	Inappropriate words	Persistent cries and screams	Persistent inappropriate crying and/or screaming	3
	Incomprehensible sounds	Grunts	Grunts, agitated, and restless	2
	No response	No response	No response	1
TOTAL PEDIATRIC GLASGOW COMA SCORE (3-15):				



LUCAS DEVICE



LUCAS[®] 3

CHEST COMPRESSION SYSTEM

QUICK REFERENCE GUIDE

	Rescuer 1 (LUCAS device operator)	Rescuer 2
   Manual positioning of the Suction Cup   	1. Power on the LUCAS device • Push ON/OFF to start self-test and power up the LUCAS device • The device will be ready and in the ADJUST mode	 • Provide manual CPR
	2. Place the LUCAS BACK PLATE • Pause manual CPR briefly • Put the BACK PLATE under the patient, immediately below the armpits	 • Assist BACK PLATE placement • Resume manual CPR
	3. Attach the UPPER PART • Pull the RELEASE RINGS once to open CLAW LOCKS. Then let go of the rings • Stop manual CPR briefly while attaching the UPPER PART to the BACK PLATE. Listen for "CLICK" sound • Pull up once to assure attachment	 • Continue manual CPR as long as possible • Help to attach the UPPER PART
	4. Push down SUCTION CUP. Adjust position if needed. • Push down the SUCTION CUP • The lower edge of SUCTION CUP should be immediately above the end of the sternum • Adjust if necessary (stay in ADJUST mode)	 • Assist
	5. Lock position. Start compressions. • Push PAUSE to lock START POSITION • Push ACTIVE (continuous) or ACTIVE (30:2) to start compressions	 • Assist
Attach stabilization strap. Follow CPR protocols.		

The LUCAS 3 device is for use as an adjunct to manual CPR when effective manual CPR is not possible (e.g., transport, extended CPR, fatigue, insufficient personnel). Refer to the Instructions For Use for complete directions for use, indications, contraindications, warnings, precautions and potential adverse events.

Physio-Control is now part of Stryker.

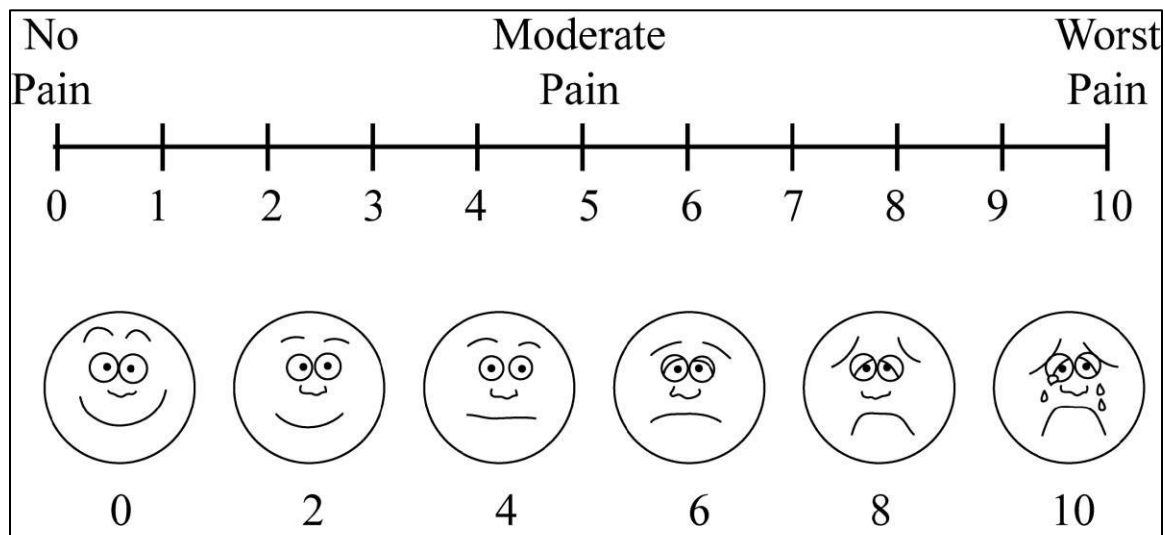
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stryker



PAIN SCALE





PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS
PEDIATRIC VITAL SIGN RANGES BY AGE



General Vital Signs and Guidelines

Age	Heart Rate (beats/min)	Blood Pressure (mmHg)	Respiratory Rate (breaths/min)
Premature	110-170	SBP 55-75 DBP 35-45	40-70
0-3 months	110-160	SBP 65-85 DBP 45-55	35-55
3-6 months	110-160	SBP 70-90 DBP 50-65	30-45
6-12 months	90-160	SBP 80-100 DBP 55-65	22-38
1-3 years	80-150	SBP 90-105 DBP 55-70	22-30
3-6 years	70-120	SBP 95-110 DBP 60-75	20-24
6-12 years	60-110	SBP 100-120 DBP 60-75	16-22
> 12 years	60-100	SBP 110-135 DBP 65-85	12-20



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS
EPINEPHRINE 1:100000 (PUSH DOSE)



EPINEPHRINE

Has alpha and beta_{1/2} effects so it is an inopressor

Do not give cardiac arrest doses (1 mg) to patients with a pulse

Mixing Instructions:

- Take a 10 ml syringe with 9 ml of normal saline
- Into this syringe, draw up 1 ml of epinephrine from the cardiac amp (Cardiac amp contains Epinephrine 100 mcg/ml)
- Now you have 10 mls of Epinephrine 10 mcg/ml

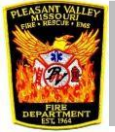
Onset-1 minute

Duration-5-10 minutes

Dose-0.5-2 ml every 2-5 minutes (5-20 mcg)

Route: IV/IO



**RECOGNIZED ABBREVIATIONS****RECOGNIZED ABBREVIATIONS**

AAA	Abdominal Aortic Aneurysm
Abd	Abdomen
APAP	Acetaminophen
AIDS	Acquired Immune Deficiency Syndrome
AMI	Acute myocardial infarction
ACLS	Advanced cardiac life support
ALS	Advanced life support
A	Airway (in primary survey)
ABC	Airway, Breathing, Circulation
A&Ox3	Alert and oriented, times three
A/O	Alert/Oriented
APGAR	Appearance, Pulse, Grimace, Activity, Respiration
≈	Approximately
ASHD	Arteriosclerotic heart disease
ASA	Aspirin
ASAP	As soon as possible
A-FIB	Atrial fibrillation
BVM	Bag valve mask
BCLS	Basic cardiac life support
BLS	Basic life support
B/F	Black female
B/M	Black male
BP	Blood pressure
BSA	Body surface area
BM	Bowel movement
B	Breathing (in primary survey)
CA	Cancer
CPR	Cardiopulmonary resuscitation
CATH	Catheter/Catheterization
CM	Centimeter
CNS	Central nervous system
CVA	Cerebral vascular accident
CSF	Cerebrospinal fluid
C-Collar	Cervical collar
C-Spine	Cervical spine
Δ	Changes
CP	Chest pain
CC	Chief complaint
COPD	Chronic obstructive pulmonary disease
C	Circulation (in primary survey)
C/O	Complains of
CHF	Congestive heart failure
CAO	Conscious, alert, oriented

CPAP	Continuous positive airway pressure
CABG	Coronary artery bypass graft
CAD	Coronary artery disease
D.O.A.	Dead on arrival
↓	Decreasing
DVT	Deep vein thrombosis
DEFIB	Defibrillate
DTs	Delirium tremors
Dept	Department
D.M.	Diabetes mellitus
DX	Diagnosis
D	Disability (neuro)
DC	Discontinue
DO	Doctor of Osteopathy
DNR	Do not resuscitate
Gtts	Drops
EKG	Electrocardiogram
↑	Elevated, increased
EMS	Emergency medical services
EMT	Emergency medical technician
EMT-P	Emergency medical technician paramedic
ER	Emergency room
ET	Endotracheal
EPI	Epinephrine
=	Equal
EOA	Esophageal obturator airway
ETA	Estimated time of arrival
ETOH	Ethyl alcohol
EJ	External jugular
♀	Female
FD	Fire department
1°	First, primary
FB	Foreign body
F/u	Follow up
FX	Fracture
GI	Gastrointestinal
Ga	Gauge
G#P#	Gravida # Para #
GYN	Gynecologist
HEENT	Head, ears, eyes, nose and throat
H/A	Headache
Hx	History
HPI	History of present illness
HOSP	Hospital
HIV	Human immunodeficiency virus
HTN	Hypertension
ID	Identification
↑	Increased, elevated
IDDM	Insulin dependent diabetes mellitus
ICU	Intensive care unit
ICP	Intracranial pressure

IM	Intramuscular
IO	Intraosseous
IV	Intravenous
IVPB	Intravenous piggyback
IVP	Intravenous push
JVD	Jugular vein distention
KED	Kendrick extrication device
Kg	Kilogram
LAC	Laceration
LR	Lactated ringers
LMP	Last menstrual cycle
LLQ	Left lower quadrant
LUQ	Left upper quadrant
<	Less than
LOC	Level of consciousness-loss of consciousness
LP 10	Lifepack 10
LP 11	Lifepack 11
LPM	Liters per minute
↓	Lower
L-Spine	Lumbar spine
♂	Male
MD	Medical doctor
MEDS	Medications
mcg	Micrograms
meq	Milliequivalent
MAST	Military anti-shock trousers
MOM	Milk of Magnesia
mg	Milligram
MI	Myocardial infarction
ml	Milliliter
mm	Millimeter
>	More than
AM	Morning
MVA	Motor vehicle accident
MAE	Moves all extremities
MOM	Milk of Magnesia
NC	Nasal cannula
NG	Nasogastric
N&V	Nausea and vomiting
NVD	Nausea, vomiting, diarrhea
-	Negative
NTG	Nitroglycerin
NKA	No known allergies
Ø	None
NIDDM	Non-insulin dependent diabetic mellitus
NRB	Non-rebreather mask
NS	Normal saline
NSR	Normal sinus rhythm
≠	Not equal
#	Number
N.H.	Nursing home

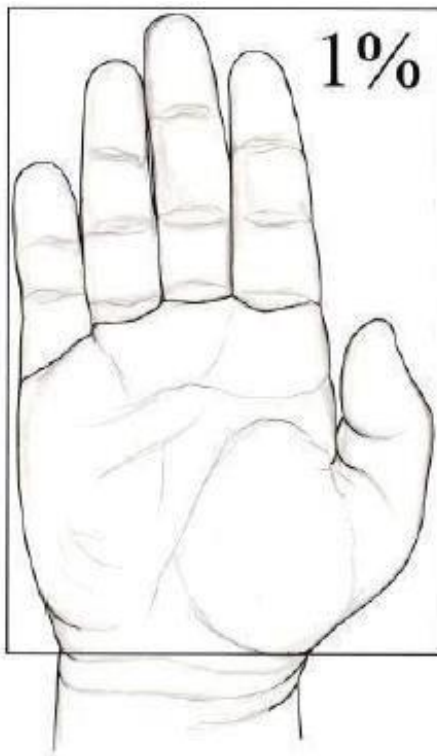
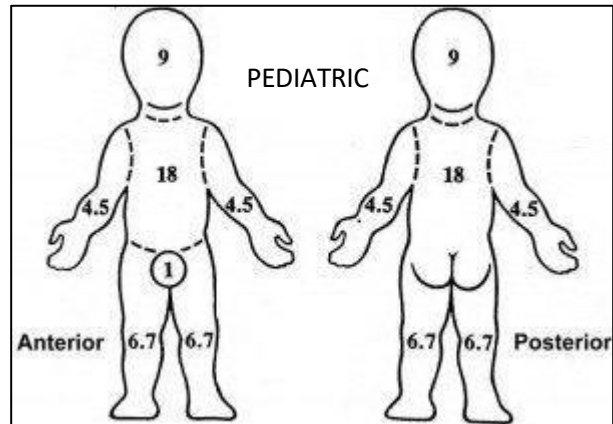
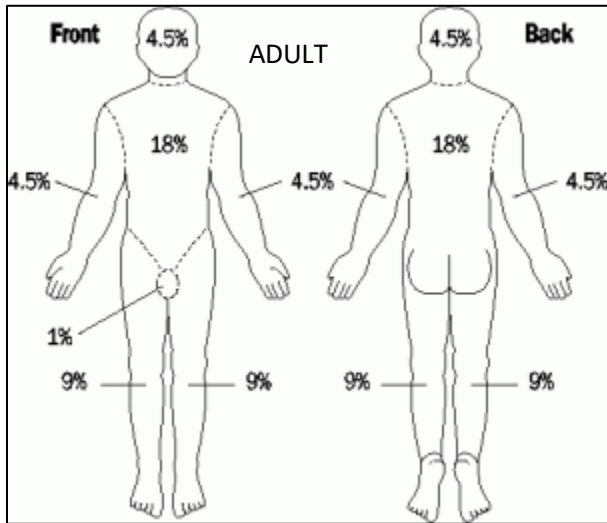
OB	Obstetrics
P.O.	Orally by mouth
oz	Ounce
OTC	Over The Counter
OD	Overdose
O2	Oxygen
SaO2	Oxygen Saturation Percentage
PMH	Past medical history
PT	Patient
PID	Pelvic inflammatory disease
PCN	Penicillin
PX	Physical exam
PASG	Pneumatic anti-shock garment
PD	Police department
+	Positive
KCL	Potassium chloride
lb	Pound
PAC	Premature atrial contraction
PJC	Premature junctional contraction
PVC	Premature ventricular contraction
RX	Prescription
1°	Primary, first
PTA	Prior to arrival
PMS	Pulse, Motor, Sensory
PERL	Pupils equal, reactive to light
ROM	Range of Motion
RN	Registered nurse
RESP	Respiration, respiratory
RLQ	Right lower quadrant
RUQ	Right upper quadrant
R/O	Rule out
2°	Second, secondary
Sz	Seizure
SAE	Sensation all extremities
SOB	Shortness of breath
S/S	Signs and symptoms
ST	Sinus tachycardia
BICARB	Sodium bicarbonate
SOP	Standard operating procedure
SIP	Status post
SQ	Subcutaneous
SL	Sublingual
SIDS	Sudden infant death syndrome
SVT	Supra ventricular tachycardia
3°	Tertiary, third
X	Times
3°	Third, tertiary
TKO	To keep open
TIA	Transient ischemic attack
TX	Treatment
TB	Tuberculosis
U/A	Upon arrival

URI	Upper respiratory infection
UTI	Urinary tract infection
VD	Venereal disease
V-FIB	Ventricular fibrillation
V-TACH	Ventricular tachycardia
VS	Vital signs
WT	Weight
c ⁻	With
W/O	Without
s ⁻	Without
Y/O	Years old



PLEASANT VALLEY FIRE DEPARTMENT EMS PROTOCOLS

TOTAL BODY SURFACE AREA



Surface of patient's palm, including fingers is equal to approximately 1% of their Total Body Surface Area

TCD DESTINATIONS

TRAUMA CENTERS	STEMI CENTERS	STROKE CENTERS		BURN CENTERS	LVAD DESTINATIONS
		Primary	Comprehensive		
Liberty Hospital	Centerpoint	Belton Regional Medical Center	KU Medical Center	Children's Mercy Hospital	KU Medical Center
North Kansas City Hospital	KU Medical Center	Centerpoint	Research Medical Center	KU Medical Center	St Lukes Plaza
Truman Medical Center Hospital Hill	Lee's Summit Medical Center	Lee's Summit Medical Center	St Lukes Plaza	Research Medical Center	
Children's Mercy Hospital	Liberty Hospital	St Joseph Medical Center			
St Lukes Plaza	Menorah Medical Center	St Lukes East			
KU Medical Center	North Kansas City Hospital	St Lukes North			
Research Medical Center	Olathe Medical Center	St Mary's Blue Springs			
Overland Park Regional Medical Center	Overland Park Regional	North Kansas City			
Centerpoint Medical Center	Providence Medical Center	Liberty Hospital			
	Research Medical Center				
	Shawnee Mission Medical Center				
	St Joseph Medical Center				
	St Lukes East				
	St Lukes Plaza				
	St Lukes North				
	St Lukes South				
	St Mary's Blue Springs				
	Truman Medical Center Hospital Hill				